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Quantitative Equity Research

Quant Matters – 2026 DM Quant Equity Mid-Year Outlook: Factors for an AI and Energy-Led Cycle

AI capex and energy volatility frame a more dispersed DM factor cycle: in the US, lean into self-funded growth and earnings delivery; in Europe, prioritize cash-flow resilience and capital discipline; in Japan, balance earnings-based value with selective growth exposure as rates remain key.

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Key Takeaways

- US: emphasizing quality growth as AI productivity cycle offsets energy headwinds, we recommend Internal Growth, Low PEG, and Up vs. Down EPS Revisions.
- Europe: positioning for energy-driven inflation, we recommend Operating Cash Flow Yield, Low YoY Asset Growth Factor, and 3m Up vs. Down EPS Revisions.
- Japan: using a combined macroeconomic and fundamental lens, we recommend Forward P/E, Low PEG, and Up vs. Down EPS Revisions.

4-Dimensions Framework Rankings (framework in [this report](#)) & Factor Views:
Europe – Up vs Down EPS Revisions (3ma) (+), Operating CF Yield (+), Low Asset Growth (+). We revise Economic Cycle from Expansion to Slowdown. Comp Value weakened (5→11). Comp Growth climbed (17→13). Comp Quality dropped to bottom (15→17). Comp Momentum surged (9→2). Low Risk rebounded (14→7). Small Size slipped (2→6).

US – Up vs Down EPS Revisions (+), Internal Growth (+), Low PEG (+). Comp Value edged up (17→15). Comp Growth rose (16→14). Comp Quality slipped (11→12). Comp Momentum weakened (9→11). Low Risk softened (5→7). Small Size surged to top (12→1).

Japan – Up vs Down EPS Revisions (+). Forward Earnings Yield (+), Low PEG (+). Comp Value weakened (2→6). Comp Growth declined (11→14). Comp Quality held steady (15). Comp Momentum advanced (16→13). Low Risk slipped (9→10). Small Size held steady (5).

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[Quantitative Equity Research: Quant Matters – QAML \(Quant + Analyst + Machine Learning\): A Preview of Our New Stock Selection Model, 29 Apr 2026](#)

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Topic of the Month: 2026 DM Quant Equity Mid-Year Outlook: Factors for an AI and Energy-Led Cycle

This month, we present our mid-year outlook for quant equities in the US, Europe and Japan. We recap the views of our economists ([here](#)) and strategists ([here](#)) as a starting point to highlight the equity factors that are likely to stand out in this environment.

The first half of 2026 has reinforced a familiar but more complex market message: macro and micro fundamentals remain supportive for risk assets, but return dispersion is likely to stay elevated. Strong earnings reports, resilient US labor markets and a powerful AI-driven capex cycle continue to support developed market equities. At the same time, energy-led volatility, driven by the Middle East conflict and uncertainty around the duration of the supply shock, has widened the range of potential macro outcomes. In this environment, the core message from our global strategists is constructive, not complacent.

DM equities remain the asset class preferred by our cross-asset strategists ([here](#)), with expected returns in the low-teens across major developed regions. While upside appears broadly similar across the US, Europe and Japan, our strategists prefer the US given its favorable bull-bear skew, stronger earnings profile and support from positive operating leverage. Europe and Japan also offer upside, but with more region-specific constraints: Europe is more exposed to energy-price volatility, faster pass-through into inflation and a different ECB reaction function, while Japan's outlook hinges on energy costs, wage growth, governance reform and sector leadership tied to AI, materials and defense.

The most important macro uncertainty is the oil-price path. Our economists note that the level and duration of the Middle East-related supply shock will dictate economic outcomes, with growth and inflation elasticities becoming increasingly non-linear as oil prices rise and remain elevated. The base case assumes de-escalation, but the range of outcomes is wider than usual. In the US, higher energy prices act as a tax on consumption, partially offsetting fiscal support, but AI capex and resilient labor markets keep the economy on a constructive path. In Europe, energy-price pass-through is faster and more direct, leaving the ECB with a different reaction function from the Fed. In Japan, higher energy prices create pressure on margins, but the broader equity story remains supported by domestic reflation, structural capex demand and improving corporate governance.

Across regions, one common thread remains central: the AI capex cycle. Hyperscaler capex expectations have been revised sharply higher, and the investment impulse from AI infrastructure is now relevant not only for technology and communication services, but also for semiconductors, industrials, utilities, power equipment, robotics, data centers and parts of credit and securitized markets. For quant investors, this reinforces the importance of factors that combine earnings delivery, quality, valuation discipline, cash-flow resilience and positive revisions.

Against this backdrop, our regional factor recommendations are as follows. In the US, we continue to emphasize Internal Growth, Low PEG, and Up vs. Down EPS Revisions, reflecting a preference for quality growth, valuation-aware earnings growth and positive earnings momentum. In Europe, we recommend Operating Cash Flow Yield, Low YoY Asset Growth, and 3-Month Up vs. Down EPS Revisions Breadth, reflecting the need for

cash-flow resilience, capital discipline and earnings selectivity in an energy-sensitive market. In Japan, we recommend Forward Earnings Yield, Low PEG, and Up vs. Down EPS Revisions, combining value exposure, valuation-aware earnings growth and bottom-up earnings momentum in a rising but more dispersed equity market.

US quant factors – Quality Growth as AI Productivity Cycle Offsets Energy Headwinds

In [Exhibit 1](#), we show the key US forecasts from our economists and strategists for 2026 and 2027. Our economists expect US growth to soften modestly through mid-2026 as higher energy prices weigh on consumption, though AI-driven capex, fiscal support and resilient labor markets should continue to support underlying momentum. On an annual basis, real GDP growth is expected to be 2.2% in 2026 and 2.5% in 2027 as energy pressures fade and the economy avoids recession. Inflation remains above target in the near term, but both headline and core inflation are expected to decelerate over the forecast horizon, allowing the Fed to remain on hold through 2026 before delivering two cuts in 1H27.

Our US equity strategists continue to expect the rolling recovery to progress. The constructive equity view is primarily an earnings story rather than a multiple-expansion story. Positive operating leverage, AI adoption, “run it lean” efficiency gains, improving pricing power and continued momentum in the AI capex cycle are expected to support earnings growth. Sector preferences remain tilted toward Financials, Industrials, Consumer Discretionary Goods and hyperscalers.

Exhibit 1: Key economist and strategist forecasts – US

Macro	Current	Estimates	
S&P 500	7444.25	8300 (2Q 2027)	
US 10Y	4.47%	4.20% (2Q 2027)	
	2025	2026E	2027E
GDP (Annual)	2.1	2.2	2.5
Inflation (Annual)	2.7	3.5	2.1
Policy Rate	3.625	3.625	3.125

Source: Datastream, Morgan Stanley Research. Data as of 13 May 2026.

Reflecting this combination of resilient earnings, AI-led productivity gains and persistent but moderating inflation pressure, we maintain our preference for Growth with a quality bias. We recommend positioning via three equity factors: **Internal Growth, Low PEG, and Up vs. Down EPS Revisions**. Together, these factors capture companies with the ability to fund growth internally, combine sustainable growth with valuation discipline, and benefit from positive analyst revisions as productivity gains and pricing power are incorporated into earnings forecasts.

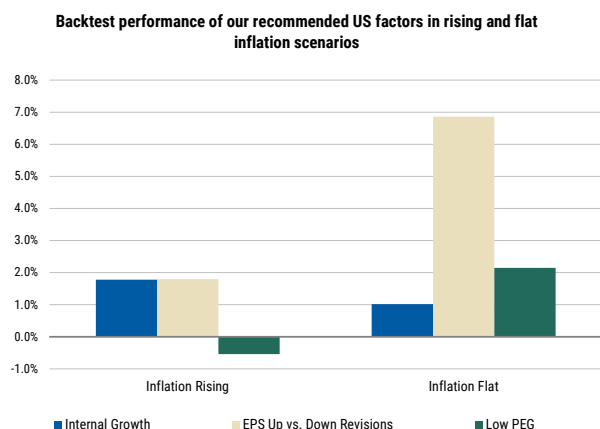
We evaluate this factor mix through three lenses. First, in [Exhibit 2](#), we present the backtested performance of our recommended US factors across rising and flat inflation regimes. Internal Growth and Up vs. Down EPS Revisions remain resilient across both regimes, while Low PEG performs particularly well in flatter inflation environments. This is consistent with the current macro setup: inflation remains a risk, but the direction of travel should become more benign as energy and tariff effects fade, allowing investors to

re-focus on earnings quality and growth durability.

Second, in [Exhibit 3](#), we evaluate the performance of our recommended US factors across normal and high Brent oil-volatility regimes. Internal Growth and Low PEG delivered positive returns in both regimes, with stronger performance during high oil volatility, highlighting the resilience of quality-growth and valuation-disciplined growth when energy uncertainty is elevated. Up vs. Down EPS Revisions performed positively in normal oil-volatility regimes but was weaker during high-volatility episodes, potentially because analyst estimates would lag the initial shock as earnings, margin and demand assumptions are recalibrated. This supports a balanced US factor basket: Internal Growth and Low PEG provide resilience during the energy-shock phase, while Up vs. Down EPS Revisions becomes more relevant as oil volatility stabilizes and earnings visibility improves.

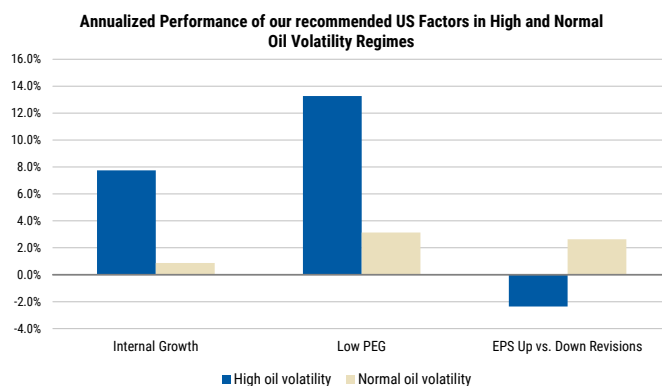
Third, in [Exhibit 4](#), we assess the thematic alignment of our recommended factors with the AI & Tech Diffusion main theme and AI Infrastructure sub-theme, measured as the mean exposure of top-quintile stocks minus bottom-quintile stocks. The exhibit shows positive exposure to the AI themes, reinforcing the alignment of our factor recommendations with our US equity strategists' emphasis on AI as a key driver of the earnings cycle. The exposure to AI Infrastructure is particularly relevant as AI-related capex remains an important global theme, with hyperscaler capex expectations revised sharply higher for 2026 and 2027.

Exhibit 2: Backtest performance of our recommended US factors in rising and flat inflation scenarios – Internal Growth and EPS Revisions Breadth remain resilient across regimes, while Low PEG benefits from flatter inflation environments



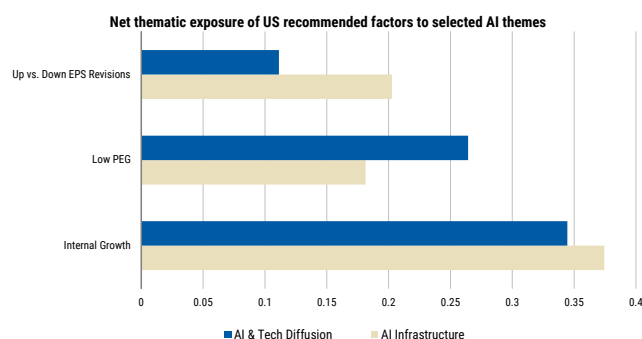
Source: Factset, Haver, Morgan Stanley Research. Inflation regimes based on spread between 1M and 6M moving averages of lagged inflation. Regimes classified as Rising when spread exceeds +0.15 and Flat when between -0.15 and +0.15. Factor returns are shown on an annualized basis. Data as of Dec 2024.

Exhibit 3: Internal Growth and Low PEG remain resilient in high oil volatility, while EPS Revisions improve as volatility normalizes



Source: Datastream, Morgan Stanley Research. Oil-volatility regimes are defined using weekly Brent prices. Realized oil volatility is calculated as the annualized 12-week (~3M) rolling standard deviation of weekly Brent returns. High and normal oil-volatility regimes correspond to periods above the 75th percentile and between the 25th and 75th percentiles, respectively. Factor returns are shown on an annualized basis.

Exhibit 4: Recommended US Factors Capture AI Diffusion and Infrastructure Capex Exposure



Source: FactSet, Datastream, Morgan Stanley Research. Each stock's sensitivity to thematic movements is estimated in two steps: we first obtain the thematic-specific return by regressing the thematic portfolio's weekly return on MSCI ACWI Equal-Weight (EQL) and defining the theme-specific component as the regression intercept plus the residual. We then regress each stock's weekly return on this thematic-specific return, controlling for EQL, to estimate its thematic beta. All regressions are run over a rolling two-year weekly estimation horizon (approximately 104 observations). Data as of April 30, 2026.

Having evaluated the recommended factors across the broader macro and thematic lenses, we now take a more granular view of the supportive characteristics behind each signal. We continue to favor a quality-oriented approach to growth investing, positioning Internal Growth as our preferred factor within the US equity landscape. Internal Growth is explicitly defined as:

$$\text{Internal Growth} = \text{Retention Ratio} \times \text{ROE}$$

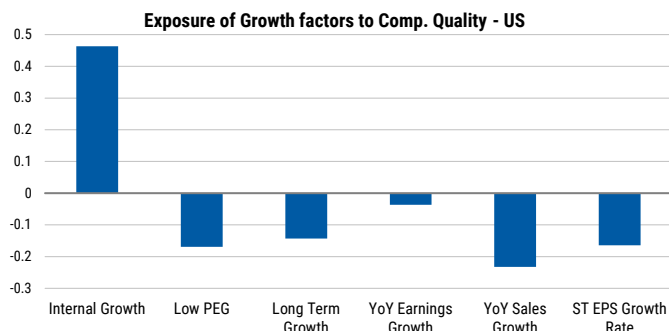
Among the growth factors we analyze, Internal Growth distinctly exhibits the strongest positive exposure to quality characteristics, measured as the difference between median Composite Quality scores of top quintile (Q1) and bottom quintile (Q5) portfolios (see [Exhibit 5](#)). This notable quality tilt underscores the factor's strategic appeal, identifying companies capable of sustainably funding growth internally through disciplined profitability and effective reinvestment.

To further illustrate the fundamental strengths underlying Internal Growth, we employ the DuPont decomposition framework to expand ROE into its component drivers:

$$\text{ROE} = \text{Profit Margin} \times \text{Asset Turnover} \times \text{Financial Leverage}$$

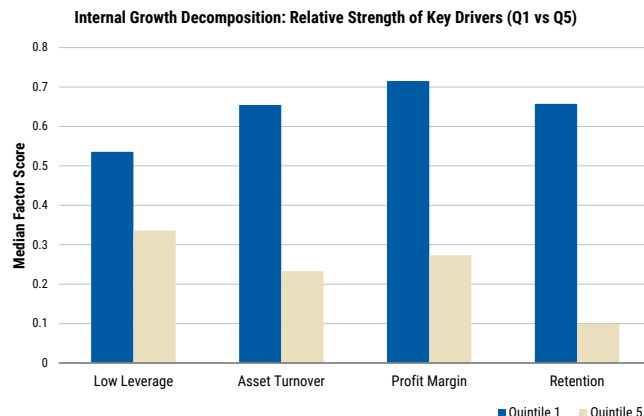
In [Exhibit 6](#), we apply this decomposition by comparing median fundamental exposures of top quintile (Q1) versus bottom quintile (Q5) firms. The top-quintile companies consistently demonstrate superior fundamentals across all key drivers – including notably stronger profit margins and greater asset efficiency, partly offset by lower financial leverage – highlighting their ability to sustain internally funded growth. This aligns with our US equity strategists' view that the rolling recovery is being driven by earnings durability, positive operating leverage and AI-enabled efficiency gains. Internal Growth is well suited to that backdrop because it captures companies with the profitability and retained earnings capacity to fund expansion internally.

Exhibit 5: Internal Growth factor stands out with strongest quality exposure, highlighting companies with sustainable profitability and disciplined reinvestment



Source: Factset, Morgan Stanley Research. Data as of April 30, 2026.

Exhibit 6: Top-quintile stocks in EPS revisions factor show superior operating leverage, margin momentum, and R&D intensity, aligned with strategists' rolling recovery outlook.



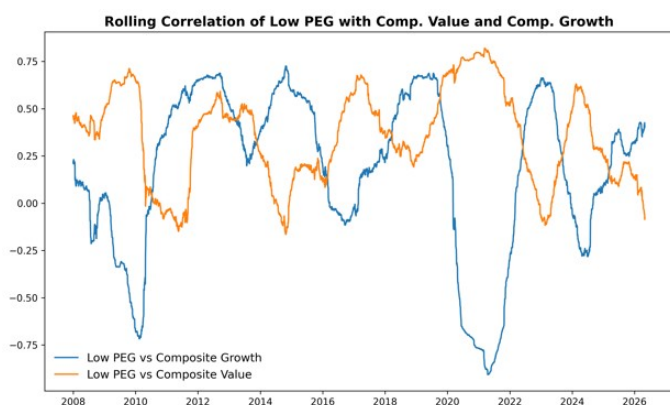
Source: Factset, Morgan Stanley Research. Data as of April 30, 2026.

As a complement to Internal Growth, **Low PEG** adds valuation discipline to the quality-growth framework. The Low PEG factor identifies companies with attractive valuations relative to their earnings growth trajectories.

Exhibit 7 shows the rolling 252-day correlation of Low PEG with Composite Value and Composite Growth. Low PEG has recently shown a higher correlation with Composite Growth than Composite Value, which is particularly relevant in the current US backdrop. Our strategists expect modest multiple compression, with the S&P 500 forward P/E declining to 20.5x from 21.2x, making their bullish index view an earnings story rather than multiple-expansion (see [report](#)). In this environment, a stronger relationship with Growth is preferable to a Value tilt, because returns are expected to come more from earnings delivery than from a broad re-rating of cheap stocks.

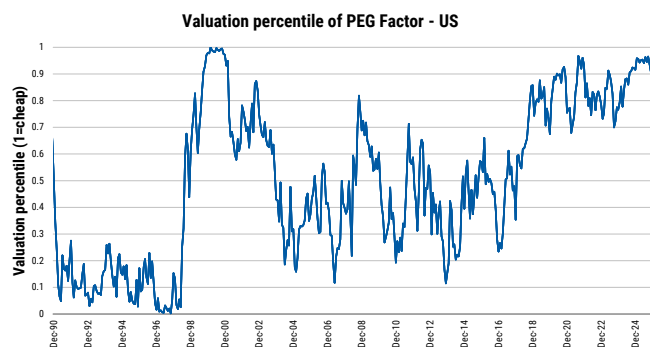
The factor is currently screening at an attractive valuation (**Exhibit 8**), trading at the 12th percentile of its historical range. In our view, the combination of growth sensitivity, valuation discipline and supportive starting valuations makes Low PEG well suited for an earnings-led market environment.

Exhibit 7: Low PEG's recent correlation profile shows greater alignment with growth than value



Source: Factset, Morgan Stanley Research. Data as of April 30, 2026.

Exhibit 8: Low PEG remains attractively valued versus history



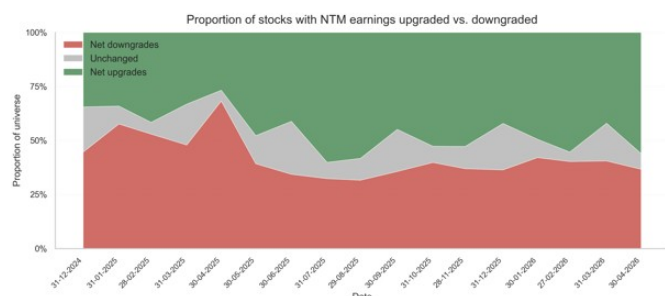
Source: Factset, Morgan Stanley Research. Data as of April 30, 2026.

Finally, **Up vs. Down EPS Revisions** adds a forward-looking earnings momentum signal to our US factor recommendation. The factor measures analyst upgrades relative to downgrades, normalized by the total number of analyst estimates, and captures the bottom-up channel through which improving earnings expectations are reflected in stock selection. This factor remains highly relevant as consensus incorporates three competing forces into forecasts: AI-driven productivity gains, improving pricing power, and inflation shocks from energy. In our view, revisions should remain one of the most direct channels through which the earnings-led recovery is reflected in quant performance.

In [Exhibit 9](#), we track the proportion of US stocks experiencing upward revisions in NTM earnings relative to downgrades on a monthly basis. 56% of stocks now have net positive earnings revisions, compared with 37% facing net downgrades, signaling broadening optimism in analyst sentiment and earnings expectations.

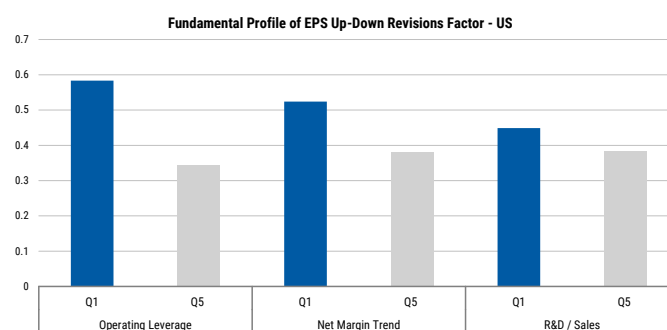
[Exhibit 10](#) provides additional support for this fundamental outlook, illustrating median factor scores for key attributes by comparing top-quintile stocks (Q1) versus bottom-quintile stocks (Q5). Stocks in the top quintile demonstrate notably higher operating leverage, stronger net margin trends, and greater investment intensity in R&D. Specifically, elevated operating leverage – defined as the 12-month change in EBIT divided by the 12-month change in sales – and superior net margin momentum – defined precisely as (Current Net Margin – 5-year average Net Margin) divided by the 5-year standard deviation of Net Margin – reinforce these companies' capacity to benefit directly from recovering pricing power and operational efficiencies. Additionally, higher R&D investment (relative to sales) among top-quintile firms aligns directly with our strategists' emphasis on innovation and AI-driven productivity enhancements, further underpinning their competitive advantage and sustainable earnings growth potential.

Exhibit 9: Proportion of US stocks with upward earnings revisions rises to 56%, signaling improving analyst sentiment



Source: Factset, Morgan Stanley Research. Data as of April 30, 2026.

Exhibit 10: Top-quintile stocks in EPS revisions factor show superior operating leverage, margin momentum, and R&D



Source: Factset, Morgan Stanley Research. Data as of April 30, 2026.

Overall, our US recommendations are designed to capture the earnings-led rolling recovery while remaining mindful of inflation and energy risks. Internal Growth provides the high-quality expression of AI-enabled productivity; Low PEG balances growth with valuation discipline; and Up vs. Down EPS Revisions captures the broadening of earnings strength as the recovery continues.

Europe quant factors – Positioning for Energy-Driven Inflation With Cash-Flow Resilience, Capital Discipline and Earnings Revisions

In [Exhibit 11](#), we show the key forecasts from our economists and strategists in Europe for 2026 and 2027. Our economists expect a temporary euro area growth slowdown driven by higher energy prices, with Europe more exposed than the US given the region's faster pass-through of energy costs into inflation and activity. However, they do not expect a prolonged downturn in the baseline scenario, assuming energy volatility gradually subsides and growth stabilizes into 2027. Europe's quick pass-through of energy costs to core inflation and growth means it may see the largest near-term sensitivity to the energy shock. Persistent dual exposure to oil and LNG is likely to keep core inflation stubbornly elevated, broadly unchanged at 2.2% in both 2026 and 2027. The ECB is expected to deliver two additional 25bp rate hikes in 2026 before policy gradually converges back toward neutral in 2027.

Against this macro backdrop, our European equity strategists continue to see upside for MSCI Europe through mid-2027, with a target of 2,700. However, European equities are expected to remain choppy in the near term, particularly as sentiment remains tied to developments around the Strait of Hormuz and energy markets. While earnings have so far proven resilient, upside is more back-end loaded and more dependent on Energy, sector selection and stock-specific momentum than in the US. At the same time, our strategists continue to see underappreciated AI productivity opportunities in Europe, particularly through Semis and select AI-related pockets of Capital Goods and Materials & Mining. Our strategists note that Germany's fiscal spend is tracking below annual targets in both Defence and Infrastructure, while a potential restart of more tangible Russia/Ukraine negotiations remains worth monitoring as an upside risk into year-end. From a sector perspective, our strategists prefer European Semis, Utilities, Tobacco, Energy, Banks and Telecoms.

Exhibit 11: Key economist and strategist forecasts – Europe

Macro	Current	Estimates		
MSCI Europe	2427.61	2700 (2Q 2027)		
DBR 10Y	3.11%	2.45% (2Q 2027)		
	2025	2026E	2027E	
GDP (Annual)	1.5	0.6	1.0	
Inflation (Annual)	2.1	3.1	2.2	
Policy Rate	2.0	2.5	2.0	

Source: Datastream, Morgan Stanley Research. Data as of 13 May 2026.

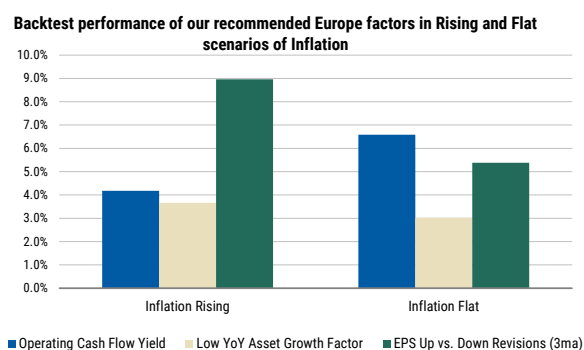
Against this backdrop, we recommend three quantitative equity factors for Europe: **Operating Cash Flow Yield, Low YoY Asset Growth, and 3-Month Up vs. Down Earnings Revisions**. These factors reflect a more defensive and selective approach than in the US, consistent with an environment shaped by energy-driven inflation, slower near-term growth and more hawkish ECB policy. The focus is on companies with resilient cash generation, disciplined capital allocation and positive earnings momentum in a market where energy inflation and central bank policy can quickly pressure margins.

We evaluate the recommended factors through three lenses: inflation, oil volatility and energy-theme exposure. First, in [Exhibit 12](#), we show performance across rising and flat inflation regimes. All three factors deliver positive returns in both regimes, supporting a balanced factor positioning for an environment where inflation and policy uncertainty remain tied to energy prices.

Second, [Exhibit 13](#) evaluates performance across normal and high Brent oil-volatility regimes. This is directly relevant to the Europe outlook, where our strategists note that earnings have so far proven resilient, but that a persistent Strait of Hormuz disruption could broaden downside risks. Operating Cash Flow Yield performs positively in both regimes and is strongest when oil volatility is high, reinforcing its role as a cash-flow resilience factor when energy costs can pressure margins and activity. Low YoY Asset Growth also performs positively across both regimes, consistent with the value of capital discipline in an energy-uncertain backdrop. 3M Up vs. Down EPS Revisions is weaker in high oil volatility but performs strongly in normal regimes, suggesting that revisions become more useful as volatility stabilizes and analysts have better visibility on margins, demand and earnings momentum.

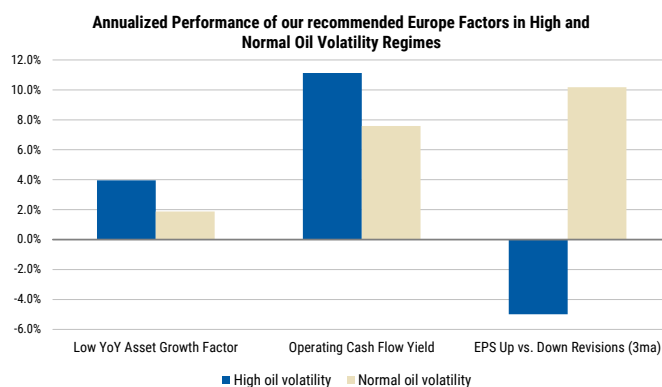
Finally, [Exhibit 14](#) shows net thematic exposure to the Future of Energy main theme and Energy Security sub-theme, calculated as top-quintile mean exposure minus bottom-quintile mean exposure. This is particularly relevant for Europe, where energy sensitivity and supply disruption risk remain central to our strategists' outlook. Positive exposure across the recommended factors supports their relevance in an energy-sensitive market, while adding alignment with longer-term themes around energy resilience and security of supply.

Exhibit 12: Recommended Europe factors deliver positive returns across rising and flat inflation regimes



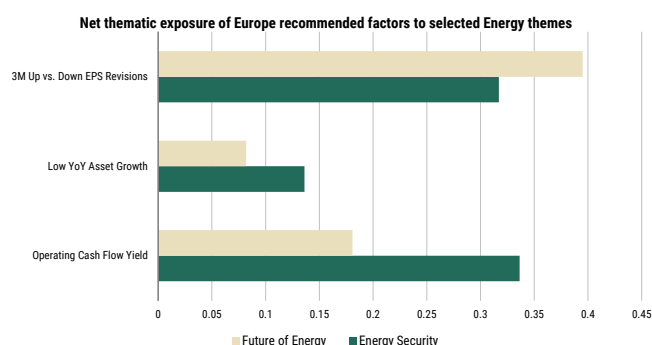
Source: Factset, Haver, Morgan Stanley Research. Inflation regimes based on spread between 1M and 6M moving averages of lagged inflation. Regimes classified as Rising when spread exceeds +0.15 and Flat when between -0.15 and +0.15. Data as of Dec 2024.

Exhibit 13: Operating Cash Flow Yield and Low Asset Growth lead in high oil volatility; Revisions improve as oil volatility normalizes



Source: Datastream, Morgan Stanley Research. Oil-volatility regimes are defined using weekly Brent prices. Realized oil volatility is calculated as the annualized 12-week (~3M) rolling standard deviation of weekly Brent returns. High and normal oil-volatility regimes correspond to periods above the 75th percentile and between the 25th and 75th percentiles, respectively. Factor returns are shown on an annualized basis.

Exhibit 14: Recommended Europe Factors Show Positive Exposure to Future of Energy and Energy Security Themes



Source: Factset, Morgan Stanley Research. Each stock's sensitivity to thematic movements is estimated in two steps: we first obtain the thematic-specific return by regressing the thematic portfolio's weekly return on MSCI ACWI Equal-Weight (EQL) and defining the theme-specific component as the regression intercept plus the residual. We then regress each stock's weekly return on this thematic-specific return, controlling for EQL, to estimate its thematic beta. All regressions are run over a rolling two-year weekly estimation horizon (approximately 104 observations). Data as of April 30, 2026.

Having evaluated the recommended factors across inflation, oil-volatility and energy-theme lenses, we now take a more granular view of the supportive characteristics behind each signal.

The **Operating Cash Flow Yield** factor targets companies with superior cash-generation capabilities, stable margins and resilient operating profiles. In Europe, where energy-price volatility can affect both demand and input costs, cash-flow durability is especially valuable. The factor is well suited to an environment in which broad-based earnings upside may be constrained, but companies with strong internal cash generation can continue to defend margins, fund operations and maintain shareholder returns.

In [Exhibit 15](#), we show the factor's cumulative performance and 126-day rolling Sharpe ratio. Operating Cash Flow Yield delivered strong cumulative returns through 2025 and early 2026, followed by a modest dip around March, before beginning to recover into April. The rolling Sharpe ratio remains elevated, reinforcing the factor's defensive quality in a market where energy costs and macro uncertainty can pressure margins and activity. In [Exhibit 16](#), we show the factor's current valuation relative to history. Operating Cash Flow Yield is trading at the 23rd percentile relative to its history since 1990, suggesting that valuation support remains reasonable despite recent performance strength.

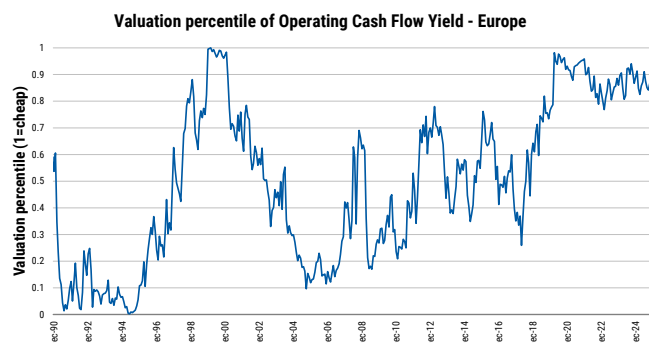
In [Exhibit 17](#) we show the factor's sector composition by comparing the number of top-quintile and bottom-quintile stocks across sectors. The factor shows positive net exposure to Communication Services, Energy, Consumer Discretionary, Consumer Staples, Utilities and Materials, and negative net exposure to Industrials, Information Technology and Health Care. Positive exposure to Utilities and Consumer Staples highlights the factor's defensive, cash-resilient profile, while positive exposure to Energy reinforces its relevance in a market shaped by energy sensitivity and oil-volatility risk.

Exhibit 15: Operating Cash Flow Yield Has Delivered Strong Risk-Adjusted Performance



Source: Factset, Morgan Stanley Research. Rolling Sharpe ratio is calculated using a 126-day lookback window. Data as of April 30, 2026.

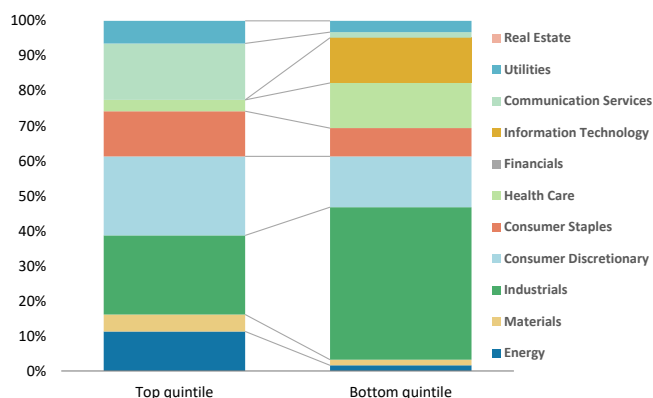
Exhibit 16: Operating Cash Flow Yield Remains Reasonably Valued Versus History



Source: Factset, Morgan Stanley Research. Data as of April 30, 2026.

Exhibit 17: Operating Cash Flow Yield Shows Positive Net Exposure to Defensive and Energy-Sensitive Areas

Quintile sector exposures - Operating Cash Flow Yield (Europe)



Source: Factset, Morgan Stanley Research. Data as of April 30, 2026.

We also favor **Low YoY Asset Growth** as a capital-discipline factor within our Europe recommendations. The factor is defined as the annual percentage change in total assets, with a preference for lower values. From a fundamental perspective, it captures companies that avoid aggressive balance-sheet expansion and invest more selectively. In an environment where energy volatility can pressure margins, demand and funding conditions, markets are likely to place greater value on disciplined capital allocation and a more cautious approach to growth.

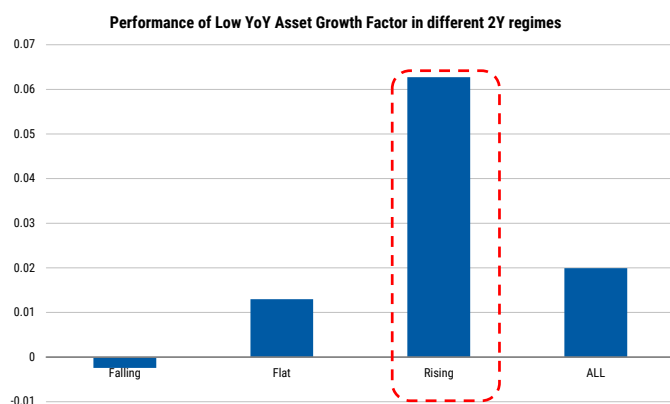
In [Exhibit 18](#), we evaluate Low YoY Asset Growth across 2-year yield regimes. The factor performs best when 2-year yields are rising, consistent with the idea that capital discipline becomes more valuable as front-end rates move higher and the cost of capital increases. This is particularly relevant in Europe, where our economists expect the ECB to deliver two additional 25bp rate hikes in 2026 before policy gradually converges back toward neutral in 2027.

In [Exhibit 19](#), we show the current average cost of equity across quintiles of the Asset Growth factor. While the relationship is not perfectly monotonic, the broader pattern is

clear: currently firms in higher asset-growth buckets tend to face a higher cost of equity than low asset-growth firms. This supports the case for disciplined asset growth in a higher-rate environment, where aggressive balance-sheet expansion is more likely to be penalized.

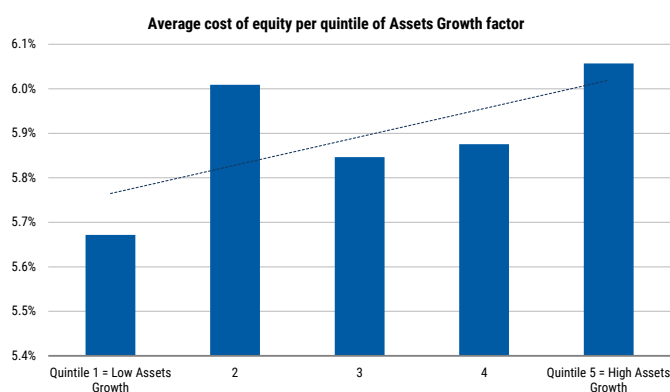
Finally, the factor's supportive tilts toward Energy and Materials (see [Exhibit 20](#)) are also notable in the current European context, where energy-related earnings and cash flows are more relevant to regional index-level earnings growth than in the US.

Exhibit 18: Low YoY Asset Growth rewards capital discipline in a higher cost-of-capital environment



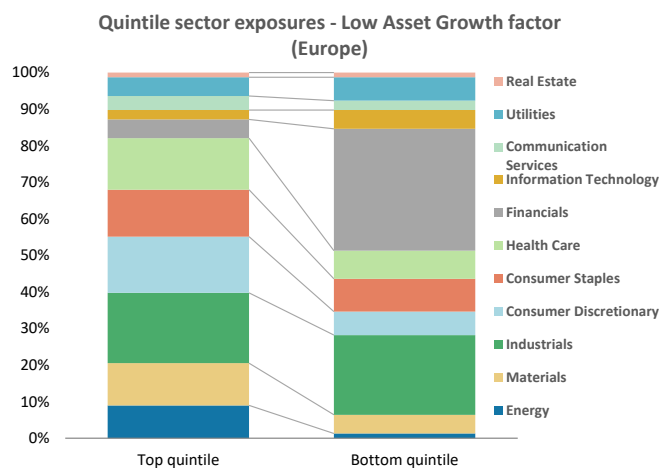
Source: Factset, Datastream, Haver, Morgan Stanley Research.

Exhibit 19: Cost of equity varies across asset growth, with low growth firms advantaged



Source: Factset, Morgan Stanley Research. Data as of April 30, 2026.

Exhibit 20: Low YoY Asset Growth shows positive exposure to energy-sensitive sectors



Source: Factset, Morgan Stanley Research. Data as of April 30, 2026.

Finally, we maintain our preference for **3-Month Up vs. Down EPS Revisions** as the forward-looking earnings-momentum component of our Europe recommendations. This is particularly relevant as consensus incorporates inflation-related pressure on margins and earnings, creating greater differentiation between companies with improving expectations and those facing downgrade risk.

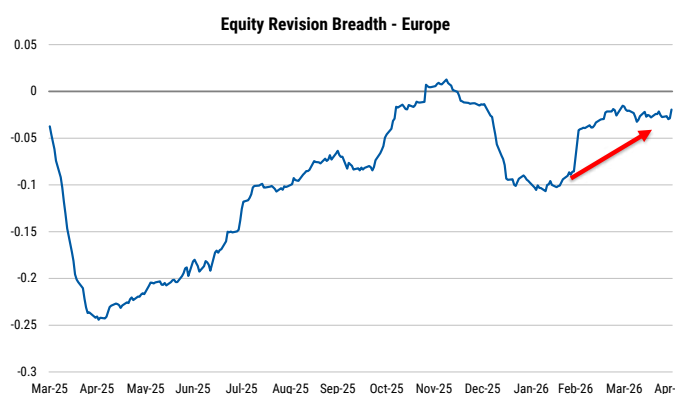
In [Exhibit 21](#), we show that European equity revision breadth has rebounded meaningfully

from its early-2026 lows. While the series remains below zero, the recovery suggests that downgrade pressure has eased and analyst sentiment is improving from depressed levels. This supports the case for a revisions-based signal as inflation and margin risks become increasingly reflected in consensus forecasts.

In [Exhibit 22](#), we show the factor's cumulative performance and rolling 126-day Sharpe ratio over the past year. The factor has delivered steady cumulative performance, with a clear recovery from the early-2026 drawdown and a strong move higher into April. The rolling Sharpe ratio has also improved from early-year lows, reinforcing the factor's role as an earnings-confirmation signal as revisions breadth recovers.

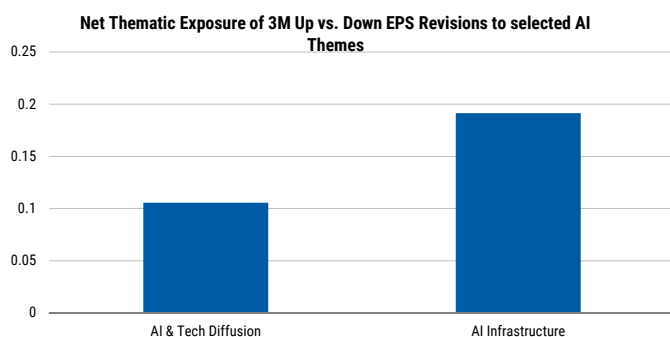
Finally, in [Exhibit 23](#), we show the factor's net thematic exposure to selected global AI themes. While energy remains the dominant macro driver for Europe, the factor also provides positive exposure to AI & Tech Diffusion and AI Infrastructure. This links the signal to the broader AI capex cycle highlighted in the outlook and supports its role as a forward-looking earnings indicator that can capture improving consensus expectations from both easing energy pressure and global AI-linked earnings momentum.

Exhibit 21: European equity revision breadth has rebounded from early-2026 lows



Source: Factset, Morgan Stanley Research. Data as of April 30, 2026.

Exhibit 23: 3M Up vs. Down EPS Revisions shows positive net exposure to selected ai themes



Source: Factset, Morgan Stanley Research. Each stock's sensitivity to thematic movements is estimated in two steps: we first obtain the thematic-specific return by regressing the thematic portfolio's weekly return on MSCI ACWI Equal-Weight (EQL) and defining the theme-specific component as the regression intercept plus the residual. We then regress each stock's weekly return on this thematic-specific return, controlling for EQL, to estimate its thematic beta. All regressions are run over a rolling two-year weekly estimation horizon (approximately 104 observations). Data as of April 30, 2026.

Exhibit 22: 3M Up vs. Down EPS Revisions has delivered strong cumulative performance as revision breadth recovers



Source: Factset, Morgan Stanley Research. Rolling Sharpe ratio is calculated using a 126-day lookback window. Data as of April 30, 2026.

Overall, our European factor recommendations reflect the need to balance upside participation with resilience in a market shaped by slower near-term growth, energy-related inflation pressure and relatively more hawkish ECB policy compared to other central banks. Operating Cash Flow Yield provides a defensive cash-flow anchor; Low YoY Asset Growth captures capital discipline in a higher-rate environment; and 3-Month Up vs. Down EPS Revisions identifies companies where earnings expectations are improving despite macro uncertainty.

Japan quant factors – A Dual Approach Integrating Macroeconomic and Micro-Level Insight

In [Exhibit 24](#), we show the key forecasts from our economists and strategists in Japan for 2026 and 2027. Our equity strategists expect the rally in Japanese equities to continue, with TOPIX projected to reach 4,300 by June 2027. The market remains supported by Japan's ongoing transition toward a more reflationary environment, structural labor shortages, wage growth, productivity-enhancing capex, corporate governance reform and strategic investments tied to AI, semiconductors, power infrastructure, defense, robotics and telecoms.

Meanwhile, our macro strategists expect long-term interest rates, which have been trending higher on the back of Bank of Japan rate hikes, to see a modest short-term decline but remain broadly range-bound over the medium term. This creates a supportive backdrop for value factors, particularly if rates do not decline materially. At the same time, higher oil prices can pressure corporate profitability, increasing the importance of valuation metrics tied directly to earnings and of bottom-up revisions as a stock-selection signal.

Exhibit 24: Key economist and strategist forecasts – Japan

Macro	Current	Estimates		
TOPIX	3919.48	4300 (2Q 2027)		
JGB 10Y	2.60%	2.25% (2Q 2027)		
	2025	2026E	2027E	
GDP (Annual)	1.2	0.4	1.0	
Inflation (Annual)	3.2	2.2	1.4	
Policy Rate	0.75	1.00	1.25	

Source: Morgan Stanley Research.

In this environment, we recommend the following three factors for Japanese equities: **Up vs. Down EPS Revisions, Forward Earnings Yield, and Low PEG**. The first is primarily driven by company-level micro factors, while the latter two are strategies focused mainly on macro considerations.

Up vs. Down EPS Revisions

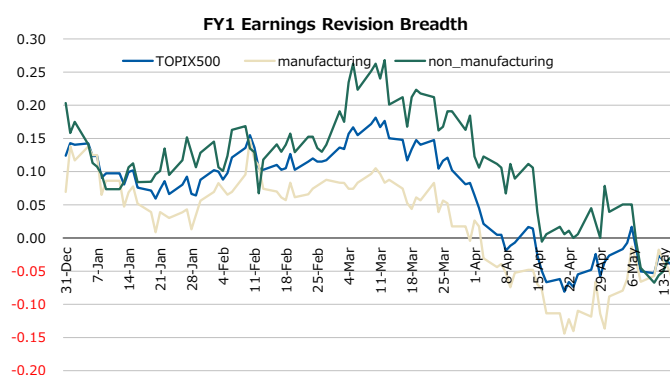
The Up vs. Down EPS Revisions factor involves taking long positions in stocks that analysts have upgraded and short positions in those that have been downgraded in terms of earnings forecasts.

Japan's earnings revision breadth turned downward following the U.S.-Iran conflict in

March 2026 and has recently been hovering around zero ([Exhibit 25](#)). Guidance for the new fiscal year also shows increasing divergence, with a clear split between companies that are effectively managing rising oil prices and those that are not ([Japan Is Back, Still Climbing: The Summit Lies Ahead \(12 May, 2026\)](#)). In other words, the gap between these groups of companies at the earnings level has become more pronounced, suggesting that earnings-driven stock selection is becoming increasingly important. Note that the earnings revision breadth refers to FY1 here, as the most recent guidance from companies with March fiscal year-ends is reflected in FY1 estimates.

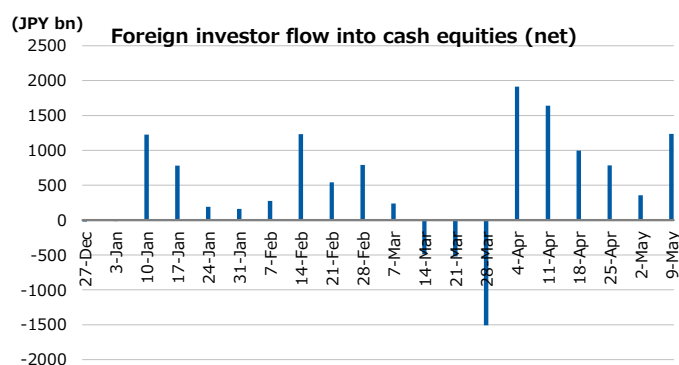
Meanwhile, foreign investor flows into Japanese equities remain robust ([Exhibit 26](#)). Based on our analysis, there is a positive correlation between foreign investor flows into cash equity and the momentum factor ([Quant Matters Japan \(Jun 2025\) \(18 Jun, 2025\)](#)). The revision factor, which broadly falls within the momentum category, is no exception. Against the backdrop of Japan's ongoing normalization, we expect continued strong foreign inflows into Japanese equities, and therefore believe the revision factor is likely to deliver solid performance over the medium term.

Exhibit 25: Earnings Revisions Breadth for Japanese firms



Source: Bloomberg, Morgan Stanley Research
Note: TOPIX 500 universe; ERB refers to FY1, as the most recent guidance from companies with March fiscal year-ends is reflected in FY1 estimates

Exhibit 26: Foreign investor flows into cash equities

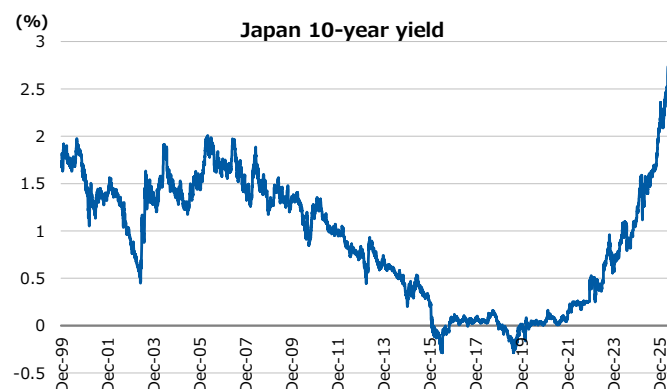


Source: JPX, Morgan Stanley Research
Note: net trading value of stocks in Tokyo and Nagoya markets

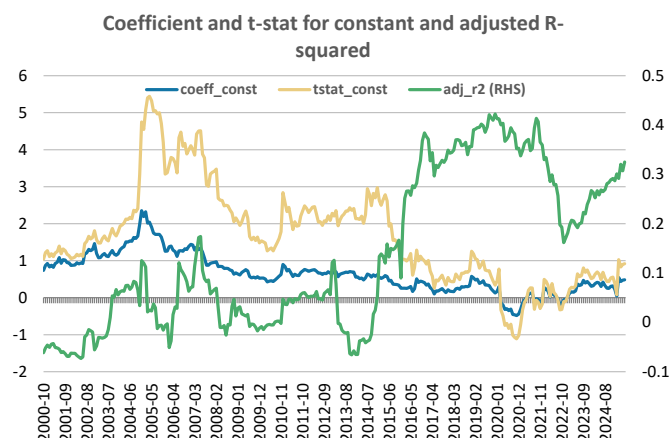
Forward Earnings Yield

As noted above, our macro strategists expect Japan's long-term interest rates to remain broadly range-bound over the medium term. Based on our analysis ([Quant Matters Japan \(Aug 2025\): Value and Rates / Earnings Revisions Acceleration \(19 Aug 2025\)](#)), such an environment favors value over growth. Even if yields remain flat, we find that Japanese equities continue to exhibit a persistent value premium ([Exhibit 28](#)).

Within the value factors, we favor Forward Earnings Yield. Amid the growing impact of higher oil prices on corporate profitability, valuation metrics based on profits tend to attract greater attention. In addition, our strategists expect the rally in Japanese equities to continue, projecting TOPIX at 4,300 by June-2027. The Forward Earnings Yield factor exhibits a more positive sensitivity to TOPIX ([Exhibit 29](#)) than other value factors, meaning it tends to outperform other value factors when the market is rising.

Exhibit 27: Japan 10-year yield

Source: Bloomberg, Morgan Stanley Research.

Exhibit 28: Rolling 60-month regression analysis of Composite Value factor against monthly changes in Japanese and US short- and long-term interest rates: Coefficient and t-statistic of the intercept, and adjusted r-squared

Source: Bloomberg, FactSet, QUICK Workstation, Morgan Stanley Research

Exhibit 29:

Regression analysis results, with value factor as the dependent variable and macro variable as the independent variable

	Fwd P/E		FCF YLD		P-to-OCF		EV/EBITDA		P/Book		P/Sales	
	coef	t-stat	coef	t-stat	coef	t-stat	coef	t-stat	coef	t-stat	coef	t-stat
const	0.94	1.75 *	0.82	2.27 **	0.79	1.56	0.60	1.16	0.53	0.94	0.49	0.93
d_JGB3M	-1.15	-0.17	0.61	0.17	-2.25	-0.37	0.23	0.03	-0.11	-0.02	-1.09	-0.16
d_JGB10Y	18.13	3.39 ***	5.81	1.35	11.52	2.29 **	10.39	1.81 *	24.25	4.16 ***	14.26	2.74 ***
d_UST3M	-0.78	-0.40	-0.36	-0.29	-1.18	-0.73	-0.47	-0.27	-0.95	-0.38	-1.20	-0.52
d_UST10Y	3.89	2.14 **	-0.19	-0.16	3.05	1.95 *	2.49	1.32	4.71	2.52 **	3.80	2.03 **
d_TPX	0.27	1.24	0.08	0.70	0.06	0.31	0.20	1.04	-0.09	-0.41	0.01	0.06
Adj. R-squared	0.24		-0.04		0.11		0.07		0.33		0.14	

	Trail P/E		Shiller P/E		Div Yield		Net BB Yield		TOT Yld		Comp Value	
	coef	t-stat	coef	t-stat	coef	t-stat	coef	t-stat	coef	t-stat	coef	t-stat
const	0.73	1.59	0.68	1.28	1.15	2.39 **	0.23	0.87	0.86	1.96 *	0.84	1.54
d_JGB3M	-1.57	-0.25	-4.17	-0.53	-2.51	-0.39	-1.03	-0.25	-1.79	-0.28	-0.38	-0.06
d_JGB10Y	13.38	3.01 ***	21.59	3.82 ***	15.98	2.99 ***	3.76	1.67	12.84	2.89 ***	23.83	4.33 ***
d_UST3M	-1.44	-0.86	-0.60	-0.26	-1.59	-1.00	-0.28	-0.33	-1.36	-0.92	-1.15	-0.58
d_UST10Y	2.72	1.69 *	4.19	2.19 **	3.66	2.24 **	2.83	4.31 ***	4.35	3.08 ***	4.75	2.65 **
d_TPX	0.23	1.29	-0.14	-0.64	-0.05	-0.25	0.06	1.00	0.06	0.34	0.15	0.67
Adj. R-squared	0.17		0.27		0.24		0.21		0.24		0.34	

Source: Bloomberg, FactSet, QUICK Workstation, Morgan Stanley Research

Note: *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The sample period is from Jan 2021 to Jun 2025. For details, see Quant Matters Japan (19 Aug 2025)

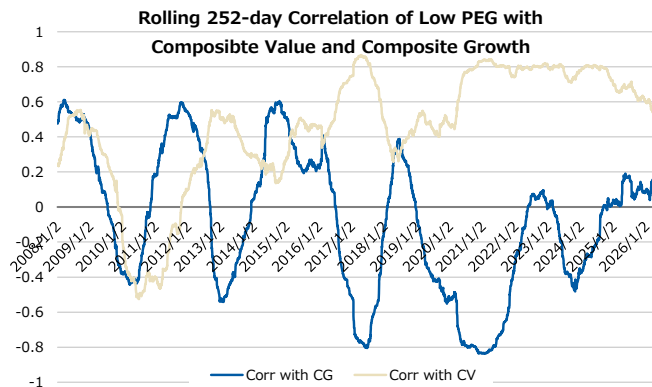
Low PEG

As noted above, we recommend value over growth in the mid-term; however, given the similarities between the current market environment and the dot-com bubble ([Are We Going Back to 1999? \(14 May, 2026\)](#)), we believe that growth stocks – particularly those linked to AI and semiconductors – still have a meaningful probability of rallying over the near term.

Looking at the return correlations between Low PEG and Composite Value and Composite Growth ([Exhibit 30](#)), while correlations with value have remained high in recent years,

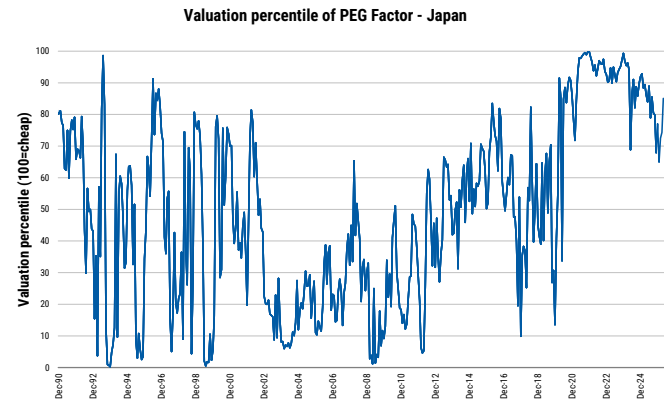
they have been declining recently, whereas correlations with growth have turned positive. In addition, the valuation percentile for Low PEG is 85th ([Exhibit 31](#)), which is attractive even compared to other growth factors.

Exhibit 30: Rolling correlation of Low PEG with Composite Value and Composite Growth



Source: FactSet, Morgan Stanley Research

Exhibit 31: Valuation percentile of Low PEG



Source: FactSet, Morgan Stanley Research

Overall, our Japan factor recommendations combine bottom-up earnings confirmation with macro-aware valuation discipline. Up vs. Down EPS Revisions captures widening earnings dispersion and continued foreign investor support; Forward Earnings Yield provides exposure to earnings-based value in a range-bound rate environment; and Low PEG balances that value tilt with selective growth exposure as AI- and semiconductor-linked stocks remain capable of leading near-term rallies.

Stock Ideas Based On Our Latest Factor Recommendations

In this section, we present stock screens based on our recommended factors for the three regions. Specifically, within the MSCI indices for the US, Europe, and Japan, we calculate overall scores by averaging the factor score percentiles. We exclude stocks that fall into the bottom third for any of the recommended factors and highlight up to 30 stocks with the highest overall scores.

US

Exhibit 32: Stock ideas based on our latest factor recommendations – US

No.	RIC	Company Name	Country	GICS Sector	MCap (\$mn)	Price (LCL)	MS rating	Top Pick	Factor Scores			Overall
									Internal Growth	Low PEG	EPS Up vs Down Revisions	
1	STX.O	SEAGATE TECHNOLOGY	United States	Information Technology	146,767	733.4	OW	✓	0.99	0.92	0.98	1.00
2	GEV.N	GE VERNOVA	United States	Industrials	293,965	1011.8	OW		0.96	0.79	0.84	0.99
3	TRV.N	TRAVELERS COS (THE)	United States	Financials	68,066	305.5	EW		0.73	1.00	0.85	0.99
4	WDC.O	WESTERN DIGITAL	United States	Information Technology	148,562	455.8	OW		0.94	0.85	0.77	0.99
5	CF.N	CF INDUSTRIES HOLDINGS	United States	Materials	19,372	127.9	EW		0.79	0.96	0.80	0.98
6	LVS.N	LAS VEGAS SANDS CORP	United States	Consumer Discretionary	36,924	49.9	EW		0.98	0.72	0.83	0.98
7	LRCX.O	LAM RESEARCH CORP	United States	Information Technology	323,880	273.4	OW		0.94	0.59	0.98	0.98
8	LITE.O	LUMENTUM HOLDINGS	United States	Information Technology	63,975	890.1	EW		0.84	0.91	0.75	0.98
9	VRT.N	VERTIV HOLDINGS A	United States	Industrials	125,594	322.6	OW		0.89	0.63	0.97	0.97
10	TER.O	TERADYNE	United States	Information Technology	53,790	321.5	EW		0.80	0.71	0.94	0.97
11	INCY.O	INCYTE CORP	United States	Health Care	18,704	95.6	EW		0.81	0.82	0.82	0.97
12	NFLX.O	NETFLIX	United States	Communication Services	396,656	89.3	OW		0.93	0.67	0.82	0.97
13	EXPE.O	EXPEDIA GROUP	United States	Consumer Discretionary	30,433	215.0	EW		0.97	0.87	0.56	0.96
14	STLD.O	STEEL DYNAMICS	United States	Materials	33,392	223.4	EW		0.53	0.91	0.93	0.96
15	LLY.N	LILLY (ELI) & COMPANY	United States	Health Care	883,556	1021.4	OW		0.97	0.77	0.61	0.96
16	AXP.N	AMERICAN EXPRESS	United States	Financials	222,534	309.3	EW		0.82	0.67	0.84	0.96
17	NVDA.O	NVIDIA	United States	Information Technology	4,849,550	220.6	OW	✓	0.97	0.87	0.47	0.95
18	MU.O	MICRON TECHNOLOGY	United States	Information Technology	582,068	698.7	OW		0.86	0.96	0.49	0.95
19	V.N	VISA A	United States	Financials	633,956	329.9	OW		0.93	0.47	0.91	0.95
20	ADBE.O	ADOBE	United States	Information Technology	103,017	255.0	EW		0.96	0.82	0.50	0.95
21	CMCSA.O	COMCAST CORP A (NEW)	United States	Communication Services	98,531	24.8	EW		0.65	0.97	0.66	0.94
22	EBAY.O	EBAY	United States	Consumer Discretionary	46,773	114.2	OW		0.87	0.54	0.86	0.94
23	ALNY.O	ALNYLAM PHARMACEUTICALS	United States	Health Care	40,888	294.3	EW		0.91	0.86	0.47	0.93
24	SYF.N	SYNCHRONY FINANCIAL	United States	Financials	27,445	69.4	EW		0.75	0.71	0.79	0.93
25	KLAC.O	KLA CORPORATION	United States	Information Technology	229,983	1740.6	OW		0.96	0.39	0.89	0.93
26	META.O	META PLATFORMS A	United States	Communication Services	1,542,550	602.6	OW	✓	0.86	0.69	0.69	0.93
27	GOOGL.O	ALPHABET A	United States	Communication Services	4,627,920	387.7	OW		0.85	0.51	0.88	0.92
28	CPAY.N	CORPAY	United States	Financials	21,440	343.1	OW		0.82	0.81	0.60	0.92
29	VLO.N	VALERO ENERGY CORP	United States	Energy	77,039	262.6	EW		0.53	0.89	0.79	0.92
30	AMZN.O	AMAZON.COM	United States	Consumer Discretionary	2,833,550	259.3	OW		0.75	0.58	0.88	0.91

Source: FactSet, Morgan Stanley Research. Market data as at 19 May 2026.

Europe

Exhibit 33: Stock ideas based on our latest factor recommendations – Europe

No.	RIC	Company Name	Country	GICS Sector	MCap (\$mn)	Price (LCL)	MS rating	Top Pick	Factor Scores			Overall
									Operating Cash Flow Yield	Low YoY Asset Growth Factor	3M EPS Up vs Down Revisions	
1	REP.MC	REPSOL	Spain	Energy	30,153	23.3	OW		0.90	0.86	0.92	1.00
2	EQNR.OL	EQUINOR	Norway	Energy	102,189	376.0	EW		0.92	0.96	0.78	0.99
3	TEF.MC	TELEFONICA	Spain	Communication Services	25,661	4.1	EW		0.99	0.93	0.69	0.99
4	AER.N	AERCAP HOLDINGS NV	Netherlands	Industrials	26,457	136.0	EW		0.92	0.68	0.99	0.98
5	SHEL.L	SHELL	United Kingdo	Energy	257,588	3284.5	EW		0.84	0.97	0.72	0.97
6	EVKn.DE	EVONIK INDUSTRIES	Germany	Materials	9,665	17.4	OW		0.86	0.94	0.71	0.97
7	DTEGn.DE	DEUTSCHE TELEKOM	Germany	Communication Services	158,639	29.4	OW	✓	0.97	0.81	0.61	0.96
8	AVOL.S	AVOLTA	Switzerland	Consumer Discretionary	8,056	45.3	EW		1.00	0.88	0.48	0.95
9	VOD.L	VODAFONE GROUP	United Kingdo	Communication Services	37,347	113.2	EW		0.98	0.76	0.58	0.95
10	TELIA.ST	TELIA CO	Sweden	Communication Services	20,511	50.9	EW		0.81	0.66	0.85	0.94
11	SCMN.S	SWISSCOM	Switzerland	Communication Services	43,817	684.5	OW		0.88	0.76	0.67	0.94
12	BNZL.L	BUNZL	United Kingdo	Industrials	10,670	2398.0	EW		0.66	0.96	0.69	0.93
13	ABI.BR	ANHEUSER-BUSCH INBEV	Belgium	Consumer Staples	152,542	70.2	OW	✓	0.65	0.88	0.76	0.93
14	PSO.NL	PEARSON	United Kingdo	Consumer Discretionary	9,344	1138.0	EW		0.64	0.87	0.73	0.92
15	SASY.PA	SANOFI	France	Health Care	114,185	75.2	EW		0.69	0.75	0.80	0.92
16	ENI.MI	ENI	Italy	Energy	88,592	23.9	EW		0.84	0.45	0.95	0.91
17	ENT.L	ENTAIN	United Kingdo	Consumer Discretionary	4,725	533.0	OW		0.90	0.90	0.40	0.91
18	HOLN.S	HOLCIM	Switzerland	Materials	52,538	70.8	EW		0.39	0.99	0.81	0.91
19	EDV.L	ENDEAVOUR MINING (GB)	United Kingdo	Materials	14,481	4370.0	OW		0.72	0.85	0.60	0.90
20	HEIN.AS	HEINEKEN NV	Netherlands	Consumer Staples	44,690	67.7	EW		0.76	0.58	0.82	0.90
21	KNIN.S	KUEHNE & NAGEL INTL	Switzerland	Industrials	28,210	171.7	EW		0.46	0.85	0.84	0.89
22	MT.AS	ARCELORMITTAL A	France	Materials	44,765	51.3	OW		0.70	0.78	0.66	0.88
23	PUBP.PA	PUBLICIS GROUPE	France	Communication Services	23,681	83.7	OW	✓	0.80	0.56	0.77	0.87
24	HLN.L	HALEON	United Kingdo	Health Care	41,149	342.3	OW	✓	0.49	0.81	0.78	0.87
25	EONGn.DE	E. ON	Germany	Utilities	58,529	18.4	OW		0.78	0.61	0.66	0.86
26	EDP.LS	EDP SA	Portugal	Utilities	22,788	4.4	EW		0.77	0.73	0.56	0.85
27	GALP.LS	GALP ENERGIA SGPS B	Portugal	Energy	17,642	20.0	EW		0.76	0.74	0.53	0.84
28	RWEG.DE	RWE STAMM	Germany	Utilities	54,081	56.4	OW		0.68	0.41	0.94	0.83
29	TTEF.PA	TOTALENERGIES	France	Energy	205,237	79.8	OW	✓	0.79	0.50	0.73	0.83
30	BT.L	BT GROUP	United Kingdo	Communication Services	29,331	232.3	OW		0.95	0.65	0.41	0.82

Source: FactSet, Morgan Stanley Research. Market data as at 19 May 2026.

Japan

Exhibit 34: Stock ideas based on our latest factor recommendations – Japan

No.	RIC	Company Name	Country	GICS Sector	MCap (\$mn)	Price (LCL)	MS rating	Top Pick	12M Fwd Earnings Yield	Low PEG	EPS Up vs Down Revisions	Overall
1	5401.T	NIPPON STEEL CORP	Japan	Materials	19,625	540.7	OW		0.97	0.97	0.83	0.99
2	8729.T	SONY FINANCIAL GROUP	Japan	Financials	6,383	145.0	EW		0.96	0.89	0.88	0.98
3	8630.T	SOMPO HOLDINGS	Japan	Financials	34,560	6238.0	OW		0.89	0.90	0.84	0.97
4	8411.T	MIZUHO FINANCIAL GROUP	Japan	Financials	106,804	6982.0	OW		0.71	0.87	0.91	0.96
5	7269.T	SUZUKI MOTOR CORP	Japan	Consumer Discretionary	21,921	1753.0	OW	✓	0.98	0.96	0.51	0.96
6	4503.T	ASTELLAS PHARMA	Japan	Health Care	25,776	2295.0	EW		0.77	0.91	0.67	0.95
7	8309.T	SUMITOMO MITSUI TRUST	Japan	Financials	23,308	5643.0	EW		0.84	0.79	0.72	0.94
8	7182.T	JAPAN POST BANK CO	Japan	Financials	61,361	3075.0	OW		0.57	0.84	0.87	0.93
9	7186.T	YOKOHAMA FINANCIAL GROU	Japan	Financials	10,847	1653.0	EW		0.66	0.86	0.67	0.92
10	3407.T	ASAHI KASEI CORP	Japan	Materials	13,378	1673.5	OW		0.73	0.81	0.63	0.91
11	6178.T	JAPAN POST HOLDINGS CO	Japan	Financials	34,585	2110.0	OW		0.70	0.73	0.69	0.89
12	8308.T	RESONA HOLDINGS	Japan	Financials	28,540	2028.5	EW		0.59	0.85	0.64	0.88
13	4507.T	SHIONOGI & CO	Japan	Health Care	17,997	3035.0	EW		0.69	0.78	0.58	0.84
14	8750.T	DAIICHI LIFE GROUP	Japan	Financials	34,010	1656.0	EW		0.79	0.57	0.66	0.82
15	1812.T	KAJIMA CORP	Japan	Industrials	20,636	5567.0	OW		0.53	0.67	0.72	0.75
16	6723.T	RENESAS ELECTRONICS CORP	Japan	Information Technology	38,342	3500.0	OW		0.36	0.56	0.93	0.75
17	4901.T	FUJII FILM HOLDINGS CO	Japan	Information Technology	22,940	3143.0	OW		0.72	0.45	0.65	0.73
18	1801.T	TAISEI CORP	Japan	Industrials	19,818	13655.0	OW		0.44	0.38	0.98	0.72
19	6479.T	MINEBEA MITSUMI	Japan	Industrials	8,514	3730.0	EW		0.52	0.69	0.47	0.68
20	3382.T	SEVEN & I HOLDINGS CO	Japan	Consumer Staples	31,097	1943.5	OW		0.47	0.66	0.50	0.64
21	8002.T	MARUBENI CORP	Japan	Industrials	64,351	5374.0	EW		0.45	0.41	0.75	0.61
22	9432.T	NTT CORP	Japan	Communication Services	88,121	156.5	EW		0.79	0.42	0.39	0.61
23	7735.T	SCREEN HOLDINGS CO	Japan	Information Technology	12,453	10545.0	OW		0.44	0.68	0.46	0.59
24	9843.T	NITORI HOLDINGS CO	Japan	Consumer Discretionary	8,094	2492.5	EW		0.62	0.39	0.55	0.58
25	8136.T	SANRIO CO	Japan	Consumer Discretionary	7,457	871.9	OW		0.39	0.55	0.61	0.56
26	6098.T	RECRUIT HOLDINGS CO	Japan	Industrials	72,884	9653.0	OW		0.38	0.72	0.45	0.54
27	8801.T	MITSUMI FUDOSAN CO	Japan	Real Estate	30,262	1508.0	OW		0.51	0.40	0.50	0.41
28	3391.T	TSURUHA HOLDINGS	Japan	Consumer Staples	5,964	2083.0	EW		0.34	0.62	0.39	0.36

Source: FactSet, Morgan Stanley Research. Market data as at 20 May 2026.

Cluster Ranking Changes

In the following sections, we provide our monthly updates based on our 4-Dimensions framework for investment factors, and highlight key changes in factor ranking against those in the previous month.

Cluster Rankings: Europe

In Europe, we revise Economic Cycle from Expansion to Slowdown.

Within our 17-factor framework, Revisions surged to 1st place, rising from 3rd to claim the top position and underscoring strong momentum support this month. At the opposite end of the rankings, **Composite Quality** fell from 15th to 17th, becoming the least favoured exposure in the framework.

Across the value complex, **Composite Value** weakened significantly, falling from 5th to 11th — one of the largest downward moves in the framework this month. Current Value strengthened notably, climbing from 6th to 3rd. Deep Value improved from 10th to 8th. Defensive Value softened modestly, declining from 7th to 9th.

Within the growth segment, **Composite Growth** improved sharply from 17th to 13th, marking a meaningful recovery from the bottom of the rankings. Growth held steady at 12th.

Quality-oriented measures were notably weaker overall in the framework. Earnings Quality softened from 8th to 10th. Low Leverage declined from 13th to 16th, indicating reduced preference for balance sheet-focused factors. Management Quality, while still highly ranked, slipped from 1st to 5th — a notable softening for this typically favoured factor. Operating Quality weakened from 11th to 14th. Profitability improved marginally from 16th to 15th.

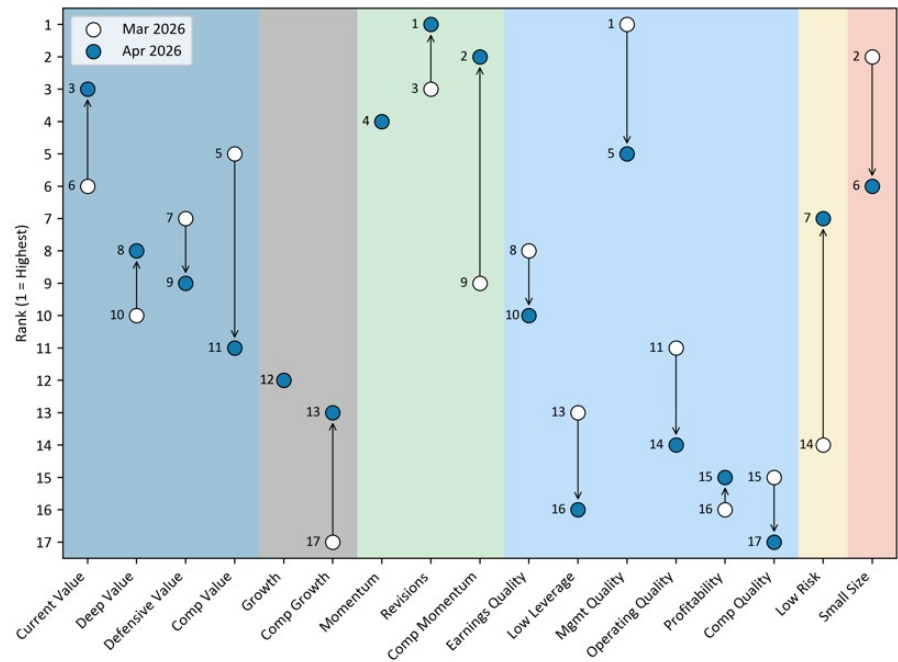
Momentum indicators strengthened substantially in the rankings. **Composite Momentum** surged from 9th to 2nd — the largest upward move in the framework this month. Price Momentum held steady at 4th.

Low Risk rebounded meaningfully, rising from 14th to 7th, indicating renewed preference for defensive exposures.

Small Size slipped from 2nd to 6th, though it remains in the upper half of the rankings, suggesting continued but moderating preference for smaller-capitalisation stocks.

In Europe, our favored factors are: (1) Operating Cash Flow Yield, (2) Low Asset Growth, and (3) Up vs. Down EPS Revisions (3-month moving average).

Exhibit 35: Europe – Changes in Cluster Rankings



This figure illustrates ranking changes of factor clusters from the end of the previous month to the most recent month. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The x-axis represents the clusters, while the y-axis indicates the ranks. Sources: FactSet, Morgan Stanley Research.

Cluster Rankings: US

In the US, the regimes remain unchanged from the previous month.

Within our 17-factor framework, **Small Size** surged to 1st place, rising dramatically from 12th to claim the top position — the largest upward move in the framework this month. At the opposite end of the rankings, Deep Value fell from 15th to 17th, becoming the least favoured exposure in the framework.

Across the value complex, **Composite Value** edged up in the rankings from 17th to 15th. Current Value slipped from 7th to 10th. Defensive Value moved lower from 3rd to 5th.

Within the growth segment, **Composite Growth** rose in the rankings from 16th to 14th. Growth declined from 14th to 16th.

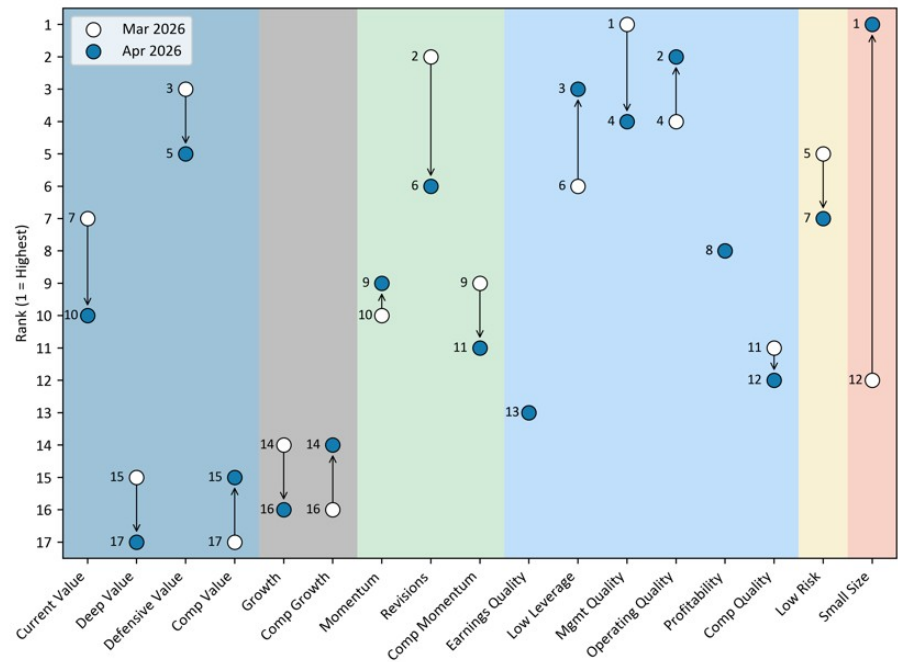
Quality-oriented measures showed notable divergence in the rankings. **Composite Quality** slipped slightly from 11th to 12th. Earnings Quality held steady at 13th. Low Leverage advanced sharply in the rankings from 6th to 3rd, indicating stronger preference for balance sheet-focused factors. Management Quality, while still highly ranked, fell from 1st to 4th — a notable decline for this typically dominant factor. Operating Quality rose strongly from 4th to 2nd, securing one of the highest-ranked positions in the framework. Profitability held steady at 8th.

Momentum indicators moved lower in the rankings overall. **Composite Momentum** slipped from 9th to 11th. Price Momentum edged up in the rankings from 10th to 9th. Revisions declined from 2nd to 6th.

Low Risk moved lower in the rankings from 5th to 7th, indicating slightly reduced preference for defensive exposures.

Our current factor recommendations for the US include 1) Up vs. Down EPS Revisions, (2) Internal Growth, and (3) Low PEG.

Exhibit 36: US – Changes in Cluster Rankings



This figure illustrates ranking changes of factor clusters from the end of the previous month to the most recent month. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The x-axis represents the clusters, while the y-axis indicates the ranks. Sources: FactSet, Morgan Stanley Research.

Cluster Rankings: Japan

In Japan, the regimes remain unchanged from the previous month.

Within our 17-factor framework, **Management Quality** reclaimed 1st place, rising from 4th to the top position, reasserting its dominance within the quality complex. At the opposite end of the rankings, Profitability remains the least favoured exposure, holding steady at 17th place.

Across the value complex, **Composite Value** moved lower in the rankings from 2nd to 6th. Current Value edged up from 3rd to 2nd. Deep Value slipped from 1st to 3rd, relinquishing its dominant position in the framework. Defensive Value rose modestly in the rankings from 5th to 4th.

Within the growth segment, **Composite Growth** declined in the rankings from 11th to 14th, indicating reduced support for broad-based growth exposures. Growth edged up in the rankings from 13th to 12th.

Quality-oriented measures presented a mixed picture. **Composite Quality** held steady at 15th. Earnings Quality slipped marginally in the rankings from 7th to 8th. Low Leverage moved lower from 14th to 15th. Management Quality rose notably from 4th to 1st, reclaiming the top position in the framework. Operating Quality advanced in the rankings from 8th to 7th.

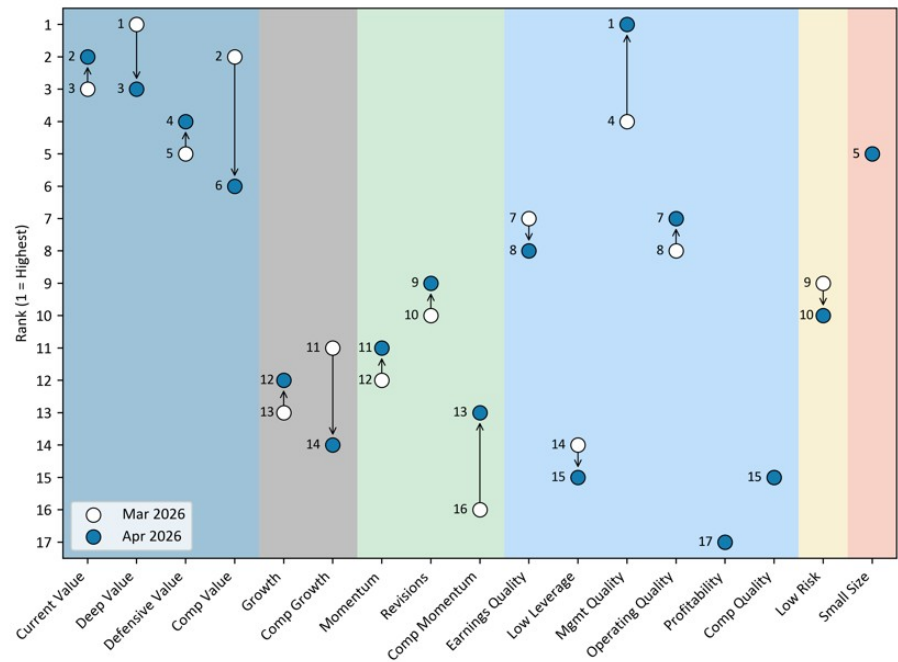
Momentum indicators rose in the rankings overall. **Composite Momentum** advanced from 16th to 13th, indicating renewed support for broad-based momentum measures. Price Momentum edged up in the rankings from 12th to 11th. Revisions rose modestly from 10th to 9th.

Low Risk slipped slightly in the rankings from 9th to 10th, indicating slightly reduced preference for defensive exposures.

Small Size held steady at 5th, maintaining its position in the upper half of the framework and underscoring continued preference for smaller-capitalisation stocks in the Japanese market.

Our preferred factors for Japan are 1) Up vs. Down EPS Revisions, 2) 12M Fwd Earnings Yield, and 3) Low PEG.

Exhibit 37: Japan – Changes in Cluster Rankings



This figure illustrates ranking changes of factor clusters from the end of the previous month to the most recent month. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The x-axis represents the clusters, while the y-axis indicates the ranks. Sources: FactSet, Morgan Stanley Research.

Cluster Rankings by Region

Exploring the 4-Dimensions Framework across regions, in [Exhibit 38](#) we present the current ranking of our 17 factor groups.

Exhibit 38: Cluster Rankings by Region

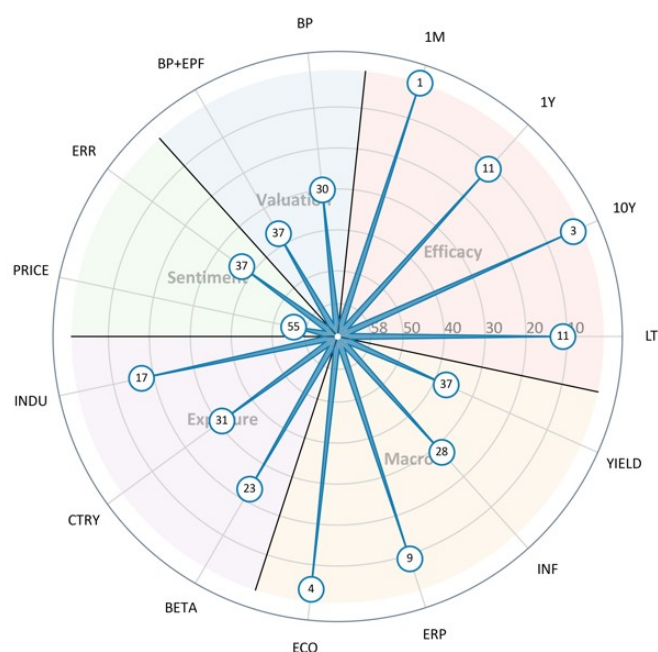
		Europe	US	Japan	World	EM	AxJ	Global All
Value	Current Value	3	10	2	4	12	8	8
	Deep Value	8	17	3	12	7	12	14
	Defensive Value	9	5	4	15	16	15	17
	Composite Value	11	15	6	16	11	13	9
Growth	Growth	12	16	12	14	15	16	15
	Composite Growth	13	14	14	10	17	17	16
Momentum	Momentum	4	9	11	3	10	11	7
	Revisions	1	6	9	8	2	1	6
	Composite Momentum	2	11	13	5	14	14	9
Quality	Earnings Quality	10	13	8	7	9	2	3
	Low Leverage	16	3	15	13	6	7	11
	Management Quality	5	4	1	1	1	4	1
	Operating Quality	14	2	7	6	5	5	3
	Profitability	15	8	17	9	3	3	13
	Composite Quality	17	12	15	17	4	10	12
Size Risk	Low Risk	7	7	10	10	8	6	5
	Small Size	6	1	5	2	13	9	2

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The factor performances are defined as the long/short returns of the top versus bottom quintile of their respective universes, the portfolios are rebalanced monthly and returns are equal weighted.

Factor Calls

Factor Calls: Europe

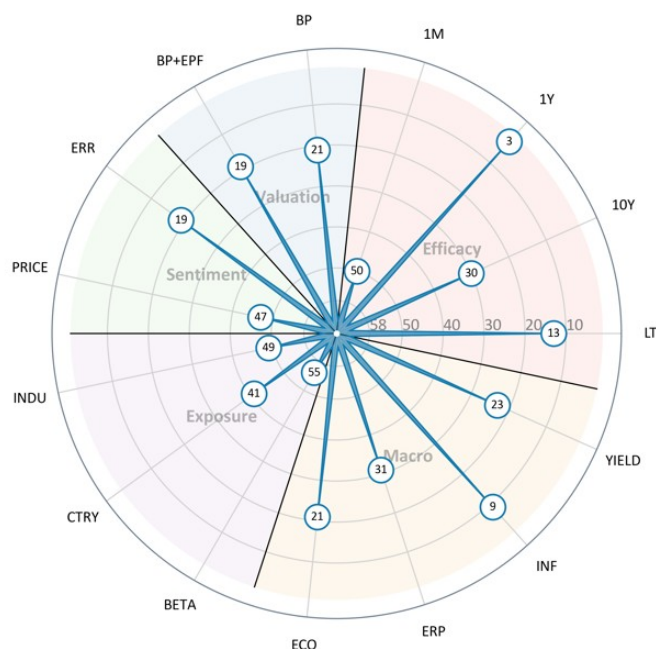
Exhibit 39: Europe – Factor Call: Up vs Down EPS Revisions (3ma)



Up vs Down EPS Revisions (3ma)

In Europe, the Up vs. Down EPS Revisions (3-month average) factor has improved its ranking from 18th to 16th (out of 58) in our framework. Amid macroeconomic uncertainty, corporate earnings remain among the most reliable indicators of underlying resilience. Additionally, this factor demonstrates positive net AI exposure, advantageous given our global expectation for continued AI-driven productivity gains. Encouragingly, the factor has exhibited strong 1-year, 10-year and long-term efficacies. With growth forecasts for 2026 still subdued, the ability to identify companies poised to deliver positive earnings surprises becomes even more critical. Moreover, our analysis shows the factor's predictive power, with the forward-looking capability of identifying companies that are poised to deliver positive earning surprises (beats) reinforcing its relevance as a timely and effective signal within the current investment landscape.

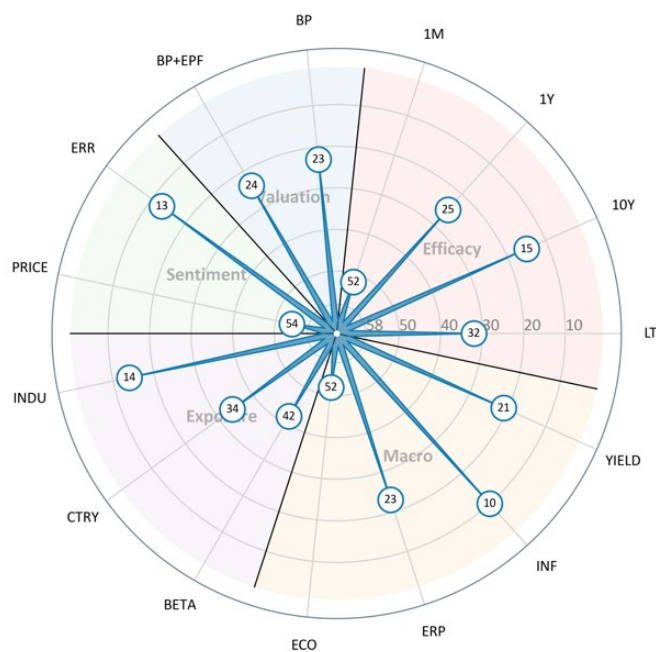
Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework. 1 = factor having top rank in a particular dimension evaluation. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI Europe, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Exhibit 40: Europe – Factor Call: Operating CF Yield

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework. 1 = factor having top rank in a particular dimension evaluation. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI Europe, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Operating CF Yield

In Europe, the Operating Cash Flow Yield factor has improved its ranking from 38th to 27th (out of 58) in our framework. Historically, this factor performs well in range-bound inflation environments, delivering an average annualised return of up to 6.8%. It offers a compelling mix of defensive characteristics and net defensive sector exposure. Its strength is underpinned by high efficacy, particularly over 1-month, 1-year and long-term horizons. In 2025, it was a standout performer, posting 24% returns over the year. In 2026, it has already surged by 14% (as of April 30, 2026).

Exhibit 41: Europe – Factor Call: "Low Asset Growth" Factor

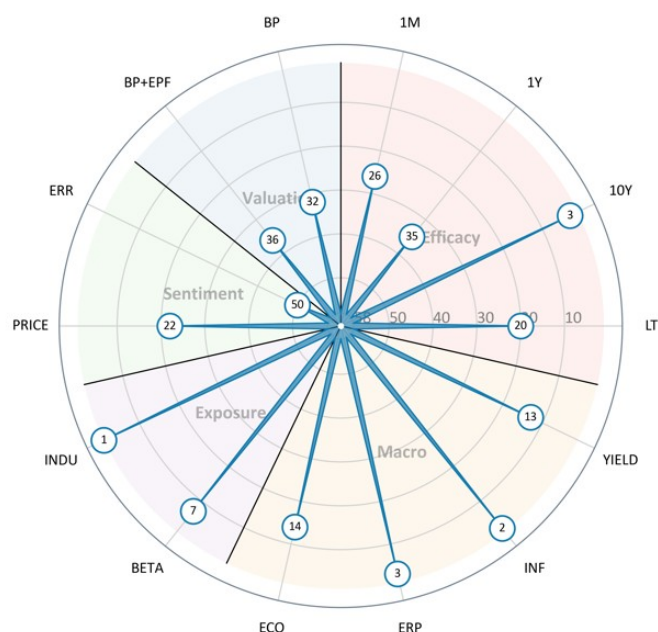
Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework. 1 = factor having top rank in a particular dimension evaluation. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI Europe, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

"Low Asset Growth" Factor

The "Low YoY Asset Growth" factor is ranked 32nd (out of 58 in our framework), reflecting a shift to a higher-for-longer rate environment. In this backdrop, the factor is well positioned, favouring companies with strong capital discipline and limited balance sheet expansion. These firms are more resilient to rising financing costs, while high asset growth companies face greater risks from overinvestment and funding pressures. It also benefits from supportive sector tilts—overweight Energy and Materials. Valuations remain reasonable, suggesting further upside.

Factor Calls: US

Exhibit 42: US – Factor Call: EPS Up-Down Revisions

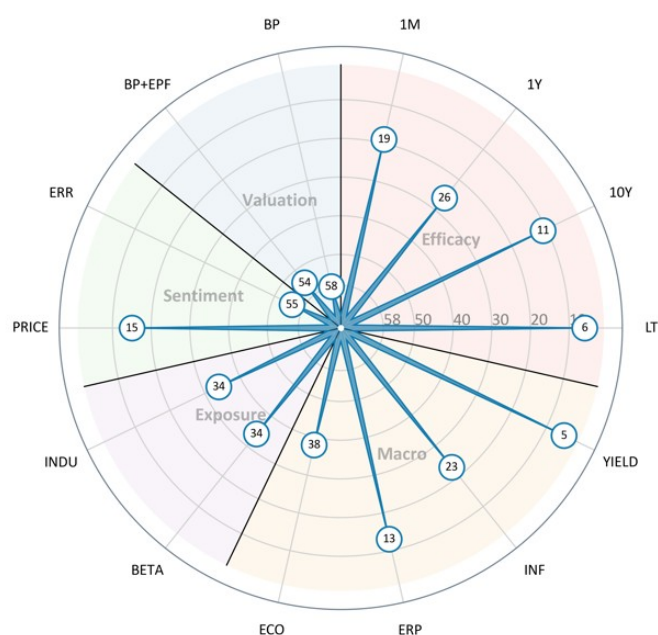


Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework. 1 = factor having top rank in a particular dimension evaluation. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI US, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Up vs. Down EPS Revisions

In the US, the EPS Up-Down Revisions factor is ranked 1st (out of 58 factors), ranking as the top factor in our framework. The factor effectively identifies companies sustaining positive earnings momentum despite a challenging macro backdrop. Moreover, our analysis highlights its strong forward-looking capability, consistently pinpointing firms poised to deliver positive earnings surprises (beats), reinforcing its credibility as a leading indicator of corporate strength. Additionally, this factor demonstrates positive net AI exposure, advantageous given our global expectation for continued AI-driven productivity gains. Its strength is further underpinned by high efficacy over the 10-year horizon. Finally, it is ranked favourably in the macro dimension of our framework, lending it further support.

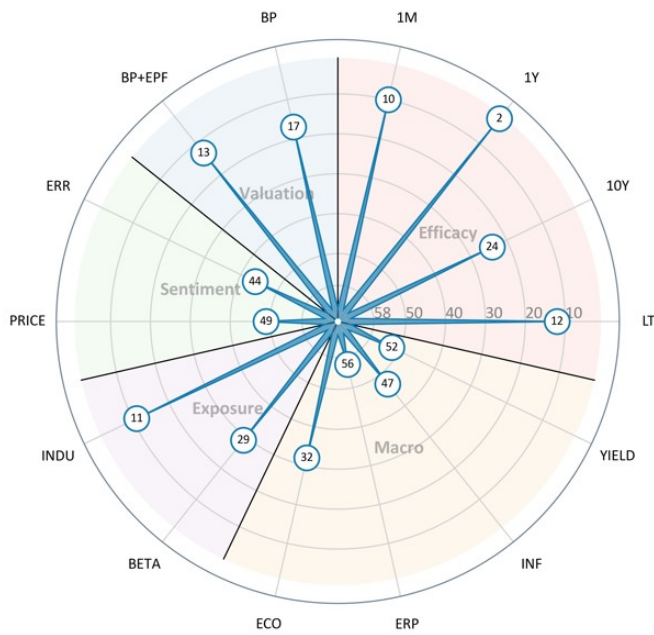
Exhibit 43: US – Factor Call: Internal Growth



Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework. 1 = factor having top rank in a particular dimension evaluation. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI US, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Internal Growth

In the US, the Internal Growth factor is ranked 49th (of 58) in our 4-D framework. 'Quality growth' is well represented by the Internal Growth factor – defined as retention $\times$ ROE – which identifies companies with the capacity to fund expansion from retained earnings, thereby combining profitability, reinvestment, and quality. The factor currently offers a balanced mix of cyclical participation and quality bias, and is exposed to the AI-driven productivity theme, aligning with our economists' expectation of efficiency gains through 2026-27.

Exhibit 44: US – Factor Call: Low PEG

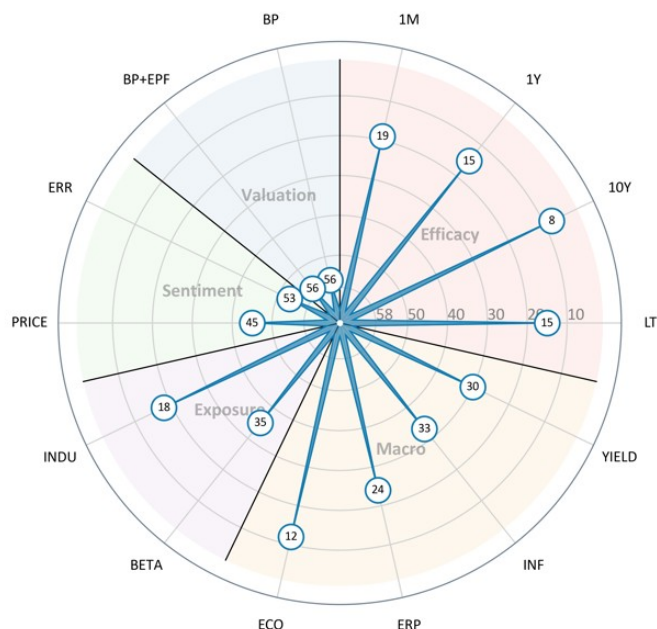
Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework. 1 = factor having top rank in a particular dimension evaluation. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI US, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Low PEG

In the US, Low PEG is ranked 14th out of 58 factors in our framework. Fundamentally, it offers a balanced blend of valuation discipline and sustainable growth exposure, positioning it well within the current market environment. Its strength is driven by three key attributes: (1) strong 1-month, 1-year and long-term efficacy, (2) a favorable placement within the Valuation dimension of our 4D framework, and (3) a healthy exposure to the AI theme.

Factor Calls: Japan

Exhibit 45: Japan – Factor Call: EPS Up-Down Revisions

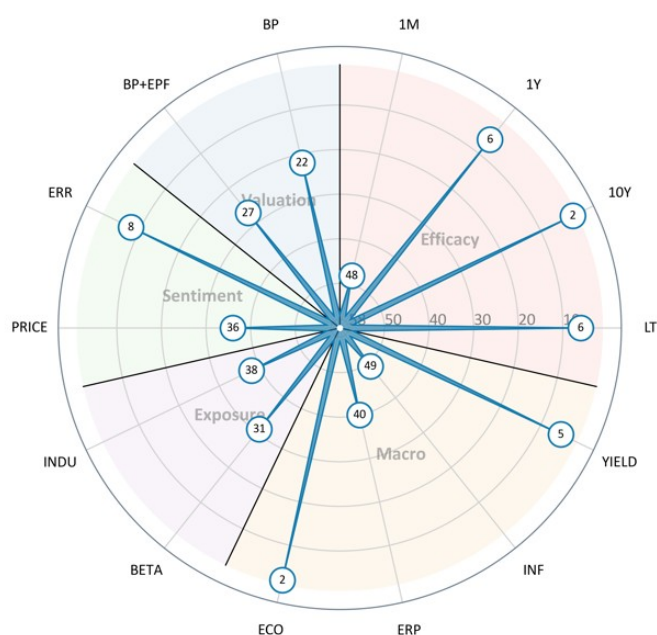


Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework. 1 = factor having top rank in a particular dimension evaluation. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI Japan, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Up vs. Down EPS Revisions

In Japan, EPS Up-Down Revisions is ranked 39th (of 58) in our framework. This factor has proven effective over 1-year to long term. As corporate earnings continue to expand steadily, we expect stock selection based on earnings performance to remain a key driver. In addition, this factor ranks well in the economic cycle dimension of our framework.

Exhibit 46: Japan – Factor Call: 12M Fwd Earnings Yield

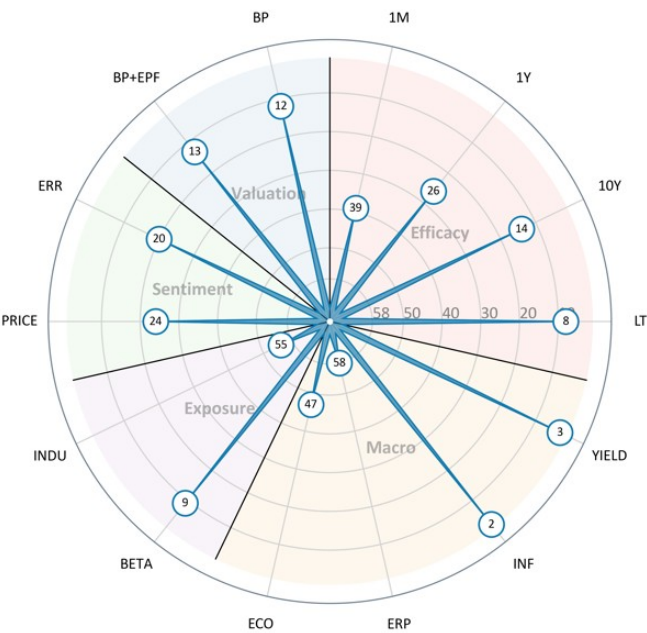


Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework. 1 = factor having top rank in a particular dimension evaluation. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI Japan, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

12M Fwd Earnings Yield

In Japan, 12M Forward Earnings Yield is ranked 9th (out of 58 factors), ranking as the top factor in our framework. Against the backdrop of rising interest rates – driven partly by fiscal expansion under the new administration – and ongoing corporate reforms, we favour the Value style in Japan. The influence of domestic rate movements on equity performance has also strengthened in recent years, particularly relative to the US. Earnings-based valuation should attract increased investor attention amid solid corporate performance, improving profitability, and a rising equity market. In addition, this factor exhibits strong efficacy across 1-year, 10-year and long-term horizon, and ranks well in the economic cycle and 10Y yield dimension of our framework.

Exhibit 47: Japan – Factor Call: PEG



Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework. 1 = factor having top rank in a particular dimension evaluation. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI Japan, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

PEG

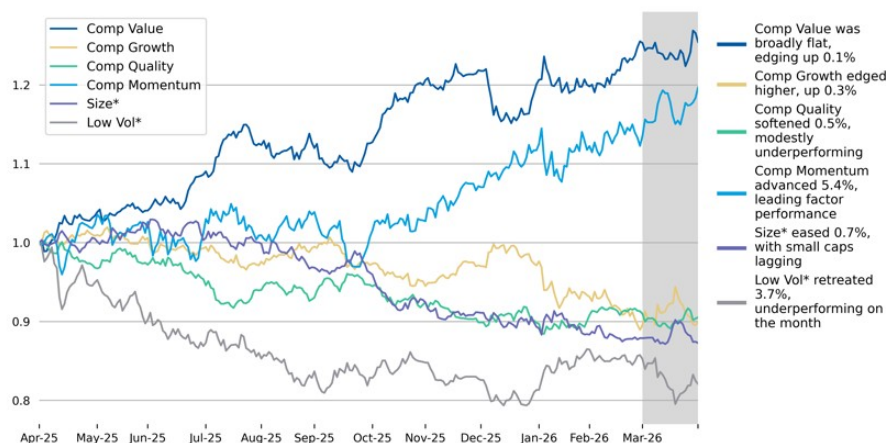
In Japan, Low PEG has improved its ranking from 27th to 12th (of 58) in our framework. Low PEG blends Value and Growth, providing an effective hedge during periods when growth stocks such as AI and semiconductor-related names rally, as seen in the latter half of 2025. Other key drivers include reasonable valuation and strong long-term efficacy. In addition, this factor ranks well in the Inflation and 10Y yield dimension of our framework.

Factor Performance

Factor Performance: Europe

European equities rebounded in April, with the Euro Stoxx rising 5.5%, supported by relief following the US-Iran ceasefire and a broader global risk-on backdrop. However, concerns around growth and inflation remained elevated amid higher energy prices. Euro Area activity data weakened sharply, with the composite PMI falling to 48.6 in April from 50.7 in March, reflecting broad-based softness across countries (see [report](#)). According to our economists, services activity is beginning to feel the impact of the energy shock, although manufacturing remained relatively resilient, with the manufacturing PMI rising to 52.2. Inflation remained mixed, with headline HICP rising to 3.0% YoY while core eased to 2.2% YoY (see [report](#)). According to our economists, further upside from energy prices remains likely. The ECB kept rates unchanged in April, while highlighting rising upside risks to inflation and downside risks to growth (see [report](#)). In the UK, activity data proved more resilient, with the composite PMI rising to 52.0 and monthly GDP increasing 0.5% m/m in February (see reports [here](#) and [here](#)).

April saw European factor performance supported by a broader global risk-on environment, despite ongoing concerns around growth and inflation. Sentiment improved early in the month following optimism around a potential Middle East ceasefire, supporting a rebound in European equities, while oil prices fell sharply before rebounding later as negotiations stalled. The recovery in markets came despite a weaker macro backdrop, with Euro Area PMIs deteriorating sharply and the ECB emphasizing rising downside risks to growth alongside persistent inflation concerns. Sector performance was broadly positive, with 9 out of 11 sectors delivering gains. Information Technology led the region with a strong +16.6% return, followed by Industrials (+11.0%) and Financials (+10.8%), while Energy (-2.1%) and Communication Services (-0.8%) lagged. Factor leadership also shifted meaningfully from March, with Momentum extending its rebound as risk appetite improved. Against this backdrop, **Composite Value** posted modest gains (+0.1%), as weakness in Energy and easing oil prices early in the month offset support from improving cyclical sentiment and rising risk appetite. Its net negative exposure to Industrials—one of the strongest-performing sectors during the month—acted as a relative headwind. In contrast, **Composite Growth** rose modestly (+0.3%), benefiting from the strong rebound in Information Technology and renewed optimism around AI and global equities, although gains remained muted relative to the US given the weaker Euro Area macro backdrop. The gap between Value and Growth narrowed significantly relative to March, when Composite Value had risen 4.4% while Composite Growth had declined 4.0% amid the energy shock and rising rates environment. **Composite Quality** declined 0.5%, as the rebound in higher-beta cyclical and technology sectors reduced investor preference for defensive and stable earnings exposures. **Composite Momentum** outperformed strongly (+5.4%), extending the positive momentum seen in March, as continued strength in technology and cyclical sectors reinforced prevailing return trends and supported trend-following exposures. **Size (Small)** fell 0.7%, with smaller caps continuing to lag as tighter financial conditions, weaker PMIs, and softer growth expectations weighed on more domestically exposed companies. **Low Volatility** fell 3.7%, as investors rotated away from defensive positioning back toward higher-beta exposures.

Exhibit 48: Europe – Performance of Main Factors

Source: FactSet, Morgan Stanley Research. The factor performances are long/short returns of the top vs. bottom quintile in MSCI Europe. The portfolios are rebalanced monthly and returns are equal weighted.

Factor Performance: US

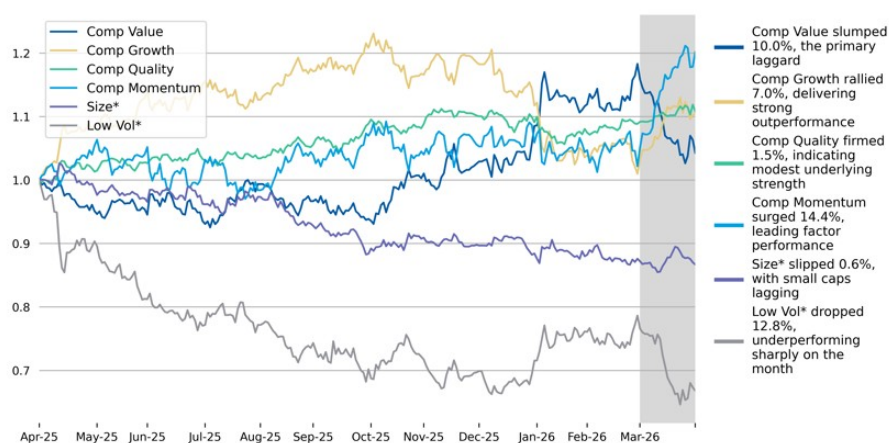
US equities rebounded strongly in April, with the S&P 500 rising 10.5% following the sharp March selloff. US data were generally firm. Payrolls rose 178k in March and the unemployment rate declined to 4.26% (see [report](#)), easing immediate Fed concerns around labor market weakness. ISM Manufacturing remained in expansionary territory at 52.7, while ISM Services moderated to 54.0 (see reports [here](#) and [here](#)). Core PCE inflation rose 0.37% m/m in February, marking a third consecutive month above 0.3% m/m, although our economists expect deceleration ahead (see [report](#)). Consumer sentiment weakened sharply, with preliminary University of Michigan sentiment falling to 47.6 (see [report](#)), while March retail sales remained strong, with headline sales rising 1.7% m/m and control group sales up 0.7% (see [report](#)). The Fed left rates unchanged, though the statement reflected a more hawkish tone and reduced easing bias (see [report](#)). Meanwhile, 1Q GDP rose 2.0%, with underlying domestic demand stronger than expected (see [report](#)).

April saw a sharp reversal in US factor leadership as improving sentiment around corporate earnings, artificial intelligence (AI), and semiconductors drove a strong rebound in risk appetite following the March selloff. Sentiment improved as macro data remained resilient and oil prices stabilized somewhat despite ongoing Middle East tensions, while a supportive earnings season further reinforced the recovery in risk appetite. Sector performance was broadly positive, with 9 out of 11 sectors delivering positive returns. Communication Services (+18.4%) led gains, followed by Information Technology (+17.4%) and Consumer Discretionary (+11.5%), while Energy (–3.5%) and Healthcare (–0.5%) lagged. Meanwhile, the 10-year US Treasury yield ended slightly higher at 4.39% versus 4.32% the previous month. Against this backdrop, **Composite Value** declined sharply (–10.0%), reversing part of its strong March outperformance as leadership rotated away from energy and defensive cyclicals toward growth-oriented sectors. Weakness in Energy, alongside Value's lower exposure to technology and AI-linked names, weighed materially on performance. In contrast, **Composite Growth** rebounded strongly (+7.0%), benefiting from renewed strength in large-cap technology, semiconductors, and AI-related equities as investors rotated back toward longer-duration exposures. **Composite Quality** gained 1.5%,

supported primarily by its strong long exposure to Information Technology, which was among the best-performing sectors during the month. Its underweight exposure to Communication Services partially offset gains, given the sector's strong performance.

Composite Momentum was the strongest performer, rising 14.4%, as the sharp rebound in technology and communication services reinforced prevailing return trends and supported trend-following exposures. **Small Size** declined 0.6%, with smaller caps continuing to lag despite the broader equity recovery, as investors remained focused on larger, more liquid growth names. **Low Volatility** fell sharply (–12.8%), as investors rotated aggressively away from defensive positioning and back into higher-beta risk exposures.

Exhibit 49: US – Performance of Main Factors



Source: FactSet, Morgan Stanley Research. The factor performances are long/short returns of the top vs. bottom quintile in MSCI US. The portfolios are rebalanced monthly and returns are equal weighted.

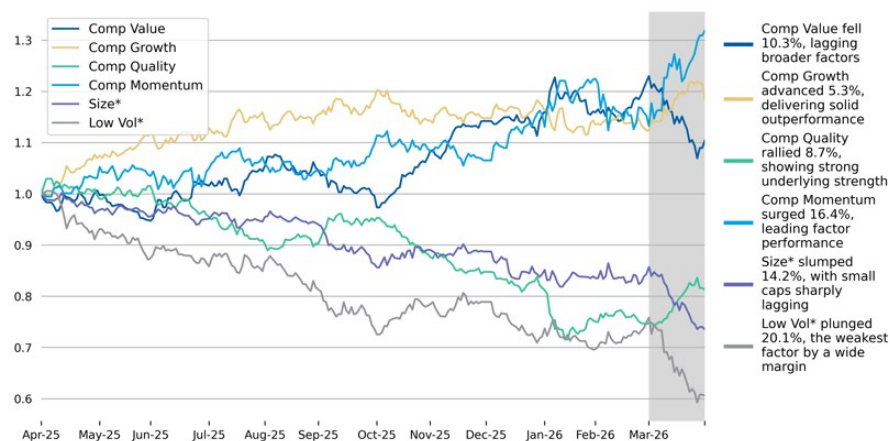
Factor Performance: Japan

Japanese equities rebounded in April following March's sharp correction, with TOPIX Total Return rising 6.6%. The recovery was supported by improving geopolitical expectations following ceasefire optimism, continued leadership from AI-linked equities, and firmer domestic rate expectations. However, soft economic data weakened materially, with both consumer sentiment indicators and Reuters Tankan manufacturing sentiment deteriorating sharply amid heightened Middle East tensions and rising supply chain concerns (see reports [here](#) and [here](#)). According to our economists, the deterioration in soft data suggests this is not an environment in which the BoJ can "raise the policy rate without hesitation." In contrast, wage data remained firm, with headline cash earnings rising 3.3% YoY and real wages up 1.9% YoY in February (see [report](#)). The BoJ delivered a hawkish hold in April, though our economists note that the outlook for a June rate hike will depend heavily on developments around the Strait of Hormuz (see [report](#)). In FX, the yen appreciated to 156.7 per US dollar by month-end. Our FX strategists maintain a neutral JPY view with a bullish skew, citing limited USD/JPY upside due to strong MoF intervention rhetoric and potential safe-haven support from prolonged geopolitical tensions (see [report](#)).

April saw a sharp reversal in Japanese factor performance as improving geopolitical expectations and optimism around a potential US-Iran ceasefire drove a strong recovery in risk appetite following March's selloff. Markets were also supported by continued

leadership from AI-linked equities and firmer domestic rate expectations. Sector performance was led by technology-related segments, with Information Technology surging 29.0%, followed by Materials (+11.3%) and Communication Services (+11.2%), while Energy (–8.0%) and Utilities (–4.1%) lagged. Against this backdrop, **Composite Value** declined sharply (–10.3%), reversing part of its strong March outperformance as weakness in Energy and the rebound in technology-related sectors weighed heavily on performance. In contrast, **Composite Growth** rose 5.3%, mirroring the broader global rebound in growth-oriented and AI-linked equities, supported by renewed strength in technology and improving market sentiment. **Composite Quality** gained strongly (+8.7%), supported primarily by its significant exposure to Information Technology, the best-performing sector during the month (+29.0%), as AI-linked equities rebounded sharply. **Composite Momentum** was the strongest performer, rising 16.4%, extending the sharp rebound following the March momentum unwind as persistent leadership in AI and technology reinforced prevailing return trends. **Size (Small)** declined sharply (–14.2%), with smaller domestic-oriented names lagging materially as investors rotated back toward larger-cap growth and technology exposures. **Low Volatility** fell 20.1%, sharply reversing its strong March outperformance as defensive positioning unwound aggressively amid improving risk appetite and the recovery in higher-beta equities. We note that year-to-date moves in Composite Momentum and Volatility factors in Japan have been particularly outsized, reflecting a narrowing and increasingly concentrated market led by high-beta names, particularly those within foreign investor universes, where investors are effectively forced to chase a shrinking pool of stocks to keep pace in rising markets (see [Japan Quantitative Strategy: Beta Polarization, 22 Apr 2026](#)).

Exhibit 50: Japan – Performance of Main Factors



Source: FactSet, Morgan Stanley Research. The factor performances are long/short returns of the top vs. bottom quintile in MSCI Japan. The portfolios are rebalanced monthly and returns are equal weighted.

Factor Contribution to Market Performance – April 2026

In this section, we analyse the contribution of key equity factors to market performance using both factor sensitivities (betas) and a realised return decomposition framework. We estimate factor exposures by regressing monthly total returns (in USD) of the MSCI US, MSCI Europe and MSCI Japan indices against six key factor return series over a rolling 36-month window, providing a more robust view of each market's structural factor tilts ([Exhibit 51](#)).

We then translate these exposures into realised contributions by combining factor betas with the latest month's factor returns, allowing us to assess how each style factor influenced overall market performance (see [Exhibit 52](#)). The residual non-style effects – defined as the market return minus the sum of the contributions from all six key factors – capture the portion of returns driven by broader market, sectoral and idiosyncratic effects.

In April, global equity markets rebounded strongly following the sharp declines seen in March, supported by easing geopolitical concerns and improving risk sentiment. In USD terms, the MSCI Europe gained 7.3%, the MSCI US rose 10.5%, and the MSCI Japan advanced 9.1%.

In **Europe**, factor contributions were broadly positive but relatively muted compared to other regions. Momentum was the primary positive driver, suggesting that recent outperformers continued to lead during the recovery. Low Volatility also contributed meaningfully, consistent with continued demand for defensive positioning even as markets rallied. Quality added modestly, while Value and Growth had negligible contributions. Size (Small) was a minor detractor, indicating limited headwinds from smaller-cap exposure. The residual non-style component remained the largest contributor, reflecting the dominance of broader market and idiosyncratic effects.

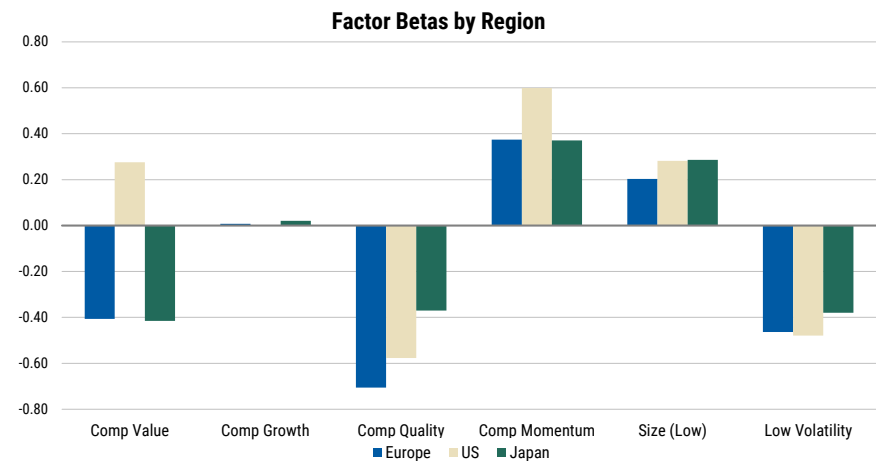
The **US** market displayed the strongest factor-driven performance profile. Momentum emerged as the dominant positive contributor, reflecting strong trend persistence and supportive factor returns during the month. Low Volatility also contributed substantially, suggesting that defensive characteristics were well-rewarded despite the risk-on environment. In contrast, Value was the primary detractor, indicating that valuation dynamics weighed on performance, while Quality also dragged modestly. Size (Small) had a limited negative impact, and Growth contributions were negligible. Notably, the residual component was slightly negative, suggesting that systematic factor effects dominated market movements in the US this month.

In **Japan**, factor contributions were pronounced, with notable divergence between positive and negative drivers. Low Volatility was the strongest positive contributor, reflecting meaningful outperformance of lower-volatility segments. Momentum also added substantially, indicating continued trend-following dynamics. Value contributed positively, suggesting a supportive environment for lower-valuation stocks. Growth was a marginal positive. However, Size (Small) was a significant detractor, indicating that smaller-cap stocks underperformed meaningfully during the period. Quality was also a drag, consistent with challenges for higher-quality segments. The residual component was modestly negative, suggesting that systematic factor exposures largely explained performance.

Overall, while factor exposures highlight persistent structural tilts across regions, the

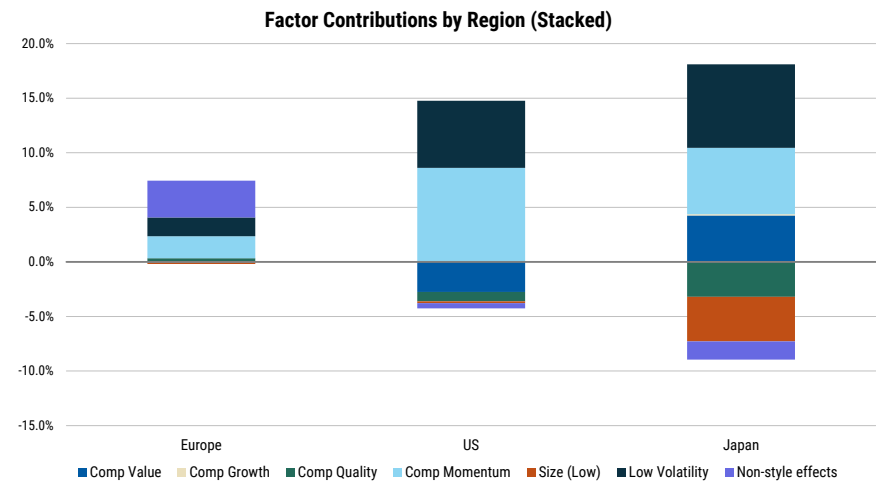
realised contributions emphasise a notable shift from March, with Momentum and Low Volatility emerging as key positive drivers across all three markets. Regional divergence remains evident: Europe saw more balanced contributions with a significant residual component; the US benefited from exceptionally strong Momentum effects offset by Value weakness; and Japan experienced pronounced factor contributions in both directions – with Low Volatility, Momentum and Value providing support, while Size and Quality posed meaningful headwinds.

Exhibit 51: Market Performance of the Three Regions



Source: Datastream, Morgan Stanley Research. Due to the small sample size, the coefficients are not statistically significant for a majority of the factors.

Exhibit 52: Factor Regression Coefficients

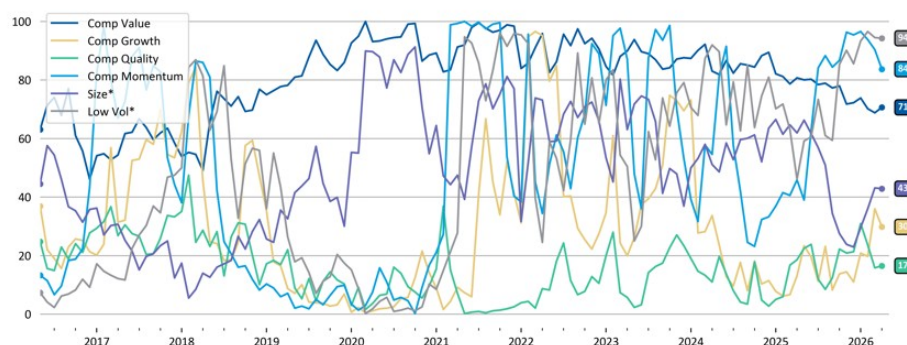


Source: FactSet, Datastream, Morgan Stanley Research.

Factor Valuation

Factor Valuation: Europe

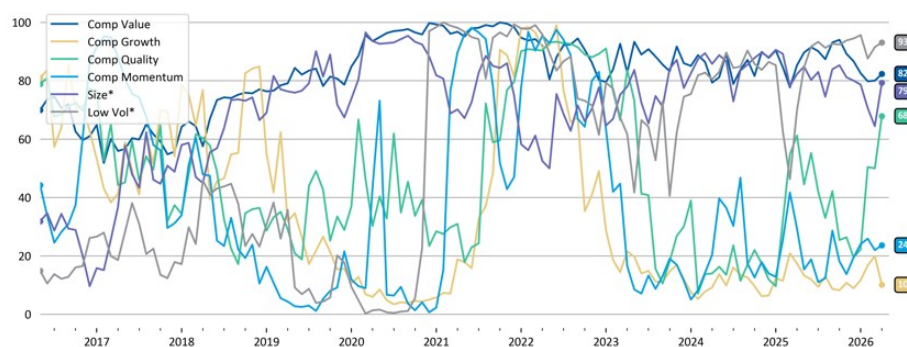
Exhibit 53: Europe – Valuation Percentiles of Main Factors



Source: FactSet, Morgan Stanley Research. This panel presents the cross-temporal valuation percentile series derived from the Book-to-Price Ratio and Forward Earnings Yield. For each factor, we calculate the ratio of valuation median of factor Q1 to valuation median of factor Q5. The series is averaged across metrics and converted to its full-sample percentile rank. Although the data spans January 1990 to the present, the horizontal axis is truncated to the last ten years to emphasise contemporary dynamics while maintaining trend smoothing. Values closer to 100 denote greater relative cheapness ("higher = more attractive").

Factor Valuation: US

Exhibit 54: US – Valuation Percentiles of Main Factors



Source: FactSet, Morgan Stanley Research. FactSet. This panel presents the cross-temporal valuation percentile series derived from the Book-to-Price Ratio and Forward Earnings Yield. For each factor, we calculate the ratio of valuation median of factor Q1 to valuation median of factor Q5. The series is averaged across metrics and converted to its full-sample percentile rank. Although the data spans January 1990 to the present, the horizontal axis is truncated to the last ten years to emphasise contemporary dynamics while maintaining trend smoothing. Values closer to 100 denote greater relative cheapness ("higher = more attractive").

Factor Valuation: Japan

Exhibit 55: Japan – Valuation Percentiles of Main Factors



Source: FactSet, Morgan Stanley Research. This panel presents the cross-temporal valuation percentile series derived from the Book-to-Price Ratio and Forward Earnings Yield. For each factor, we calculate the ratio of valuation median of factor Q1 to valuation median of factor Q5. The series is averaged across metrics and converted to its full-sample percentile rank. Although the data spans January 1990 to the present, the horizontal axis is truncated to the last ten years to emphasise contemporary dynamics while maintaining trend smoothing. Values closer to 100 denote greater relative cheapness ("higher = more attractive").

Factor Rankings

Factor Rankings: Europe

Cluster Rankings

Exhibit 56: Europe – Cluster Rankings

	EFFICACY				VALUATION		SENTIMENT		EXPOSURE			MACRO				RANK
	LT	10Y	1Y	1M	BP	BP+EPF	ERR	PRICE	INDU	CTRY	BETA	ECO	ERP	INF	YIELD	
Value																
Current Value	2	8	3	11	5	5	3	12	14	15	12	7	10	9	4	3
Deep Value	13	6	2	12	9	8	5	10	13	12	6	11	15	3	2	8
Defensive Value	1	4	6	9	7	5	9	15	15	13	15	2	14	8	14	9
Composite Value	4	6	1	8	8	9	7	11	17	17	13	12	9	5	3	11
Growth																
Growth	14	12	10	5	14	14	12	4	3	4	13	10	5	11	6	12
Composite Growth	8	9	13	5	17	15	17	1	6	1	16	3	12	15	9	13
Momentum																
Momentum	12	5	4	3	3	3	11	14	11	5	10	9	3	14	13	4
Revisions	3	1	5	2	4	7	13	16	4	7	5	4	1	7	11	1
Composite Momentum	6	3	7	1	2	2	16	17	10	14	3	5	2	16	8	2
Quality																
Earnings Quality	11	10	9	7	10	10	10	13	2	6	7	8	13	6	15	10
Low Leverage	10	11	11	14	13	16	2	5	8	16	2	15	11	12	10	16
Management Quality	5	2	8	16	6	4	8	9	5	10	9	17	8	4	7	5
Operating Quality	15	13	12	4	12	12	4	6	16	9	11	14	7	10	5	14
Profitability	9	15	17	13	16	13	15	2	9	11	4	6	4	13	12	15
Composite Quality	7	14	14	9	15	17	14	7	12	2	8	16	16	2	17	17
Risk																
Low Risk	16	16	16	17	1	1	6	8	7	8	17	1	6	17	16	7
Size																
Small Size	17	17	15	15	11	11	1	3	1	3	1	13	17	1	1	6

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI Europe, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Factor Rankings

Exhibit 57: Europe – Factor Rankings

	EFFICACY				VALUATION		SENTIMENT		EXPOSURE			MACRO				
	1T	10Y	1Y	1M	BP	BP+EPF	ERR	PRICE	INDU	CTRY	BETA	ECO	ERP	INF	YIELD	RANK
Value																
Current Value																
12M Fwd Earnings Yield	20	17	6	20	22	26	29	33	53	50	12	29	24	27	7	23
Free Cash Flow Yield	1	27	36	42	17	15	3	25	32	36	38	13	40	57	11	9
Operating CF Yield	13	30	3	50	21	19	19	47	49	42	55	21	31	9	23	27
EBITDA-to-EV	12	34	8	47	19	20	6	48	46	56	47	37	43	23	22	35
Deep Value																
Book-to-Price	46	31	4	33	28	29	12	34	48	52	15	42	34	8	8	28
Sales-to-Price	48	37	2	49	27	23	10	46	26	55	22	36	49	11	9	29
Trailing Earnings Yield	14	14	21	31	26	22	27	29	50	10	2	6	47	17	19	2
Shiller P/E*	37	18	16	48	20	21	20	40	44	26	45	51	51	7	14	41
Defensive Value																
Dividend Yield	6	23	19	30	29	28	40	53	52	19	46	12	45	26	33	40
Total Yield	5	16	23	44	12	12	16	32	45	57	50	11	42	31	50	21
Composite Value																
Composite Value	19	25	5	29	25	25	23	38	58	52	40	38	33	20	15	37
Growth																
Growth																
Long Term Growth	49	33	34	16	53	55	43	22	7	2	51	55	15	46	13	51
YoY Earnings Growth	34	38	52	56	18	17	24	2	1	4	30	8	41	41	20	2
YoY Sales Growth	55	51	37	24	57	57	25	7	40	47	48	40	46	33	16	56
P/E-to-Growth*	17	21	1	22	39	41	58	37	22	38	24	32	25	6	10	34
ST EPS Growth Rate	57	39	33	11	46	49	49	26	8	4	44	57	7	38	5	48
Internal Growth	27	36	50	39	48	38	15	1	23	11	43	2	30	19	49	15
Composite Growth																
Composite Growth	26	28	45	28	49	47	50	3	20	1	53	14	36	50	31	39
Momentum																
Momentum																
1M Momentum Reversal*	43	40	44	17	37	43	7	17	42	20	58	56	57	1	12	45
3M Momentum	51	32	18	27	35	40	57	35	30	33	56	39	3	35	55	55
6M Momentum	47	26	15	21	14	10	51	42	25	6	1	24	21	47	53	14
12M Momentum	29	20	12	12	8	9	39	44	39	22	34	23	18	55	46	19
6x1M Momentum	31	24	14	7	13	13	26	50	28	6	49	30	22	36	52	18
12x1M Momentum	21	22	17	6	6	6	31	51	36	21	13	25	20	54	38	10
12x1M Residual Mom	2	2	10	10	2	3	34	45	36	22	31	28	8	58	27	4
Revisions																
Earnings Revisions	7	12	24	23	11	27	22	39	15	15	35	10	5	22	42	8
EPS Up-Down Revs	4	5	20	13	32	45	35	36	6	29	32	17	4	13	45	20
Sales Revisions	36	29	28	19	9	16	38	31	33	40	8	18	6	14	34	13
Sales Up-Down Revs	8	4	13	14	16	18	53	41	11	24	33	31	2	16	36	12
3M EPS Up-Down Revs	11	3	11	1	30	37	37	55	17	31	23	4	9	28	37	16
3M Sales Up-Down Revs	18	8	9	2	7	5	33	49	29	46	6	15	14	48	41	1
Earnings Momentum	33	9	29	3	24	11	32	56	10	28	7	16	12	56	26	7
Composite Momentum																
Composite Momentum	22	11	22	4	15	14	48	57	31	39	19	19	19	51	29	24
Quality																
Earnings Quality																
Dividend Payouts	40	46	31	25	46	52	47	58	8	18	52	3	35	29	51	52
Short Term Accruals*	9	13	7	15	3	7	52	43	3	16	20	48	53	4	40	11
Accruals*	3	1	30	58	2	2	21	28	34	5	21	54	16	42	28	6
Tax-to-Pretax Income	54	52	32	34	56	58	30	52	12	37	10	49	39	15	43	57
Earnings Volatility*	52	55	58	9	38	46	17	16	5	43	25	5	48	49	47	42
Low Leverage																
Net Debt-to-Equity*	39	49	43	43	44	56	9	14	54	45	14	34	58	24	48	53
Chg LT Debt-to-Assets*	24	19	40	45	42	51	14	19	2	48	5	53	13	43	25	44
Management Quality																
Net Buyback Yield	15	7	35	55	6	1	8	15	21	58	17	41	37	32	32	5
YoY Asset Growth*	32	15	25	52	23	24	13	54	14	34	42	52	23	10	21	32
Change in Headcount*	16	10	27	38	31	31	56	30	18	14	36	50	32	12	24	38
Operating Quality																
Asset Turnover	42	42	38	8	55	42	28	23	51	17	27	33	10	21	4	33
Inventory-to-Sales*	53	58	42	18	10	8	36	24	57	49	54	22	1	18	56	43
Capex-to-Sales*	41	44	53	36	43	44	4	6	55	30	37	47	52	44	1	46
Capex-to-Depreciation*	38	53	56	35	54	50	5	8	43	27	26	58	29	39	17	49
R&D-to-Market Cap	35	6	26	5	33	30	11	27	56	51	41	43	55	30	6	31
Profitability																
Return on Equity	28	43	54	46	36	32	45	4	16	25	4	7	44	40	35	26
Return on Invest Capital	10	35	39	26	58	53	46	11	37	44	16	27	27	45	39	47
Net Margin	44	50	55	53	40	39	41	5	24	13	11	9	38	37	44	30
EBITDA Margin	30	48	41	57	50	48	55	18	47	42	39	20	26	52	58	58
Gross Profit-to-Assets	45	54	57	32	51	35	54	10	41	36	29	44	11	25	18	50
Return on Assets	23	45	51	41	52	34	44	9	13	54	18	26	17	34	30	36
Composite Quality																
Composite Quality	25	41	47	37	47	54	42	20	38	8	28	46	50	5	57	54
Risk																
Low Risk																
Low Vol (6M Ret Vol)*	50	47	48	54	4	4	18	21	28	32	57	1	28	53	54	22
Size																
Small Size																
Size (Free Float)*	58	57	46	51	34	32	1	12	4	12	9	35	54	2	2	16
Size*	56	56	49	40	41	36	2	13	20	9	3	45	56	3	3	25

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI Europe, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Factor Values

Exhibit 58: Europe – Factor Values

	EFFICACY				VALUATION		SENTIMENT		EXPOSURE			MACRO			
	LT	10Y	1Y	1M	BP	BP+EPF	ERR	PRICE	INDU	CTRY	BETA	ECO	ERP	INF	YIELD
Value	SR	SR	SR	SR	Pctil	Pctil	Pctil	Z	%	%	Spread	T-Stat	T-Stat	T-Stat	T-Stat
Current Value															
12M Fwd Earnings Yield	0.5	0.4	2.4	0.5	0.7	0.7	0.5	1.3	0.8	0.5	0.1	0.2	0.3	-0.2	0.9
Free Cash Flow Yield	0.9	0.2	0.0	-1.0	0.8	0.8	0.0	-0.0	0.5	0.4	0.2	1.2	-0.1	-1.0	0.7
Operating CF Yield	0.6	0.2	2.8	-1.8	0.7	0.8	0.2	1.9	0.7	0.4	0.4	0.6	0.1	0.4	0.2
EBITDA-to-EV	0.6	0.1	2.1	-1.6	0.7	0.8	0.0	1.9	0.7	0.6	0.2	-0.1	-0.2	-0.1	0.2
Deep Value															
Book-to-Price	0.1	0.2	2.8	-0.2	0.6	0.7	0.1	1.5	0.7	0.5	0.1	-0.3	0.0	0.5	0.8
Sales-to-Price	-0.1	0.1	3.1	-1.8	0.7	0.7	0.1	1.9	0.5	0.6	0.1	-0.1	-0.4	0.3	0.8
Trailing Earnings Yield	0.5	0.4	1.3	-0.1	0.7	0.8	0.5	0.6	0.7	0.3	0.0	1.9	-0.3	0.1	0.3
Shiller P/E*	0.2	0.4	1.7	-1.6	0.7	0.8	0.3	1.6	0.6	0.3	0.2	-0.8	-0.5	0.6	0.6
Defensive Value															
Dividend Yield	0.7	0.3	1.4	-0.1	0.6	0.7	0.8	2.0	0.8	0.3	0.2	1.5	-0.3	-0.1	-0.5
Total Yield	0.9	0.4	1.2	-1.2	0.9	0.9	0.2	1.3	0.7	0.6	0.3	1.5	-0.2	-0.3	-1.1
Composite Value															
Composite Value	0.5	0.3	2.4	0.1	0.7	0.7	0.5	1.6	1.0	0.5	0.2	-0.1	0.1	-0.0	0.6
Growth															
Growth															
Long Term Growth	0.1	0.1	0.3	1.1	0.2	0.2	0.8	-0.7	0.3	0.2	0.3	-1.0	0.5	-0.6	0.6
YoY Earnings Growth	0.2	0.0	-2.1	-2.2	0.8	0.8	0.5	-2.2	0.2	0.2	0.1	1.7	-0.2	-0.5	0.3
YoY Sales Growth	-0.2	-0.2	-0.1	0.4	0.1	0.1	0.5	-1.9	0.5	0.5	0.2	-0.2	-0.3	-0.3	0.5
P/E-to-Growth*	0.5	0.3	3.9	0.5	0.4	0.4	1.0	1.6	0.4	0.4	0.1	-0.0	0.3	0.6	0.8
ST EPS Growth Rate	-0.3	-0.0	0.3	1.9	0.3	0.2	0.9	0.3	0.3	0.2	0.2	-1.1	0.9	-0.4	1.0
Internal Growth	0.3	0.1	-1.6	-0.8	0.3	0.4	0.2	-2.5	0.4	0.3	0.2	2.2	0.2	0.0	-0.9
Composite Growth															
Composite Growth	0.3	0.2	-1.0	0.2	0.2	0.3	0.9	-2.2	0.4	0.2	0.3	1.2	-0.1	-0.8	-0.4
Momentum															
Momentum															
1M Momentum Reversal*	0.2	-0.0	-1.0	0.9	0.4	0.4	0.1	-1.2	0.6	0.3	0.5	-1.1	-0.9	1.2	0.6
3M Momentum	0.0	0.2	1.6	0.3	0.5	0.4	1.0	1.5	0.5	0.4	0.4	-0.2	1.0	-0.4	-1.3
6M Momentum	0.1	0.3	1.8	0.5	0.9	0.9	0.9	1.7	0.4	0.2	0.0	0.5	0.4	-0.6	-1.2
12M Momentum	0.3	0.3	1.9	1.8	0.9	0.9	0.8	1.8	0.5	0.3	0.1	0.5	0.5	-0.9	-0.8
6x1M Momentum	0.3	0.3	1.8	2.7	0.9	0.8	0.5	2.0	0.5	0.2	0.3	0.1	0.4	-0.4	-1.1
12x1M Momentum	0.4	0.3	1.7	3.6	1.0	0.9	0.6	2.0	0.5	0.3	0.1	0.4	0.5	-0.9	-0.6
12x1M Residual Mom	0.9	0.9	2.0	2.1	1.0	1.0	0.7	1.8	0.5	0.3	0.1	0.3	0.9	-1.2	-0.1
Revisions															
Earnings Revisions	0.7	0.4	1.0	0.5	0.9	0.7	0.5	1.6	0.4	0.3	0.1	1.5	1.0	-0.0	-0.8
EPS Up-Down Revs	0.9	0.7	1.3	1.7	0.5	0.3	0.7	1.5	0.3	0.4	0.1	1.0	1.0	0.2	-0.8
Sales Revisions	0.2	0.2	0.6	0.5	0.9	0.8	0.7	1.2	0.5	0.4	0.0	0.9	0.9	0.2	-0.5
Sales Up-Down Revs	0.7	0.7	1.8	1.4	0.8	0.8	0.9	1.6	0.3	0.3	0.1	0.1	1.1	0.1	-0.6
3M EPS Up-Down Revs	0.6	0.8	2.0	9.8	0.6	0.4	0.7	2.2	0.4	0.4	0.1	2.1	0.8	-0.2	-0.6
3M Sales Up-Down Revs	0.5	0.6	2.1	8.9	0.9	0.9	0.7	2.0	0.5	0.5	0.0	1.2	0.6	-0.7	-0.7
Earnings Momentum	0.2	0.5	0.6	7.5	0.7	0.9	0.7	2.4	0.3	0.4	0.0	1.0	0.7	-0.9	-0.0
Composite Momentum															
Composite Momentum	0.4	0.5	1.3	6.5	0.8	0.8	0.9	2.5	0.5	0.4	0.1	0.7	0.5	-0.8	-0.4
Quality															
Earnings Quality															
Dividend Payouts	0.2	-0.1	0.4	0.4	0.3	0.2	0.8	2.7	0.3	0.3	0.3	2.1	-0.0	-0.2	-1.1
Short Term Accruals*	0.7	0.4	2.2	1.2	1.0	0.9	0.9	1.7	0.3	0.3	0.1	-0.4	-0.6	0.6	-0.7
Accruals*	0.9	1.0	0.5	-2.9	1.0	1.0	0.4	0.5	0.5	0.2	0.1	-0.9	0.5	-0.5	-0.2
Tax-to-Pretax Income	-0.2	-0.2	0.3	-0.3	0.1	0.1	0.6	2.0	0.4	0.4	0.0	-0.6	-0.1	0.1	-0.8
Earnings Volatility*	-0.0	-0.3	-2.9	2.3	0.4	0.3	0.2	-1.2	0.3	0.4	0.1	2.0	-0.3	-0.7	-0.8
Low Leverage															
Net Debt-to-Equity*	0.2	-0.1	-0.9	-1.1	0.3	0.1	0.1	-1.4	0.9	0.5	0.1	-0.0	-1.0	-0.1	-0.8
Chg LT Debt-to-Assets*	0.4	0.4	-0.8	-1.3	0.4	0.2	0.2	-1.1	0.3	0.5	0.0	-0.9	0.6	-0.5	0.0
Management Quality															
Net Buyback Yield	0.5	0.6	0.1	-2.1	1.0	1.0	0.1	-1.3	0.4	0.6	0.1	-0.2	-0.1	-0.3	-0.5
YoY Asset Growth*	0.3	0.4	0.9	-2.0	0.7	0.7	0.2	2.1	0.4	0.4	0.2	-0.9	0.4	0.4	0.3
Change in Headcount*	0.5	0.5	0.6	-0.8	0.6	0.6	1.0	1.1	0.4	0.3	0.2	-0.8	0.1	0.3	0.1
Operating Quality															
Asset Turnover	0.2	-0.1	-0.3	2.3	0.1	0.4	0.5	-0.4	0.7	0.3	0.1	-0.0	0.7	-0.0	1.0
Inventory-to-Sales*	-0.1	-0.7	-0.8	0.8	0.9	0.9	0.7	-0.1	0.9	0.5	0.3	0.5	1.4	0.1	-1.5
Capex-to-Sales*	0.2	-0.1	-2.2	-0.5	0.4	0.4	0.0	-2.0	0.9	0.4	0.2	-0.4	-0.5	-0.5	1.5
Capex-to-Depreciation*	0.2	-0.2	-2.6	-0.3	0.2	0.2	0.0	-1.8	0.6	0.4	0.1	-1.3	0.2	-0.4	0.4
R&D-to-Market Cap	0.2	0.6	0.7	4.3	0.5	0.6	0.1	0.4	0.9	0.5	0.2	-0.3	-0.8	-0.3	0.9
Profitability															
Return on Equity	0.3	-0.1	-2.2	-1.5	0.5	0.5	0.8	-2.2	0.4	0.3	0.0	1.7	-0.2	-0.5	-0.6
Return on Invest Capital	0.6	0.1	-0.6	0.3	0.1	0.2	0.8	-1.7	0.5	0.4	0.1	0.3	0.3	-0.6	-0.6
Net Margin	0.2	-0.2	-2.5	-2.1	0.4	0.4	0.8	-2.1	0.4	0.3	0.0	1.6	-0.1	-0.4	-0.8
EBITDA Margin	0.3	-0.1	-0.8	-2.3	0.2	0.3	0.9	-1.1	0.7	0.4	0.2	0.7	0.3	-0.8	-1.7
Gross Profit-to-Assets	0.1	-0.3	-2.7	-0.2	0.2	0.5	0.9	-1.7	0.5	0.4	0.1	-0.3	0.7	-0.1	0.4
Return on Assets	0.4	-0.1	-2.0	-0.9	0.2	0.5	0.8	-1.7	0.4	0.6	0.1	0.3	0.5	-0.3	-0.4
Composite Quality															
Composite Quality	0.4	-0.1	-1.2	-0.6	0.3	0.2	0.8	-0.9	0.5	0.3	0.1	-0.4	-0.4	0.6	-1.7
Risk															
Low Risk															
Low Vol (6M Ret Vol)*	0.1	-0.1	-1.4	-2.1	1.0	0.9	0.2	-0.9	0.5	0.4	0.5	2.2	0.3	-0.9	-1.2
Size															
Small Size															
Size (Free Float)*	-0.3	-0.6	-1.2	-1.9	0.5	0.5	0.0	-1.5	0.3	0.3	0.0	-0.1	-0.6	0.8	1.1
Size*	-0.3	-0.4	-1.5	-0.9	0.4	0.4	0.0	-1.5	0.4	0.3	0.0	-0.3	-0.8	0.8	1.1

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI Europe, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Factor Rankings: US

Cluster Rankings

Exhibit 59: US – Cluster Rankings

	EFFICACY				VALUATION		SENTIMENT		EXPOSURE		MACRO				RANK
	LT	10Y	1Y	1M	BP	BP+EPF	ERR	PRICE	INDU	BETA	ECO	ERP	INF	YIELD	
Value															
Current Value	2	10	12	17	3	5	4	5	13	14	13	14	5	11	10
Deep Value	14	15	8	15	4	3	12	12	11	12	15	16	12	5	17
Defensive Value	11	9	13	14	2	2	7	10	6	15	11	8	3	3	5
Composite Value	7	11	10	16	6	6	6	8	12	15	16	15	10	7	15
Growth															
Growth	12	6	4	3	16	15	15	13	5	10	5	12	11	12	16
Composite Growth	5	5	5	4	17	17	17	6	9	13	3	2	9	16	14
Momentum															
Momentum	15	7	3	5	9	11	3	16	14	8	6	3	4	8	9
Revisions	13	2	6	2	15	16	10	14	3	4	2	4	6	6	6
Composite Momentum	6	4	2	1	12	14	14	17	8	11	1	5	2	15	11
Quality															
Earnings Quality	10	14	16	13	13	12	9	4	7	7	14	9	8	2	13
Low Leverage	9	12	7	6	7	10	8	9	4	6	4	11	13	13	3
Management Quality	8	13	9	11	5	4	13	11	2	9	8	13	14	1	4
Operating Quality	4	8	14	9	10	7	5	3	17	3	9	10	15	9	2
Profitability	3	3	11	10	14	13	11	7	10	5	7	6	7	14	8
Composite Quality	1	1	1	7	11	9	16	15	15	1	12	7	16	17	12
Risk															
Low Risk	17	16	17	12	1	1	1	2	16	17	10	1	1	10	7
Size															
Small Size	16	17	15	8	8	8	2	1	1	2	17	17	17	4	1

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI US, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Factor Rankings

Exhibit 60: US – Factor Rankings

	EFFICACY				VALUATION		SENTIMENT		EXPOSURE		MACRO				RANK
Value	LT	10Y	1Y	1M	BP	BP+EPF	ERR	PRICE	INDU	BETA	ECO	ERP	INF	YIELD	
Current Value															
12M Fwd Earnings Yield	15	29	20	57	11	22	14	30	42	38	47	48	12	33	35
Free Cash Flow Yield	1	30	48	56	2	3	12	1	44	35	35	10	26	47	3
Operating CF Yield	10	44	42	48	10	10	13	21	51	57	49	49	16	32	42
EBITDA-to-EV	21	47	30	50	16	18	35	27	41	55	46	51	31	35	54
Deep Value															
Book-to-Price	53	53	17	47	19	16	47	41	35	31	52	54	48	16	57
Sales-to-Price	32	45	31	49	12	14	26	26	46	51	37	55	36	29	51
Trailing Earnings Yield	19	39	25	55	4	7	30	25	27	47	41	34	19	38	20
Shiller P/E*	37	49	27	45	15	12	51	38	26	52	53	50	45	3	56
Defensive Value															
Dividend Yield	47	43	37	43	13	17	26	35	32	54	36	26	13	11	44
Total Yield	22	23	41	54	5	5	24	23	6	44	45	31	25	20	8
Composite Value															
Composite Value	28	38	32	51	20	21	23	28	38	49	50	47	34	24	53
Growth															
Growth															
Long Term Growth	49	25	10	5	38	38	58	48	22	53	6	44	53	43	55
YoY Earnings Growth	35	33	4	16	51	57	53	42	4	13	18	21	14	57	45
YoY Sales Growth	56	40	38	15	47	50	16	16	21	41	4	28	29	58	37
P/E-to-Growth*	12	24	2	10	17	13	44	49	11	29	32	56	47	52	14
ST EPS Growth Rate	50	8	15	4	50	45	57	46	15	45	1	46	40	34	52
Internal Growth	6	11	26	19	58	54	55	15	34	34	38	13	23	5	49
Composite Growth															
Composite Growth	25	20	18	14	46	48	56	20	28	46	13	17	33	50	50
Momentum															
Momentum															
1M Momentum Reversal*	45	46	54	27	21	6	9	8	39	56	39	53	58	28	27
3M Momentum	52	32	8	23	33	41	28	50	49	22	20	1	8	2	40
6M Momentum	46	17	3	20	26	29	10	52	55	4	15	4	5	7	6
12M Momentum	36	28	6	2	28	34	6	56	43	48	12	6	9	37	22
6x1M Momentum	40	19	13	22	7	9	1	53	56	33	11	8	18	9	4
12x1M Momentum	33	26	11	6	29	39	4	57	40	39	9	16	20	46	26
12x1M Residual Mom	13	13	5	12	41	52	2	55	37	14	17	33	27	51	23
Revisions															
Earnings Revisions	30	14	22	9	52	56	34	43	13	9	2	12	10	23	13
EPS Up-Down Revs	20	3	35	26	32	36	50	22	1	7	14	3	2	13	1
Sales Revisions	51	12	19	7	54	58	21	54	20	27	16	5	24	41	46
Sales Up-Down Revs	38	22	14	11	31	32	15	47	18	12	22	43	38	6	9
3M EPS Up-Down Revs	24	10	23	13	44	46	41	33	9	18	3	25	15	18	10
3M Sales Up-Down Revs	54	21	39	8	48	53	31	29	17	25	19	19	30	21	38
Earnings Momentum	29	7	9	3	42	44	45	51	14	30	10	30	32	42	29
Composite Momentum															
Composite Momentum	26	18	12	1	35	42	46	58	25	40	8	20	17	48	43
Quality															
Earnings Quality															
Dividend Payouts	44	41	45	39	9	11	22	34	33	50	33	7	7	19	24
Short Term Accruals*	17	55	57	53	36	26	3	3	23	24	29	39	50	12	11
Accruals*	4	42	47	35	39	30	42	19	2	26	58	38	41	10	31
Tax-to-Pretax Income	42	31	44	37	44	33	40	18	30	10	54	45	54	1	48
Earnings Volatility*	48	48	58	58	49	50	54	4	24	28	51	14	3	30	58
Low Leverage															
Net Debt-to-Equity*	41	27	46	21	8	22	32	12	29	32	21	36	55	40	19
Chg LT Debt-to-Assets*	18	54	1	18	37	35	27	45	3	6	7	29	22	44	5
Management Quality															
Net Buyback Yield	14	16	40	46	14	20	33	14	19	16	34	41	37	14	7
YoY Asset Growth*	39	56	33	40	23	15	36	37	8	43	43	40	44	8	39
Change in Headcount*	34	51	21	38	3	2	49	40	5	36	31	37	42	4	12
Operating Quality															
Asset Turnover	9	4	43	25	56	47	18	13	50	5	26	42	49	22	21
Inventory-to-Sales*	43	50	56	52	1	1	12	2	47	42	5	2	11	31	2
Capex-to-Sales*	11	37	55	29	40	31	18	6	58	17	42	32	43	27	33
Capex-to-Depreciation*	23	36	53	36	27	26	38	7	54	2	56	27	52	39	47
R&D-to-Market Cap	7	5	16	17	22	8	18	32	57	23	57	52	51	26	15
Profitability															
Return on Equity	8	6	36	42	57	55	39	24	10	21	23	15	4	49	28
Return on Invest Capital	2	1	24	28	55	37	43	17	45	3	30	22	21	56	16
Net Margin	27	9	34	33	34	50	52	31	16	11	24	11	6	54	36
EBITDA Margin	31	34	28	41	18	19	20	39	36	37	28	35	28	45	32
Gross Profit-to-Assets	5	35	52	31	43	43	29	5	53	20	25	18	39	25	34
Return on Assets	16	15	29	34	53	40	37	36	31	19	27	23	35	55	41
Composite Quality															
Composite Quality	3	2	7	24	30	28	48	44	48	1	44	24	46	53	30
Risk															
Low Risk															
Low Vol (6M Ret Vol)*	57	52	51	44	6	4	6	11	52	58	40	9	1	36	25
Size															
Small Size															
Size (Free Float)*	58	58	49	30	26	25	8	9	6	15	48	57	56	17	17
Size*	55	57	50	32	24	24	7	10	12	8	55	58	57	15	18

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI US, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Factor Values

Exhibit 61: US – Factor Values

	EFFICACY				VALUATION		SENTIMENT		EXPOSURE		MACRO			
	LT	10Y	1Y	1M	BP	BP+EPF	ERR	PRICE	INDU	BETA	ECO	ERP	INF	YIELD
	SR	SR	SR	SR	Pct	Pct	Pct	Z	%	Spread	T-Stat	T-Stat	T-Stat	T-Stat
Value														
Current Value														
12M Fwd Earnings Yield	0.3	0.1	0.6	-4.1	0.9	0.8	0.2	0.6	0.6	0.4	-0.9	-0.7	0.8	-0.3
Free Cash Flow Yield	0.6	0.1	-0.6	-4.0	1.0	0.9	0.2	-3.3	0.6	0.4	-0.5	0.9	0.5	-0.6
Operating CF Yield	0.4	-0.0	-0.1	-3.4	0.9	0.9	0.2	-0.1	0.7	0.9	-0.9	-0.7	0.7	-0.3
EBITDA-to-EV	0.2	-0.1	0.3	-3.6	0.9	0.9	0.5	0.4	0.6	0.8	-0.9	-0.9	0.4	-0.4
Deep Value														
Book-to-Price	-0.1	-0.2	0.8	-3.3	0.9	0.9	0.7	1.2	0.5	0.3	-1.0	-1.3	-0.6	0.1
Sales-to-Price	0.1	-0.1	0.3	-3.5	0.9	0.9	0.4	0.4	0.6	0.7	-0.6	-1.3	0.2	-0.2
Trailing Earnings Yield	0.2	0.0	0.4	-4.0	1.0	0.9	0.5	0.4	0.5	0.6	-0.7	0.0	0.6	-0.4
Shiller P/E*	0.1	-0.1	0.3	-3.0	0.9	0.9	0.8	1.0	0.5	0.7	-1.1	-0.7	-0.3	0.9
Defensive Value														
Dividend Yield	-0.0	-0.0	-0.0	-2.7	0.9	0.9	0.4	0.8	0.5	0.7	-0.5	0.5	0.8	0.3
Total Yield	0.2	0.2	-0.1	-4.0	1.0	0.9	0.4	0.0	0.3	0.5	-0.8	0.2	0.5	0.0
Composite Value														
Composite Value	0.1	0.0	0.3	-3.7	0.9	0.8	0.4	0.5	0.6	0.6	-0.9	-0.6	0.3	-0.1
Growth														
Growth														
Long Term Growth	-0.0	0.2	1.1	18.1	0.3	0.3	0.9	2.0	0.4	0.7	1.5	-0.4	-1.1	-0.5
YoY Earnings Growth	0.1	0.1	1.6	6.3	0.1	0.0	0.9	1.3	0.2	0.1	0.8	0.7	0.7	-1.0
YoY Sales Growth	-0.1	0.0	-0.0	7.4	0.1	0.1	0.3	-0.8	0.4	0.5	1.5	0.4	0.4	-1.1
P/E-to-Growth*	0.3	0.2	2.0	9.7	0.9	0.9	0.7	2.2	0.4	0.3	-0.2	-1.9	-0.6	-0.8
ST EPS Growth Rate	-0.0	0.4	0.9	18.2	0.1	0.1	0.9	1.7	0.4	0.6	2.0	-0.5	0.2	-0.3
Internal Growth	0.4	0.4	0.3	3.8	0.0	0.1	0.9	-0.9	0.5	0.4	-0.6	0.9	0.5	0.7
Composite Growth														
Composite Growth	0.2	0.2	0.7	7.7	0.1	0.1	0.9	-0.4	0.5	0.6	1.0	0.9	0.3	-0.7
Momentum														
Momentum														
1M Momentum Reversal*	0.0	-0.1	-1.7	0.7	0.8	0.9	0.2	-1.5	0.6	0.8	-0.6	-1.2	-1.5	-0.1
3M Momentum	-0.1	0.1	1.3	2.5	0.5	0.3	0.4	2.2	0.6	0.2	0.5	1.4	0.9	1.0
6M Momentum	0.0	0.3	1.6	3.7	0.8	0.6	0.2	2.3	0.7	0.0	1.0	1.3	1.0	0.7
12M Momentum	0.1	0.1	1.4	21.2	0.6	0.3	0.1	3.4	0.6	0.6	1.1	1.1	0.9	-0.4
6x1M Momentum	0.0	0.3	0.9	3.3	1.0	0.9	0.0	2.4	0.7	0.4	1.1	1.0	0.6	0.6
12x1M Momentum	0.1	0.2	1.1	16.7	0.6	0.3	0.1	3.4	0.6	0.5	1.2	0.9	0.6	-0.6
12x1M Residual Mom	0.3	0.4	1.6	9.3	0.2	0.1	0.0	3.1	0.6	0.1	0.9	0.1	0.4	-0.7
Revisions														
Earnings Revisions	0.1	0.4	0.5	11.3	0.0	0.1	0.5	1.4	0.4	0.1	1.8	0.9	0.9	-0.0
EPS Up-Down Revs	0.2	0.6	0.2	0.7	0.5	0.3	0.8	-0.1	0.2	0.0	1.0	1.4	1.2	0.3
Sales Revisions	-0.1	0.4	0.7	16.6	0.0	0.0	0.3	2.6	0.4	0.3	1.0	1.2	0.5	-0.4
Sales Up-Down Revs	0.0	0.2	0.9	9.6	0.5	0.4	0.3	1.9	0.4	0.1	0.4	-0.3	0.2	0.7
3M EPS Up-Down Revs	0.2	0.4	0.5	8.3	0.2	0.1	0.7	0.7	0.3	0.1	1.7	0.6	0.7	0.1
3M Sales Up-Down Revs	-0.1	0.2	-0.1	11.6	0.1	0.1	0.5	0.5	0.4	0.2	0.5	0.8	0.4	0.0
Earnings Momentum	0.1	0.4	1.2	18.7	0.2	0.2	0.7	2.2	0.4	0.3	1.2	0.2	0.3	-0.5
Composite Momentum														
Composite Momentum	0.1	0.3	1.1	21.7	0.4	0.2	0.7	3.7	0.5	0.5	1.3	0.8	0.6	-0.6
Quality														
Earnings Quality														
Dividend Payouts	0.0	-0.0	-0.2	-2.4	0.9	0.9	0.4	0.8	0.5	0.7	-0.4	1.1	0.9	0.0
Short Term Accruals*	0.3	-0.3	-1.9	-3.9	0.4	0.7	0.0	-2.5	0.4	0.2	-0.0	-0.2	-0.7	0.3
Accruals*	0.5	-0.0	-0.4	-1.5	0.3	0.5	0.7	-0.5	0.2	0.2	-2.0	-0.1	0.1	0.5
Tax-to-Pretax Income	0.0	0.1	-0.2	-1.9	0.2	0.4	0.6	-0.5	0.5	0.1	-1.1	-0.4	-1.1	1.1
Earnings Volatility*	-0.0	-0.1	-2.4	-6.1	0.1	0.1	0.9	-2.3	0.4	0.3	-0.9	0.9	1.1	-0.2
Low Leverage														
Net Debt-to-Equity*	0.0	0.1	-0.3	3.6	0.9	0.8	0.5	-1.3	0.5	0.4	0.4	-0.1	-1.2	-0.4
Chg LT Debt-to-Assets*	0.3	-0.2	2.4	3.9	0.3	0.3	0.4	1.5	0.2	0.0	1.3	0.3	0.5	-0.5
Management Quality														
Net Buyback Yield	0.3	0.3	-0.1	-3.3	0.9	0.8	0.5	-1.1	0.4	0.1	-0.5	-0.2	0.2	0.3
YoY Asset Growth*	0.0	-0.3	0.2	-2.5	0.8	0.9	0.6	0.9	0.3	0.5	-0.8	-0.2	-0.3	0.6
Change in Headcount*	0.1	-0.1	0.6	-2.2	1.0	1.0	0.8	1.1	0.3	0.4	-0.2	-0.1	0.1	0.8
Operating Quality														
Asset Turnover	0.4	0.5	-0.2	2.0	0.0	0.1	0.3	-1.2	0.7	0.0	0.0	-0.3	-0.7	-0.0
Inventory-to-Sales*	0.0	-0.1	-1.8	-3.7	1.0	1.0	0.2	-2.5	0.6	0.5	1.5	1.4	0.8	-0.2
Capex-to-Sales*	0.4	0.0	-1.7	-0.4	0.3	0.4	0.3	-2.0	0.8	0.1	-0.7	0.1	-0.2	-0.1
Capex-to-Depreciation*	0.2	0.1	-1.6	-1.7	0.7	0.7	0.6	-2.0	0.7	0.0	-1.2	0.5	-0.7	-0.4
R&D-to-Market Cap	0.4	0.5	0.9	4.5	0.8	0.9	0.3	0.7	0.8	0.2	-1.5	-1.2	-0.7	-0.1
Profitability														
Return on Equity	0.4	0.4	0.1	-2.7	0.0	0.1	0.6	0.2	0.4	0.2	0.4	0.9	1.1	-0.7
Return on Invest Capital	0.5	0.7	0.5	0.2	0.0	0.3	0.7	-0.7	0.6	0.0	-0.1	0.7	0.5	-1.0
Net Margin	0.1	0.4	0.2	-1.0	0.5	0.1	0.8	0.7	0.4	0.1	0.2	0.9	1.0	-0.8
EBITDA Margin	0.1	0.1	0.3	-2.6	0.9	0.8	0.3	1.1	0.6	0.4	0.0	-0.1	0.4	-0.5
Gross Profit-to-Assets	0.4	0.1	-1.6	-0.6	0.2	0.2	0.4	-2.2	0.7	0.2	0.0	0.8	0.2	-0.1
Return on Assets	0.3	0.3	0.3	-1.1	0.0	0.3	0.6	0.8	0.5	0.1	0.0	0.7	0.2	-0.9
Composite Quality														
Composite Quality	0.5	0.6	1.3	2.3	0.5	0.7	0.7	1.4	0.6	0.0	-0.8	0.6	-0.4	-0.8
Risk														
Low Risk														
Low Vol (6M Ret Vol)*	-0.1	-0.2	-1.5	-2.8	1.0	0.9	0.1	-1.3	0.7	1.1	-0.6	0.9	1.2	-0.4
Size														
Small Size														
Size (Free Float)*	-0.2	-0.5	-1.1	-0.6	0.8	0.8	0.1	-1.5	0.3	0.1	-0.9	-2.2	-1.2	0.1
Size*	-0.1	-0.4	-1.2	-0.7	0.8	0.8	0.1	-1.4	0.4	0.0	-1.1	-2.2	-1.3	0.1

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI US, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Factor Rankings: Japan

Cluster Rankings

Exhibit 62: Japan – Cluster Rankings

	EFFICACY				VALUATION		SENTIMENT		EXPOSURE		MACRO				RANK
	LT	10Y	1Y	1M	BP	BP+EPF	ERR	PRICE	INDU	BETA	ECO	ERP	INF	YIELD	
Value															
Current Value	2	3	10	15	2	4	1	4	16	12	5	13	14	4	2
Deep Value	4	6	8	16	3	6	3	9	12	8	8	14	2	2	3
Defensive Value	3	1	11	14	4	5	5	8	3	11	12	16	17	3	4
Composite Value	1	1	6	13	7	7	6	11	17	10	4	17	7	1	6
Growth															
Growth	13	10	7	5	13	13	10	12	8	7	9	10	8	9	12
Composite Growth	16	8	2	8	14	17	11	13	1	13	7	9	15	11	14
Momentum															
Momentum	14	9	5	3	8	8	12	15	11	14	3	3	3	11	11
Revisions	5	5	4	2	15	14	13	16	7	9	2	4	10	10	9
Composite Momentum	12	10	1	1	9	10	15	17	15	16	1	1	1	13	13
Quality															
Earnings Quality	6	12	13	12	10	9	7	5	6	5	13	8	12	8	8
Low Leverage	11	14	9	7	17	15	17	10	4	3	15	7	5	14	15
Management Quality	9	4	3	10	5	3	8	14	2	4	11	6	11	6	1
Operating Quality	8	7	12	9	11	12	9	3	13	6	6	2	6	7	7
Profitability	10	13	14	6	16	16	16	7	9	2	16	5	9	15	17
Composite Quality	7	15	15	4	12	11	14	6	14	1	17	11	13	17	15
Risk															
Low Risk	17	16	16	11	1	1	2	2	10	17	14	12	16	16	10
Size															
Small Size	15	17	16	17	6	2	4	1	5	15	10	15	4	5	5

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI Japan, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Factor Rankings

Exhibit 63: Japan – Factor Rankings

	EFFICACY				VALUATION		SENTIMENT		EXPOSURE		MACRO				RANK
Value	LT	10Y	1Y	1M	BP	BP+EPF	ERR	PRICE	INDU	BETA	ECO	ERP	INF	YIELD	
Value															
Current Value															
12M Fwd Earnings Yield	6	2	6	48	22	27	8	36	38	31	2	40	49	5	9
Free Cash Flow Yield	2	1	35	55	1	6	4	8	35	40	25	41	41	22	4
Operating CF Yield	5	32	38	43	5	5	3	18	58	44	18	48	48	9	11
EBITDA-to-EV	4	31	44	49	7	11	2	3	52	51	37	57	36	11	20
Deep Value															
Book-to-Price	9	34	25	47	23	18	4	30	50	36	33	55	15	1	21
Sales-to-Price	20	36	36	53	10	12	2	19	45	32	3	45	17	4	8
Trailing Earnings Yield	18	6	28	52	8	21	14	15	40	29	38	44	14	7	15
Shiller P/E*	12	23	31	46	19	16	25	31	24	37	43	52	8	13	22
Defensive Value															
Dividend Yield	3	9	40	45	16	14	17	17	18	43	34	51	56	8	10
Total Yield	11	5	37	51	14	15	16	28	12	34	39	54	47	15	13
Composite Value															
Composite Value	1	7	22	44	18	18	18	32	50	38	19	53	26	2	17
Growth															
Growth															
Long Term Growth	54	46	13	20	30	40	55	50	37	47	17	28	57	25	58
YoY Earnings Growth	43	37	21	24	32	41	27	27	1	21	30	31	12	35	24
YoY Sales Growth	46	41	39	22	52	54	36	22	8	8	29	13	29	29	34
P/E-to-Growth*	8	14	26	39	12	13	20	24	55	9	47	58	2	3	12
ST EPS Growth Rate	50	48	30	3	52	42	22	43	35	30	16	38	6	21	35
Internal Growth	42	12	24	13	55	54	38	34	13	27	41	25	52	51	48
Composite Growth															
Composite Growth	49	30	14	27	45	48	34	37	10	42	28	32	44	42	45
Momentum															
Momentum															
1M Momentum Reversal*	7	43	43	5	4	3	6	25	46	57	44	56	58	26	26
3M Momentum	58	27	16	30	35	38	50	40	24	46	10	4	20	31	37
6M Momentum	57	25	7	17	20	20	40	46	44	52	7	5	4	39	30
12M Momentum	56	39	11	9	24	29	44	52	54	58	15	10	18	54	53
6x1M Momentum	47	16	3	1	6	4	32	51	24	26	14	18	7	45	6
12x1M Momentum	52	42	17	2	21	24	44	58	51	54	24	15	30	53	54
12x1M Residual Mom	22	24	20	28	37	33	33	47	24	17	13	21	9	46	22
Revisions															
Earnings Revisions	29	17	5	14	58	57	31	49	40	39	5	7	42	36	42
EPS Up-Down Revs	15	8	15	19	56	56	53	45	18	35	12	24	33	30	39
Sales Revisions	21	28	33	7	36	35	47	56	28	41	22	35	5	37	40
Sales Up-Down Revs	14	22	27	10	40	39	23	41	28	11	9	27	37	32	18
3M EPS Up-Down Revs	27	26	12	8	54	52	45	54	18	48	11	1	43	41	44
3M Sales Up-Down Revs	19	18	19	11	34	32	28	57	9	33	26	36	35	34	27
Earnings Momentum	34	19	2	12	44	44	38	53	14	49	1	9	21	28	29
Composite Momentum															
Composite Momentum	40	33	9	6	25	30	46	55	44	55	8	12	13	47	41
Quality															
Earnings Quality															
Dividend Payouts	25	51	46	50	29	23	29	26	4	28	50	37	40	24	32
Short Term Accruals*	10	4	32	54	9	10	11	11	21	23	27	8	51	19	1
Accruals*	16	3	4	38	27	25	21	35	3	13	21	17	25	16	2
Tax-to-Pretax Income	36	55	57	31	42	36	9	13	33	10	46	47	31	18	33
Earnings Volatility*	35	56	58	33	39	37	35	14	39	24	54	42	46	58	55
Low Leverage															
Net Debt-to-Equity*	28	49	51	32	38	34	51	10	30	4	56	30	11	57	38
Chg LT Debt-to-Assets*	51	47	10	21	58	58	58	44	2	22	42	23	38	38	56
Management Quality															
Net Buyback Yield	30	10	34	58	2	1	10	33	6	20	49	14	54	23	7
YoY Asset Growth*	24	29	1	34	31	26	30	39	26	18	36	19	3	10	16
Change in Headcount*	32	20	8	23	13	8	26	48	5	12	23	46	53	17	5
Operating Quality															
Asset Turnover	23	15	29	37	49	53	40	23	35	16	6	3	32	27	31
Inventory-to-Sales*	53	53	53	42	28	22	24	6	47	45	53	2	55	44	50
Capex-to-Sales*	31	13	45	41	46	50	56	2	56	15	4	16	22	20	36
Capex-to-Depreciation*	17	11	49	36	26	28	14	7	7	1	35	11	16	33	3
R&D-to-Market Cap	13	35	18	15	11	18	12	42	57	25	20	43	1	6	14
Profitability															
Return on Equity	45	21	23	18	50	50	54	38	12	19	45	26	10	40	47
Return on Invest Capital	44	38	47	25	48	49	58	12	30	2	52	34	28	43	52
Net Margin	37	40	42	29	43	43	48	21	28	3	55	29	19	48	46
EBITDA Margin	39	45	41	4	41	45	37	29	48	6	57	20	27	50	51
Gross Profit-to-Assets	33	52	52	35	51	47	49	9	54	7	48	6	45	56	57
Return on Assets	38	44	48	26	46	46	52	16	15	14	51	22	34	49	49
Composite Quality															
Composite Quality	26	50	50	16	33	31	42	20	42	5	58	33	39	55	43
Risk															
Low Risk															
Low Vol (6M Ret Vol)*	55	54	55	40	3	2	6	5	32	56	40	39	50	52	28
Size															
Small Size															
Size (Free Float)*	48	57	56	57	14	7	19	4	18	53	32	50	24	14	25
Size*	41	58	54	56	17	9	13	1	18	50	31	49	23	12	19

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI Japan, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Factor Values

Exhibit 64: Japan – Factor Values

	EFFICACY				VALUATION		SENTIMENT		EXPOSURE		MACRO			
	LT	10Y	1Y	1M	BP	BP+EPF	ERR	PRICE	INDU	BETA	ECO	ERP	INF	YIELD
	SR	SR	SR	SR	Pct	Pct	Pct	Z	%	Spread	T-Stat	T-Stat	T-Stat	T-Stat
Value														
Current Value														
12M Fwd Earnings Yield	0.6	0.7	1.9	-3.6	0.8	0.6	0.1	0.9	0.6	0.3	1.7	-0.3	-0.8	1.6
Free Cash Flow Yield	0.7	1.1	0.0	-4.6	1.0	0.9	0.0	-1.6	0.6	0.3	0.5	-0.4	-0.5	0.2
Operating CF Yield	0.6	0.2	-0.1	-3.4	0.9	0.9	0.0	-1.0	1.0	0.4	0.8	-0.9	-0.8	1.0
EBITDA-to-EV	0.6	0.2	-0.5	-3.8	0.9	0.9	0.0	-2.4	0.8	0.5	0.1	-1.3	-0.4	1.0
Deep Value														
Book-to-Price	0.5	0.2	0.6	-3.6	0.8	0.8	0.0	0.2	0.8	0.3	0.3	-1.2	0.0	2.4
Sales-to-Price	0.2	0.1	-0.0	-4.2	0.9	0.9	0.0	-0.9	0.7	0.3	1.6	-0.7	0.0	1.7
Trailing Earnings Yield	0.3	0.5	0.5	-4.2	0.9	0.8	0.1	-1.1	0.6	0.3	0.1	-0.7	0.1	1.0
Shiller P/E*	0.5	0.3	0.3	-3.6	0.8	0.8	0.4	0.3	0.6	0.3	-0.5	-1.1	0.3	0.9
Defensive Value														
Dividend Yield	0.6	0.5	-0.2	-3.5	0.9	0.8	0.1	-1.0	0.5	0.4	0.3	-1.1	-1.0	1.0
Total Yield	0.5	0.5	-0.0	-4.1	0.9	0.8	0.1	-0.0	0.5	0.3	0.0	-1.2	-0.8	0.9
Composite Value														
Composite Value	0.7	0.5	0.7	-3.5	0.8	0.8	0.2	0.5	0.8	0.3	0.8	-1.2	-0.2	2.1
Growth														
Growth														
Long Term Growth	-0.2	-0.1	1.6	8.2	0.4	0.1	0.9	2.6	0.6	0.4	0.8	0.4	-1.3	0.0
YoY Earnings Growth	-0.0	0.1	1.0	5.8	0.3	0.1	0.5	-0.1	0.3	0.2	0.5	0.3	0.1	-0.8
YoY Sales Growth	-0.1	0.0	-0.2	6.6	0.0	0.0	0.7	-0.7	0.4	0.0	0.5	0.8	-0.2	-0.5
P/E-to-Growth*	0.5	0.4	0.6	-2.4	0.9	0.9	0.2	-0.4	0.9	0.0	-0.6	-1.6	0.9	2.0
ST EPS Growth Rate	-0.1	-0.2	0.3	17.4	0.0	0.1	0.4	2.0	0.6	0.3	0.8	-0.3	0.3	0.3
Internal Growth	-0.0	0.4	0.6	12.3	0.0	0.0	0.7	0.9	0.5	0.2	-0.5	0.4	-0.9	-1.8
Composite Growth														
Composite Growth	-0.1	0.2	1.5	5.1	0.1	0.1	0.6	1.1	0.5	0.4	0.5	0.3	-0.7	-1.3
Momentum														
Momentum														
1M Momentum Reversal*	0.6	-0.0	-0.4	17.0	0.9	0.9	0.0	-0.3	0.8	0.7	-0.5	-1.3	-1.3	-0.3
3M Momentum	-0.3	0.3	1.4	3.0	0.3	0.2	0.8	1.5	0.6	0.4	1.0	1.1	-0.1	-0.6
6M Momentum	-0.3	0.3	1.9	10.2	0.8	0.8	0.7	2.1	0.7	0.5	1.3	1.0	0.5	-1.3
12M Momentum	-0.3	0.1	1.6	14.0	0.7	0.5	0.8	2.8	0.8	0.8	0.9	0.8	-0.0	-1.8
6x1M Momentum	-0.1	0.4	2.0	18.0	0.9	0.9	0.6	2.7	0.6	0.2	0.9	0.6	0.3	-1.5
12x1M Momentum	-0.1	0.0	1.3	17.7	0.8	0.7	0.8	3.8	0.8	0.6	0.6	0.7	-0.2	-1.8
12x1M Residual Mom	0.1	0.3	1.1	3.5	0.2	0.3	0.6	2.2	0.6	0.1	0.9	0.5	0.3	-1.5
Revisions														
Earnings Revisions	0.1	0.4	2.0	11.7	0.0	0.0	0.6	2.4	0.6	0.3	1.4	0.9	-0.6	-0.8
EPS Up-Down Revs	0.4	0.5	1.5	8.4	0.0	0.0	0.9	2.1	0.5	0.3	1.0	0.4	-0.3	-0.6
Sales Revisions	0.1	0.3	0.2	15.8	0.3	0.3	0.8	3.2	0.6	0.4	0.6	0.1	0.4	-0.9
Sales Up-Down Revs	0.4	0.3	0.6	13.8	0.1	0.1	0.4	1.8	0.6	0.1	1.0	0.4	-0.4	-0.6
3M EPS Up-Down Revs	0.1	0.3	1.6	14.9	0.0	0.0	0.8	3.1	0.5	0.4	1.0	1.5	-0.6	-1.3
3M Sales Up-Down Revs	0.3	0.4	1.2	13.8	0.3	0.3	0.5	3.5	0.5	0.3	0.5	0.1	-0.4	-0.8
Earnings Momentum	0.0	0.4	2.1	13.2	0.1	0.1	0.7	2.8	0.5	0.5	1.8	0.9	-0.1	-0.4
Composite Momentum														
Composite Momentum	-0.0	0.2	1.7	16.7	0.6	0.4	0.8	3.2	0.7	0.7	1.2	0.8	0.1	-1.6
Quality														
Earnings Quality														
Dividend Payouts	0.1	-0.3	-0.7	-3.9	0.5	0.7	0.5	-0.1	0.4	0.2	-1.1	-0.0	-0.5	0.0
Short Term Accruals*	0.5	0.5	0.3	-4.3	0.9	0.9	0.1	-1.2	0.5	0.2	0.5	0.9	-0.9	0.4
Accruals*	0.4	0.6	2.0	-2.3	0.5	0.7	0.3	0.9	0.3	0.1	0.6	0.6	-0.2	0.8
Tax-to-Pretax Income	0.0	-0.5	-2.2	2.0	0.1	0.2	0.1	-1.2	0.6	0.1	-0.6	-0.9	-0.3	0.7
Earnings Volatility*	0.0	-0.5	-2.5	1.2	0.2	0.2	0.6	-1.1	0.6	0.2	-1.5	-0.5	-0.8	-4.4
Low Leverage														
Net Debt-to-Equity*	0.1	-0.3	-1.5	1.4	0.2	0.3	0.9	-1.3	0.6	0.0	-1.7	0.3	0.2	-2.1
Chg LT Debt-to-Assets*	-0.1	-0.1	1.7	7.8	0.0	0.0	1.0	2.0	0.3	0.2	-0.5	0.4	-0.4	-1.2
Management Quality														
Net Buyback Yield	0.1	0.5	0.2	-5.2	1.0	1.0	0.1	0.7	0.4	0.2	-1.0	0.8	-0.9	0.1
YoY Asset Growth*	0.1	0.3	2.4	1.1	0.4	0.7	0.5	1.4	0.6	0.1	0.2	0.6	0.6	1.0
Change in Headcount*	0.0	0.4	1.8	6.1	0.9	0.9	0.5	2.3	0.4	0.1	0.6	-0.8	-0.9	0.8
Operating Quality														
Asset Turnover	0.1	0.4	0.3	-2.1	0.0	0.0	0.7	-0.5	0.6	0.1	1.4	1.1	-0.3	-0.3
Inventory-to-Sales*	-0.1	-0.4	-1.9	-3.4	0.5	0.8	0.4	-2.0	0.8	0.4	-1.4	1.2	-1.0	-1.4
Capex-to-Sales*	0.1	0.4	-0.6	-3.0	0.1	0.1	1.0	-2.5	0.9	0.1	1.5	0.6	-0.1	0.4
Capex-to-Depreciation*	0.3	0.5	-1.3	-1.8	0.6	0.5	0.1	-1.9	0.4	0.0	0.2	0.8	0.0	-0.7
R&D-to-Market Cap	0.5	0.2	1.2	11.4	0.9	0.8	0.1	1.9	1.0	0.2	0.7	-0.5	1.0	1.2
Profitability														
Return on Equity	-0.1	0.3	0.6	8.7	0.0	0.1	0.9	1.4	0.5	0.1	-0.6	0.4	0.2	-1.3
Return on Invst Capital	-0.0	0.1	-1.0	5.4	0.1	0.1	1.0	-1.2	0.6	0.0	-1.3	0.2	-0.2	-1.4
Net Margin	0.0	0.1	-0.3	3.4	0.1	0.1	0.8	-0.8	0.6	0.0	-1.5	0.4	-0.0	-1.6
EBITDA Margin	-0.0	-0.1	-0.2	17.4	0.1	0.1	0.7	0.0	0.8	0.0	-2.0	0.5	-0.2	-1.7
Gross Profit-to-Assets	0.0	-0.3	-1.8	0.7	0.0	0.1	0.8	-1.4	0.8	0.0	-0.7	1.0	-0.7	-1.9
Return on Assets	-0.0	-0.0	-1.0	5.2	0.1	0.1	0.9	-1.1	0.5	0.1	-1.3	0.5	-0.4	-1.6
Composite Quality														
Composite Quality	0.1	-0.3	-1.4	11.3	0.3	0.3	0.8	-0.9	0.7	0.0	-2.2	0.3	-0.5	-1.9
Risk														
Low Risk														
Low Vol (6M Ret Vol)*	-0.2	-0.4	-2.0	-2.5	1.0	1.0	0.0	-2.2	0.6	0.7	-0.3	-0.3	-0.9	-1.8
Size														
Small Size														
Size (Free Float)*	-0.1	-0.7	-2.1	-4.9	0.9	0.9	0.2	-2.4	0.5	0.6	0.3	-1.0	-0.1	0.9
Size*	-0.0	-0.7	-1.9	-4.9	0.9	0.9	0.1	-2.7	0.5	0.5	0.4	-0.9	-0.1	0.9

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI Japan, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Factor Rankings: World

Cluster Rankings

Exhibit 65: World – Cluster Rankings

	EFFICACY				VALUATION		SENTIMENT		EXPOSURE		MACRO				RANK
	LT	10Y	1Y	1M	BP	BP+EPF	ERR	PRICE	INDU	BETA	ECO	ERP	INF	YIELD	
Value															
Current Value	1	7	6	17	2	3	2	8	14	13	11	14	11	13	4
Deep Value	8	13	4	15	6	4	6	11	12	10	14	16	5	14	12
Defensive Value	3	10	9	12	5	4	5	14	11	16	17	12	16	15	15
Composite Value	2	6	2	16	7	8	4	12	17	14	15	15	7	16	16
Growth															
Growth	15	8	8	4	16	14	15	10	4	11	4	11	3	4	14
Composite Growth	14	5	11	3	15	15	16	4	6	15	1	3	1	1	10
Momentum															
Momentum	12	4	3	5	3	6	11	15	13	8	3	4	12	6	3
Revisions	5	1	5	2	17	17	13	16	3	4	5	5	9	7	8
Composite Momentum	7	2	1	1	9	10	14	17	8	12	2	2	10	2	5
Quality															
Earnings Quality	9	14	13	13	8	7	9	6	5	7	12	8	13	8	7
Low Leverage	13	15	10	7	12	12	10	7	7	6	7	9	8	12	13
Management Quality	11	11	7	10	4	2	7	13	1	9	10	13	17	11	1
Operating Quality	10	9	15	8	11	11	8	3	16	5	6	10	4	3	6
Profitability	6	3	14	9	14	16	12	2	9	3	9	7	6	5	9
Composite Quality	4	12	12	6	13	13	16	8	10	2	16	6	14	9	17
Risk															
Low Risk	16	16	17	14	1	1	3	5	15	17	8	1	15	10	10
Size															
Small Size	17	17	16	11	10	9	1	1	2	1	13	17	2	17	2

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI World, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Factor Rankings

Exhibit 66: World – Factor Rankings

	EFFICACY				VALUATION		SENTIMENT		EXPOSURE		MACRO				RANK
Value	LT	10Y	1Y	1M	BP	BP+EPF	ERR	PRICE	INDU	BETA	ECO	ERP	INF	YIELD	
Current Value															
12M Fwd Earnings Yield	12	11	6	47	15	24	10	33	50	29	30	48	9	40	16
Free Cash Flow Yield	1	14	42	58	3	4	5	2	37	39	33	36	39	33	4
Operating CF Yield	8	43	25	54	14	12	2	29	47	56	51	47	47	51	38
EBITDA-to-EV	9	45	21	55	17	20	1	28	42	55	50	50	34	54	41
Deep Value															
Book-to-Price	44	47	11	42	26	19	21	37	36	23	57	56	23	57	43
Sales-to-Price	34	46	20	52	19	22	5	31	43	44	38	43	29	55	39
Trailing Earnings Yield	10	25	12	53	4	5	18	27	44	41	41	44	18	45	15
Shiller P/E*	26	35	22	46	22	18	28	39	28	49	52	55	22	42	48
Defensive Value															
Dividend Yield	20	44	31	41	23	21	20	42	41	54	58	42	52	50	57
Total Yield	3	21	33	51	10	11	8	32	29	48	53	33	53	52	28
Composite Value															
Composite Value	11	28	10	50	20	23	13	34	55	45	48	49	25	53	47
Growth															
Growth															
Long Term Growth	50	32	24	8	50	44	58	35	16	53	2	41	2	23	44
YoY Earnings Growth	48	37	32	23	34	40	26	21	1	24	14	22	4	5	8
YoY Sales Growth	57	48	43	17	57	56	42	15	22	40	5	38	1	17	51
P/E-to-Growth*	16	20	1	13	28	25	56	47	15	37	42	58	10	44	34
ST EPS Growth Rate	54	24	29	6	54	48	56	40	27	47	1	28	21	11	52
Internal Growth	24	16	40	19	56	48	54	12	21	38	34	16	54	13	45
Composite Growth															
Composite Growth	39	26	36	14	46	45	50	18	24	46	6	10	5	1	31
Momentum															
Momentum															
1M Momentum Reversal*	42	40	52	26	24	13	14	20	33	57	56	57	8	41	37
3M Momentum	51	38	7	28	29	31	55	46	35	13	4	14	45	32	33
6M Momentum	47	27	4	18	12	14	31	49	52	4	9	5	42	19	7
12M Momentum	40	30	3	7	13	26	38	55	39	51	3	3	37	7	29
6x1M Momentum	37	18	8	16	5	7	26	50	50	35	10	11	41	24	10
12x1M Momentum	29	22	5	5	7	15	30	56	38	43	8	4	33	9	11
12x1M Residual Mom	4	2	2	15	2	7	28	52	40	3	19	19	30	21	2
Revisions															
Earnings Revisions	17	8	17	11	51	49	34	48	18	12	13	2	44	22	17
EPS Up-Down Revs	5	3	16	24	56	55	36	45	5	21	22	12	26	30	22
Sales Revisions	41	15	28	9	49	54	35	53	26	22	7	26	38	26	42
Sales Up-Down Revs	13	10	23	12	37	41	32	41	3	9	27	23	11	14	6
3M EPS Up-Down Revs	14	1	13	4	43	43	48	51	7	7	12	6	27	27	9
3M Sales Up-Down Revs	25	9	27	2	48	58	41	57	14	28	29	32	32	20	54
Earnings Momentum	36	7	15	3	42	46	51	54	17	33	18	13	28	16	36
Composite Momentum															
Composite Momentum	27	13	9	1	24	29	46	58	30	42	11	7	31	8	21
Quality															
Earnings Quality															
Dividend Payouts	46	53	37	39	8	7	17	38	31	52	54	20	58	38	46
Short Term Accruals*	2	31	45	57	11	2	14	1	11	25	28	39	19	10	1
Accruals*	6	19	30	49	16	10	8	24	9	19	40	37	55	28	3
Tax-to-Pretax Income	56	54	39	37	42	34	44	26	23	8	39	35	46	49	53
Earnings Volatility*	52	55	58	56	33	36	52	11	19	32	46	8	36	12	56
Low Leverage															
Net Debt-to-Equity*	43	50	47	22	47	42	45	14	53	31	25	18	16	39	55
Chg LT Debt-to-Assets*	31	49	18	27	32	33	12	30	2	20	26	40	40	48	13
Management Quality															
Net Buyback Yield	19	4	35	43	6	3	24	22	12	26	37	25	51	47	5
YoY Asset Growth*	45	52	26	35	18	16	11	43	6	30	44	45	57	46	30
Change in Headcount*	38	42	14	34	22	17	37	44	4	34	35	46	56	37	32
Operating Quality															
Asset Turnover	22	23	44	25	58	57	43	17	54	16	17	17	6	31	49
Inventory-to-Sales*	49	57	56	45	1	1	16	13	57	50	21	29	43	3	23
Capex-to-Sales*	30	36	55	32	39	37	29	6	58	18	16	24	3	6	24
Capex-to-Depreciation*	28	39	57	33	38	35	18	9	45	14	20	34	20	2	18
R&D-to-Market Cap	35	6	19	10	35	30	22	25	56	27	31	53	7	43	25
Profitability															
Return on Equity	21	17	49	36	53	53	40	4	20	15	24	15	14	15	20
Return on Invest Capital	7	5	41	21	52	52	48	16	32	11	23	30	15	18	27
Net Margin	32	29	48	31	36	50	53	7	25	6	47	27	35	25	50
EBITDA Margin	33	34	34	38	27	32	23	36	51	36	55	52	49	35	58
Gross Profit-to-Assets	23	41	54	30	44	39	33	8	48	17	15	9	12	4	19
Return on Assets	18	12	46	29	44	51	39	10	13	10	36	31	24	29	26
Composite Quality															
Composite Quality	15	33	38	20	40	38	50	23	34	5	49	21	48	34	40
Risk															
Low Risk															
Low Vol (6M Ret Vol)*	53	51	53	48	8	9	7	19	46	58	32	1	50	36	35
Size															
Small Size															
Size (Free Float)*	58	58	51	44	30	27	5	3	10	1	43	51	17	58	14
Size*	55	56	50	40	31	28	2	5	8	2	45	54	13	56	12

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI World, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Factor Values

Exhibit 67: World Factor Values

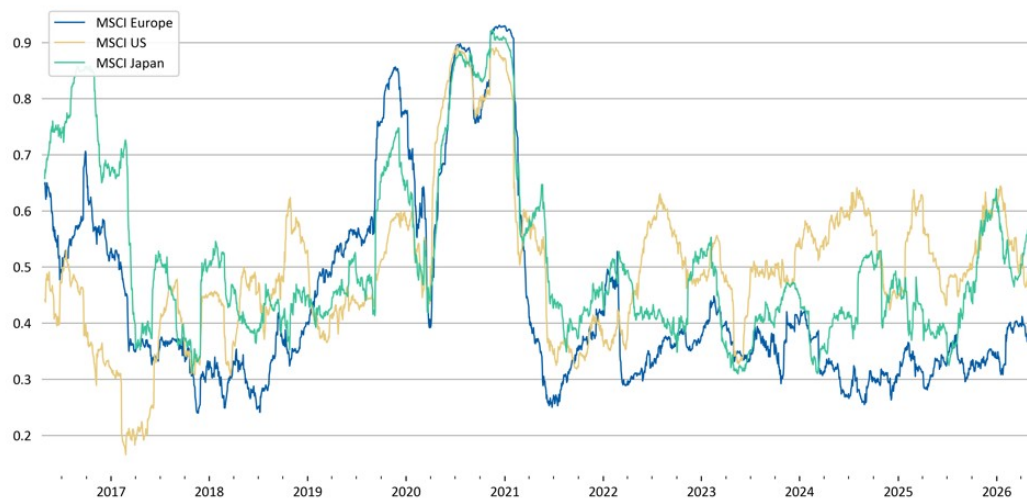
	EFFICACY				VALUATION		SENTIMENT		EXPOSURE		MACRO			
	LT	10Y	1Y	1M	BP	BP+EPF	ERR	PRICE	INDU	BETA	ECO	ERP	INF	YIELD
	SR	SR	SR	SR	Pct	Pct	Pct	Z	%	Spread	T-Stat	T-Stat	T-Stat	T-Stat
Value														
Current Value														
12M Fwd Earnings Yield	0.6	0.5	2.0	-3.4	0.9	0.8	0.1	1.1	0.6	0.2	-0.1	-0.6	0.6	-0.3
Free Cash Flow Yield	1.0	0.4	-0.4	-6.2	1.0	0.9	0.0	-2.6	0.5	0.3	-0.3	0.3	-0.7	-0.0
Operating CF Yield	0.7	0.1	0.9	-4.5	0.9	0.9	0.0	0.8	0.5	0.6	-1.1	-0.5	-1.0	-0.9
EBITDA-to-EV	0.7	0.0	1.1	-4.8	0.9	0.8	0.0	0.8	0.5	0.5	-1.0	-0.6	-0.5	-1.1
Deep Value														
Book-to-Price	0.2	-0.0	1.6	-2.5	0.8	0.8	0.2	1.2	0.5	0.1	-1.4	-1.1	-0.1	-1.2
Sales-to-Price	0.3	0.0	1.1	-4.2	0.9	0.8	0.0	1.0	0.5	0.3	-0.4	-0.1	-0.4	-1.2
Trailing Earnings Yield	0.6	0.3	1.5	-4.4	1.0	0.9	0.2	0.8	0.5	0.3	-0.5	-0.1	0.3	-0.5
Shiller P/E*	0.4	0.2	1.1	-3.3	0.8	0.8	0.5	1.3	0.4	0.4	-1.1	-1.0	0.1	-0.4
Defensive Value														
Dividend Yield	0.5	0.1	0.6	-2.5	0.8	0.8	0.2	1.5	0.5	0.5	-1.5	-0.1	-1.4	-0.7
Total Yield	0.8	0.4	0.5	-3.9	0.9	0.9	0.1	1.0	0.4	0.4	-1.1	0.4	-1.5	-1.0
Composite Value														
Composite Value	0.6	0.3	1.7	-3.8	0.8	0.8	0.1	1.1	0.6	0.3	-1.0	-0.6	-0.2	-1.1
Growth														
Growth														
Long Term Growth	0.0	0.2	0.9	10.5	0.1	0.1	1.0	1.1	0.3	0.5	1.7	0.0	1.8	0.3
YoY Earnings Growth	0.1	0.1	0.6	2.5	0.4	0.2	0.4	-0.6	0.2	0.1	1.0	0.8	1.1	0.8
YoY Sales Growth	-0.1	-0.0	-0.4	3.4	0.0	0.1	0.8	-1.4	0.3	0.3	1.4	0.2	2.1	0.4
P/E-to-Growth*	0.6	0.4	3.5	4.9	0.8	0.8	1.0	1.7	0.3	0.2	-0.6	-1.4	0.6	-0.4
ST EPS Growth Rate	-0.1	0.3	0.7	11.0	0.1	0.1	1.0	1.4	0.4	0.4	1.9	0.6	0.1	0.6
Internal Growth	0.4	0.4	-0.3	3.3	0.1	0.1	0.9	-1.6	0.3	0.3	-0.3	1.1	-1.9	0.6
Composite Growth														
Composite Growth	0.2	0.3	0.1	4.9	0.1	0.1	0.9	-1.2	0.4	0.4	1.3	1.2	0.9	1.2
Momentum														
Momentum														
1M Momentum Reversal*	0.2	0.1	-1.4	1.6	0.8	0.9	0.1	-1.2	0.5	0.6	-1.3	-1.2	0.6	-0.3
3M Momentum	0.0	0.1	1.7	1.3	0.7	0.5	0.9	1.7	0.5	0.1	1.4	1.1	-0.9	0.0
6M Momentum	0.1	0.3	2.2	3.3	0.9	0.9	0.6	1.9	0.6	0.0	1.3	1.4	-0.7	0.4
12M Momentum	0.2	0.3	2.3	10.8	0.9	0.8	0.7	2.4	0.5	0.4	1.5	1.5	-0.6	0.7
6x1M Momentum	0.2	0.4	1.7	3.6	1.0	0.9	0.4	2.1	0.6	0.2	1.2	1.2	-0.7	0.3
12x1M Momentum	0.3	0.4	2.1	11.2	1.0	0.9	0.6	2.6	0.5	0.3	1.3	1.4	-0.5	0.6
12x1M Residual Mom	0.8	0.8	2.4	4.5	1.0	0.9	0.5	2.2	0.5	0.0	0.8	1.0	-0.4	0.4
Revisions														
Earnings Revisions	0.6	0.5	1.2	6.8	0.1	0.1	0.7	1.8	0.3	0.1	1.0	1.5	-0.8	0.4
EPS Up-Down Revs	0.8	0.7	1.2	2.3	0.1	0.1	0.7	1.6	0.2	0.1	0.4	1.2	-0.2	0.1
Sales Revisions	0.2	0.4	0.7	8.2	0.1	0.1	0.7	2.3	0.4	0.1	1.3	0.8	-0.6	0.2
Sales Up-Down Revs	0.6	0.5	1.0	5.7	0.3	0.2	0.6	1.5	0.2	0.0	0.1	0.8	0.5	0.5
3M EPS Up-Down Revs	0.6	0.8	1.5	12.3	0.2	0.2	0.9	2.1	0.2	0.0	1.1	1.4	-0.4	0.2
3M Sales Up-Down Revs	0.4	0.5	0.7	14.5	0.1	0.0	0.8	2.7	0.3	0.2	0.0	0.4	-0.5	0.4
Earnings Momentum	0.2	0.6	1.3	13.0	0.2	0.1	0.9	2.4	0.3	0.2	0.8	1.2	-0.4	0.5
Composite Momentum														
Composite Momentum	0.4	0.4	1.7	14.7	0.8	0.6	0.8	2.9	0.4	0.3	1.1	1.3	-0.5	0.7
Quality														
Earnings Quality														
Dividend Payouts	0.1	-0.2	0.1	-2.3	0.9	0.9	0.2	1.3	0.4	0.5	-1.1	1.0	-2.7	-0.3
Short Term Accruals*	0.8	0.2	-0.5	-6.0	0.9	1.0	0.1	-3.2	0.3	0.1	0.1	0.2	0.2	0.6
Accruals*	0.7	0.4	0.6	-3.6	0.9	0.9	0.1	0.3	0.2	0.1	-0.5	0.3	-2.1	0.2
Tax-to-Pretax Income	-0.1	-0.2	-0.2	-2.0	0.2	0.4	0.8	0.7	0.4	0.0	-0.5	0.3	-1.0	-0.6
Earnings Volatility*	0.0	-0.4	-3.6	-4.9	0.4	0.3	0.9	-1.6	0.3	0.2	-0.8	1.3	-0.6	0.6
Low Leverage														
Net Debt-to-Equity*	0.2	-0.1	-0.9	2.6	0.1	0.2	0.8	-1.4	0.6	0.2	0.2	1.0	0.4	-0.3
Chg LT Debt-to-Assets*	0.3	-0.1	1.2	1.3	0.5	0.5	0.1	1.0	0.2	0.1	0.1	0.1	-0.7	-0.6
Management Quality														
Net Buyback Yield	0.5	0.7	0.1	-2.6	1.0	1.0	0.4	-0.6	0.3	0.1	-0.4	0.8	-1.2	-0.5
YoY Asset Growth*	0.1	-0.2	0.8	-1.7	0.9	0.9	0.1	1.5	0.2	0.2	-0.8	-0.2	-2.4	-0.5
Change in Headcount*	0.2	0.1	1.3	-1.4	0.8	0.8	0.7	1.5	0.2	0.2	-0.3	-0.3	-2.1	-0.1
Operating Quality														
Asset Turnover	0.5	0.3	-0.4	2.2	0.0	0.0	0.8	-1.2	0.6	0.1	0.8	1.1	0.7	0.0
Inventory-to-Sales*	0.0	-0.6	-2.2	-3.1	1.0	1.0	0.1	-1.5	0.7	0.4	0.7	0.6	-0.8	1.0
Capex-to-Sales*	0.3	0.1	-2.1	-0.5	0.3	0.3	0.6	-1.9	0.8	0.1	0.8	0.8	1.2	0.8
Capex-to-Depreciation*	0.4	0.1	-2.3	-0.9	0.3	0.4	0.2	-1.8	0.5	0.1	0.7	0.3	0.2	1.1
R&D-to-Market Cap	0.2	0.6	1.2	6.9	0.4	0.6	0.2	0.6	0.6	0.2	-0.3	-0.9	0.7	-0.4
Profitability														
Return on Equity	0.5	0.4	-1.0	-1.8	0.1	0.1	0.7	-2.1	0.3	0.1	0.3	1.1	0.5	0.5
Return on Invst Capital	0.7	0.7	-0.3	2.8	0.1	0.1	0.9	-1.3	0.4	0.1	0.4	0.5	0.4	0.4
Net Margin	0.3	0.3	-0.9	-0.4	0.3	0.1	0.9	-1.9	0.4	0.0	-0.9	0.7	-0.5	0.3
EBITDA Margin	0.3	0.2	0.3	-2.2	0.8	0.5	0.3	1.2	0.6	0.2	-1.2	-0.8	-1.1	-0.1
Gross Profit-to-Assets	0.4	0.1	-2.0	0.1	0.2	0.3	0.7	-1.9	0.6	0.1	0.9	1.3	0.5	1.0
Return on Assets	0.5	0.4	-0.8	0.1	0.2	0.1	0.7	-1.7	0.3	0.0	-0.4	0.4	-0.2	0.1
Composite Quality														
Composite Quality	0.6	0.2	-0.1	2.9	0.2	0.3	0.9	0.2	0.5	0.0	-1.0	0.8	-1.1	-0.1
Risk														
Low Risk														
Low Vol (6M Ret Vol)*	-0.0	-0.2	-1.6	-3.4	0.9	0.9	0.0	-1.2	0.5	0.8	-0.3	1.8	-1.2	-0.1
Size														
Small Size														
Size (Free Float)*	-0.2	-0.6	-1.2	-2.9	0.7	0.8	0.0	-2.4	0.3	0.0	-0.7	-0.8	0.4	-1.3
Size*	-0.1	-0.5	-1.0	-2.4	0.7	0.7	0.0	-2.0	0.2	0.0	-0.8	-1.0	0.5	-1.2

Source: FactSet, Morgan Stanley Research. Rankings are based on our 4-Dimensions Framework, which organises 58 investment factors into 17 clusters and further into 6 groups. The factor performances are defined as the long/short returns of the top versus bottom quintile of the MSCI World, the portfolios are rebalanced monthly and returns are equal weighted. Please refer to [Summary of the 4-Dimensions Factor Framework](#) for the methodology.

Factor Correlations

Pairwise Correlation by Region

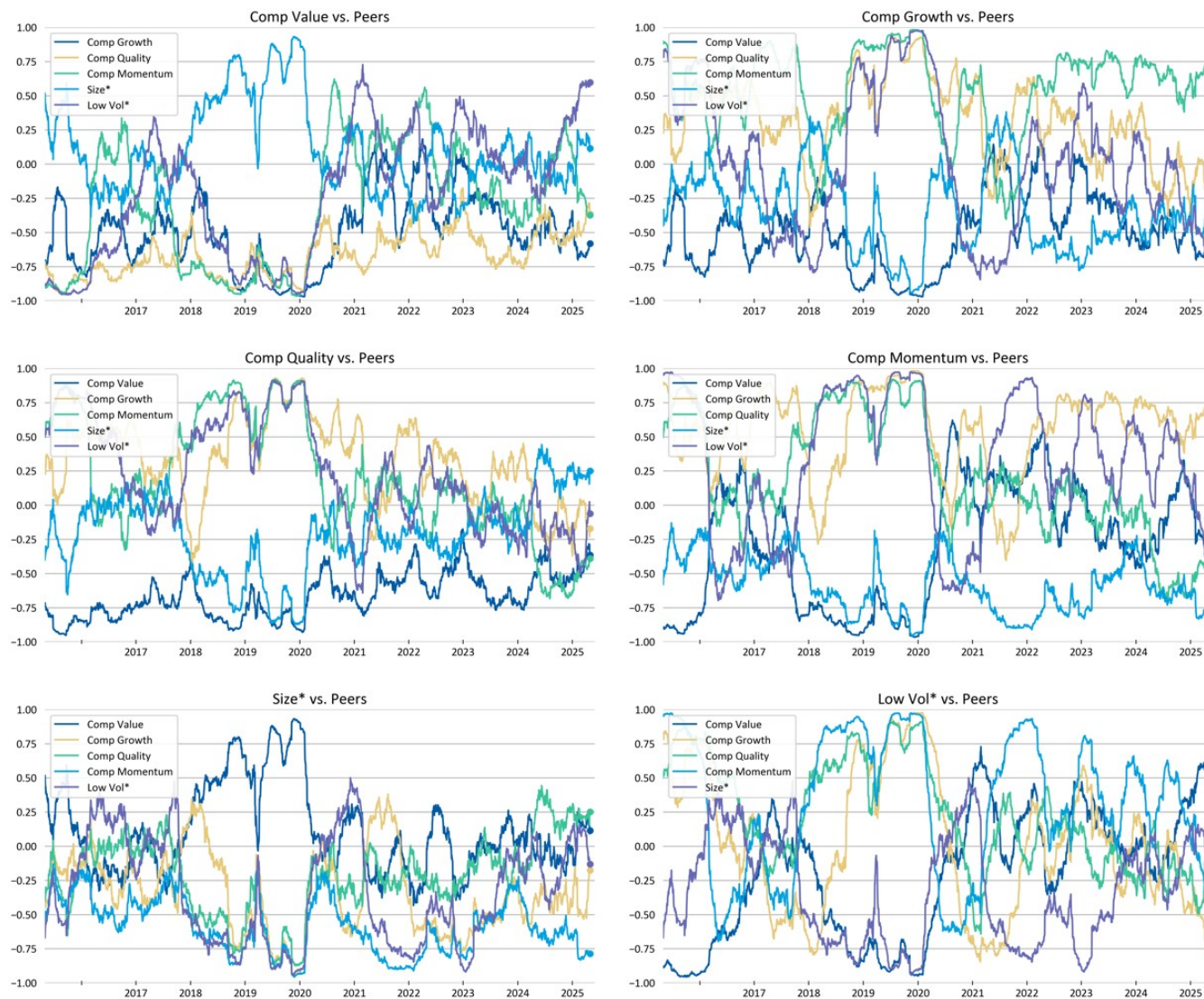
Exhibit 68: Average pairwise correlation by region (average absolute value)



Source: FactSet, Morgan Stanley Research. Average absolute value of pairwise correlation of the daily returns of Composite Value, Composite Growth, Composite Momentum, Composite Quality, Low Vol and Size.

Factor Correlations: Europe

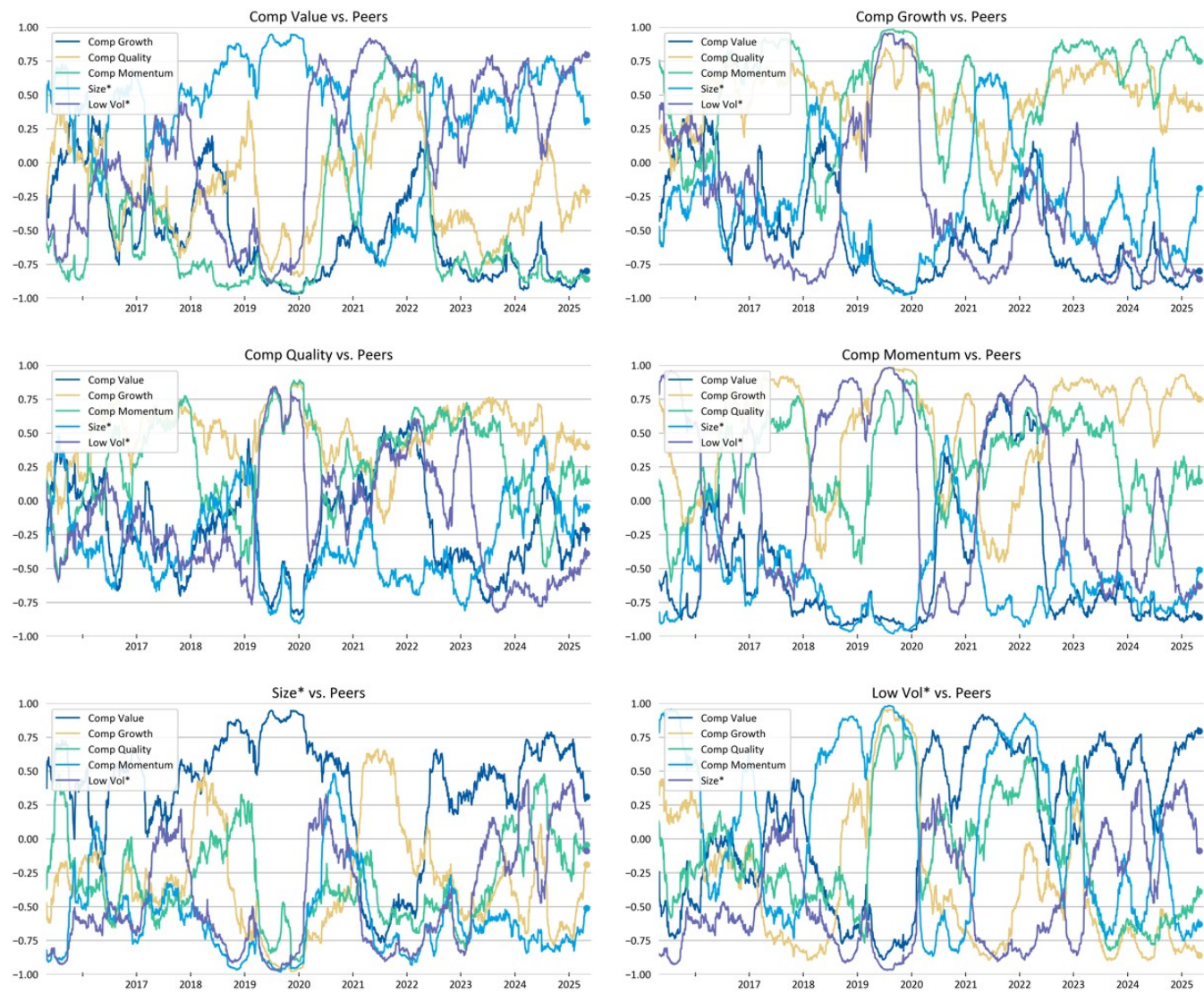
Exhibit 69: MSCI Europe Factor Correlation



Source: FactSet, Morgan Stanley Research. 3M rolling correlation of composite factors.

Factor Correlations: US

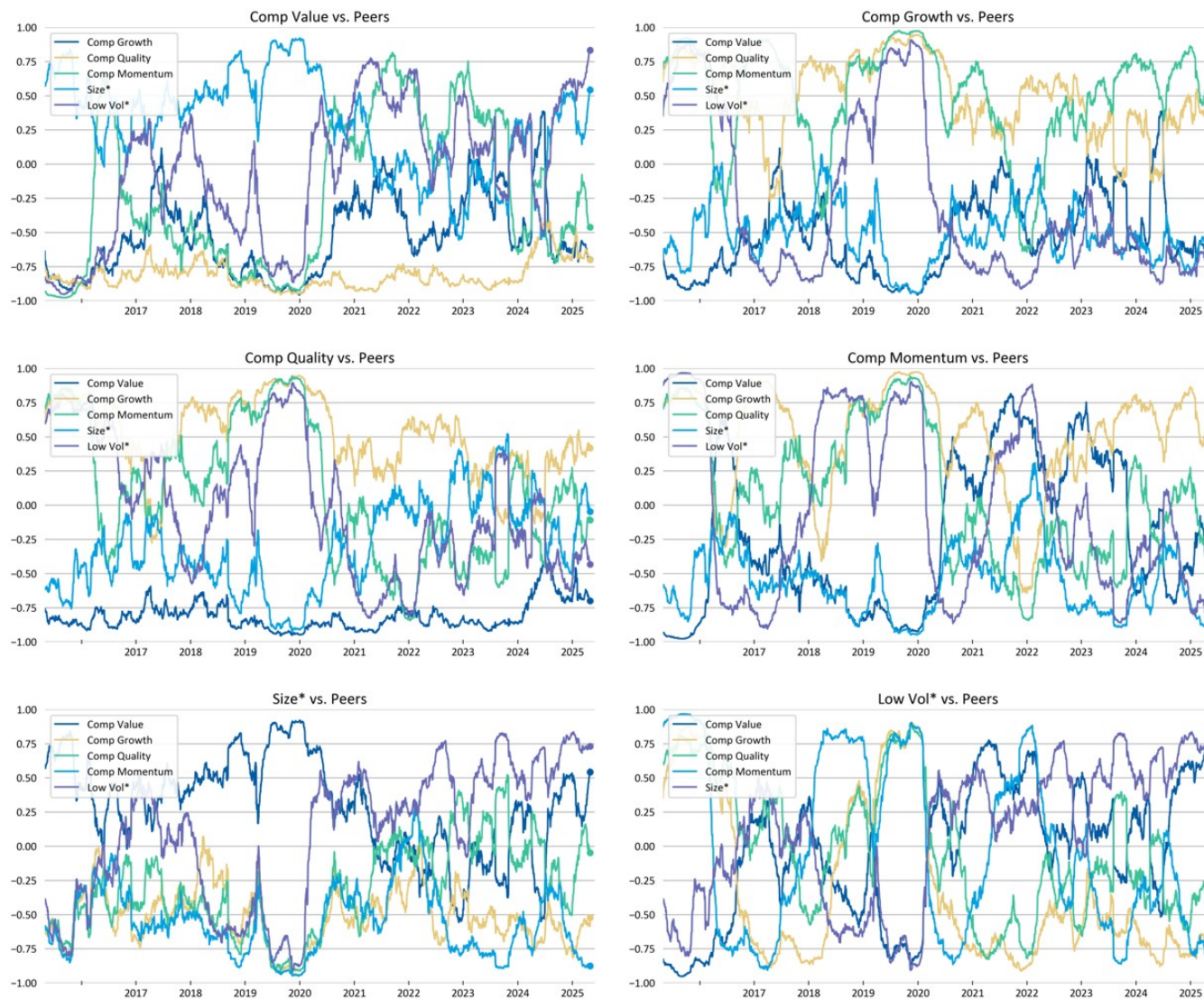
Exhibit 70: MSCI US Factor Correlation



Source: FactSet, Morgan Stanley Research. 3M rolling correlation of composite factors.

Factor Correlations: Japan

Exhibit 71: MSCI Japan Factor Correlation



Source: FactSet, Morgan Stanley Research. 3M rolling correlation of composite factors.

Methodology & Definitions

4-Dimensions Framework

I. Efficacy

We calculate the Sharpe ratios of a long-short quintile portfolio over four horizons, namely long-term (i.e. since 1990) (**LT**), past 10 years (**10Y**), past 1 year (**1Y**), and past 1 month (**1M**). We prefer factors with higher Sharpe ratios.

II. Valuation

We take the median of the price-to-book ratio of all equities in the factor long holdings divided by the median price-to-book ratio of the equities in the factor short holdings, and we calculate the same ratio based on forward P/E. The ratios of the book-to-price (**BP**) ratio and the forward earnings yield (**EPF**) are then averaged. We determine the valuation level for each factor based on its time series percentiles since 1990. We prefer factors with cheap valuation percentile.

III. Sentiment

Earnings Revision Ratio (**ERR**): The signal is defined as the 3-month average of the number of upward EPS revisions minus the number of downward EPS revisions, divided by the total number of analyst estimates for the upcoming fiscal year. The ratio is averaged for the last three months. Then we take the spread of the median ERR of the factor top quintile minus the factor bottom quintile. We determine a net ERR of a factor in terms of percentiles over the factor's own history since 1990.

Price Z-Score (**PRICE**): The signal is defined as the one-year rolling z-score of the daily factor performance. The z-score is a contrarian signal at its extremes, it detects when a factor is overbought or oversold relative to history.

IV. Macro and Exposure

Exposure: We would prefer factors with lower sensitivity to country, industry and market beta. We quantify the country (**CTRY**) and industry (**INDU**) exposures by the sum of the top four net overweight or underweight countries and industries in absolute value. We calculate the net beta (**BETA**) as the absolute spread between the top and bottom factor quintile beta.

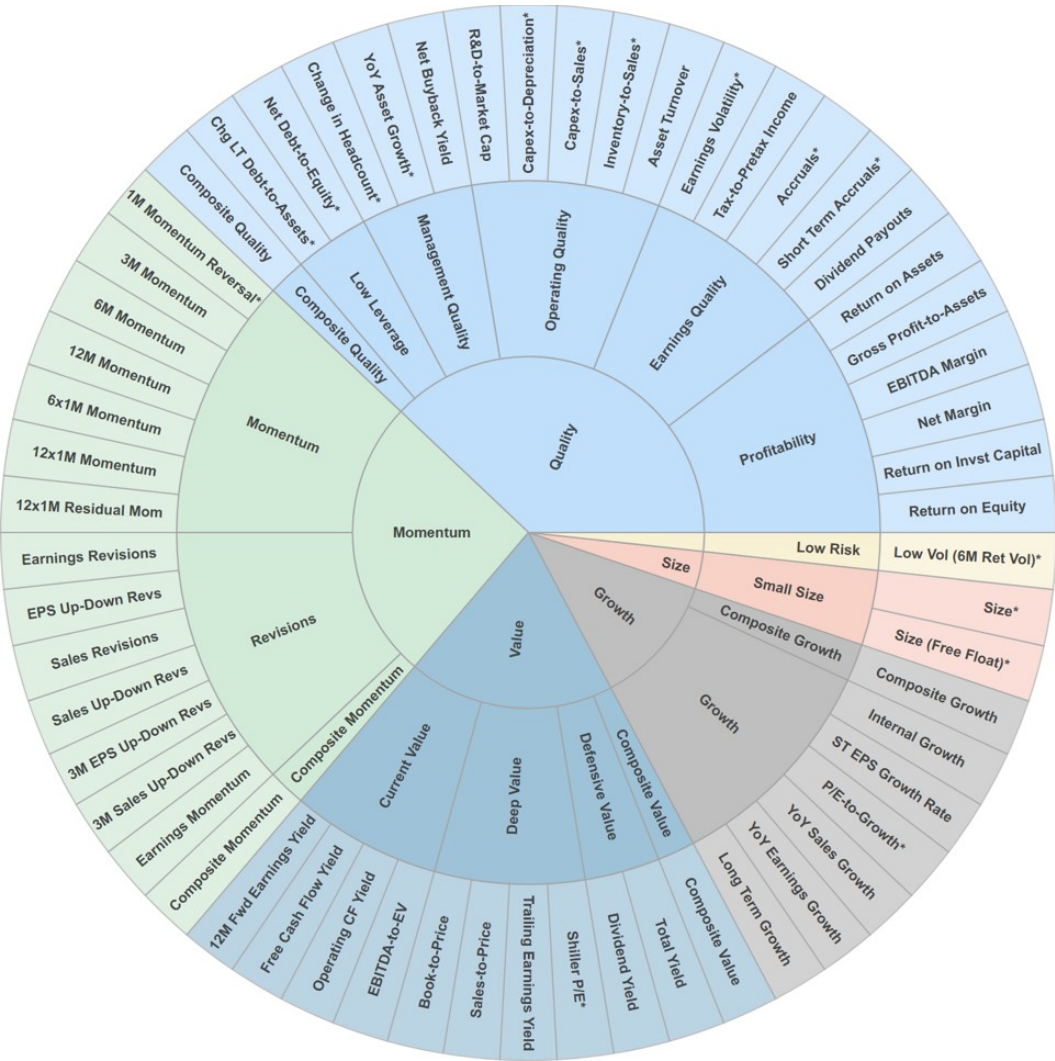
Macro: We focus on four macro indicators, namely economic cycles (**ECO**), equity risk premium (**ERP**), inflation (**INF**), and long-term interest rate (**YIELD**). For each macro indicator, we calculate the average factor return in a certain regime against the average return over the entire history. Then we perform a Welch's T-test and calculate the t-statistics to evaluate the significance of the relative performance.

Combined Ranking

Within Efficacy, Valuation, Sentiment and Exposure, each component inside carries equal weight. Within the dimension of Macro, economic cycles carry the same weight as the combination of equity risk premium, inflation and long-term interest rate. To consolidate, we determine the combined ranking by assigning 20% to each of the Efficacy, Valuation, Sentiment, Macro, and Exposures dimensions.

Factor Clusters

Exhibit 72: Classification of Factors into 6 Groups and 17 Clusters



Source: Morgan Stanley Research. This figure presents the 58 investment factors that form the foundation of our 4-Dimensions Framework. These factors are organised into 17 clusters, which are further categorised into six groups.

Factor Definitions

Exhibit 73: Factor Definitions

Group > Cluster > Factor	Definition
Value	
Current Value	
12M Fwd Earnings Yield	NTM earnings divided by current market cap
Free Cash Flow Yield	Trailing 12M free cash flow (typically operating cash flow minus capex) divided by market cap
Operating CF Yield	Trailing 12M operating cash flow divided by market cap
EBITDA-to-EV	Trailing 12M EBITDA divided by enterprise value (market cap + total debt + minority interest + preferred stock - cash and equivalents)
Deep Value	
Book-to-Price	Last reported book value of common equity divided by current market cap
Sales-to-Price	Trailing 12M total revenue divided by current market cap
Trailing Earnings Yield	Trailing 12M earnings divided by current market cap
Shiller P/E*	Current market cap divided by CAPE (avg inflation adjusted net income over the last 7 years)
Defensive Value	
Dividend Yield	Trailing 12M DPS divided by current price
Total Yield	Dividend yield plus the net buyback yield
Composite Value	
Composite Value	Composite score based on EW avg of 12M Fwd P/E, Price-to-Book, and Total Yield (Dividend Yield + Net Buyback Yield)
Growth	
Growth	
Long Term Growth	Consensus analyst estimate of LT (5Y) EPS growth
YoY Earnings Growth	Growth rate in last reported net income compared to a year earlier
YoY Sales Growth	Growth rate in last reported total revenue compared to a year earlier
P/E-to-Growth*	12M Forward Price to Earnings divided by LT Growth Estimate
ST EPS Growth Rate	Percentage change of the FY2 vs FY1 IBES EPS estimates
Internal Growth	Retention rate times ROE
Composite Growth	
Composite Growth	Composite score from EW avg of LT EPS Growth, YoY Earnings Growth and Internal Growth
Momentum	
Momentum	
1M Momentum Reversal*	Trailing 1M return
3M Momentum	Trailing 3M returns
6M Momentum	Trailing 6M returns
12M Momentum	Trailing 12M returns
6x1M Momentum	Price momentum using the last 6 months except the most recent month
12x1M Momentum	Price momentum using the last 12 months except the most recent month
12x1M Residual Mom	Residual from regression of price momentum against beta, size and value
Revisions	
Earnings Revisions	3M recency-weighted change in NTM EPS forecasts, scaled by book value per share (Δ EPS/BPS)
EPS Up-Down Revs	Net count of upward minus downward EPS revisions, divided by the total analyst estimates for NTM
Sales Revisions	3M smoothed NTM revenue revisions, scaled by book value
Sales Up-Down Revs	Net count of upward minus downward revenue revisions, divided by the total analyst estimates for NTM
3M EPS Up-Down Revs	3M avg of net count of upward minus downward EPS revisions, divided by total analyst estimates for NTM
3M Sales Up-Down Revs	3M avg of net count of upward minus downward revenue revisions, divided by total analyst estimates for NTM
Earnings Momentum	3M percentage change of the 3ma NTM Earnings estimates
Composite Momentum	
Composite Momentum	Composite score from EW avg of 12-1M Price Mom and Earnings Mom (3M pct change of the 3ma NTM earnings estimates)
Quality	
Earnings Quality	
Dividend Payouts	Trailing 12M DPS divided by trailing 12M EPS
Short Term Accruals*	YoY change in net current assets - excluding net cash - divided by last reported total assets
Accruals*	Change in net operating assets (total assets less cash/equivalents minus total liabilities less debt) from T-1, divided by total assets
Tax-to-Pretax Income	Trailing 12M taxes paid to pretax income
Earnings Volatility*	5Y SD of earnings growth YoY The growth rate in last reported net income compared to a year earlier
Low Leverage	
Net Debt-to-Equity*	Net debt to common equity
Chg LT Debt-to-Assets*	YoY change in last reported LT debt divided by last reported total assets
Management Quality	
Net Buyback Yield	12M trailing net stock repurchases (extracted from the statement of cash flows) divided by current market value
YoY Asset Growth*	YoY asset growth
Change in Headcount*	YoY change in number of employees
Operating Quality	
Asset Turnover	12M trailing revenue divided by last reported total assets
Inventory-to-Sales*	YoY change in the ratio of last reported total inventory divided by 12M trailing revenue
Capex-to-Sales*	Trailing 12M net capital expenditure divided by trailing 12M total revenue
Capex-to-Depreciation*	Capex to Depreciation: Trailing 12M net capital expenditure divided by trailing 12M depreciation expense
R&D-to-Market Cap	Rolling 3Y R&D expenditure divided by current market cap
Profitability	
Return on Equity	Trailing 12M net income divided by current book value
Return on Invest Capital	Operating income divided by Total Invested Capital (sum of Total Shareholder's Equity and Net Debt)
Net Margin	Last reported net income divided by last reported revenue
EBITDA Margin	Trailing 12M EBITDA divided by revenue
Gross Profit-to-Assets	Trailing 4 quarter sales net of COGS divided by total assets as of latest earnings reporting date
Return on Assets	Trailing 12M operating income divided by last reported total assets
Composite Quality	
Composite Quality	Composite score from EW avg of ROE, EBITDA Margin, Debt-to-Equity, Tax-to-Pretax Income, and Change in Headcount
Risk	
Low Risk	
Low Vol (6M Ret Vol)*	SD of the last 6M weekly returns
Size	
Small Size	
Size (Free Float)*	Current floating market cap
Size*	Current market cap

Abbreviations: 12-1M = Last 12 months excluding the most recent month · 12M = 12 months · 1M = 1 month · 3M = 3 months · 3Y = 3 years · 3ma = Three-month average · 5Y = 5 years · 6M = 6 months · BB = Buybacks (net buyback yield) · BPS = Book value per share · CAPE = Cyclically adjusted P/E · CF = Cash flow · COGS = Cost of goods sold · DPS = Dividends per share · E/P = Earnings-to-price · EBITDA = Earnings before interest, taxes, depreciation & amortization · EPS = Earnings per share · EV = Enterprise value · EW = Equal-weighted · FCF = Free cash flow · FY1 = Next fiscal year EPS estimate · FY2 = Second fiscal year EPS estimate · GPOA = Gross profit to assets · IBES = I/B/E/S (Institutional Brokers' Estimate System) · LT = Long term · NTM = Next Twelve Months · P/E = Price-to-earnings · PEG = Price/Earnings-to-Growth · R&D = Research & development · RD = Research & development · ROA = Return on assets · ROE = Return on equity · ROIC = Return on invested capital · SD = Standard deviation · ST = Short-term · T-1 = Previous period (t minus one) · YoY = Year over year · avg = Average · pct = Percent

Abbreviations: FY1 = Upcoming Fiscal Year, LT = Long Term, EW = Equal-Weighted, SD = Standard Deviation, YoY = Year-over-Year, T-1 = Previous Year. Source: Morgan Stanley Research

Quant Lens Framework

Exhibit 74: Quant Lens based on four dimensions

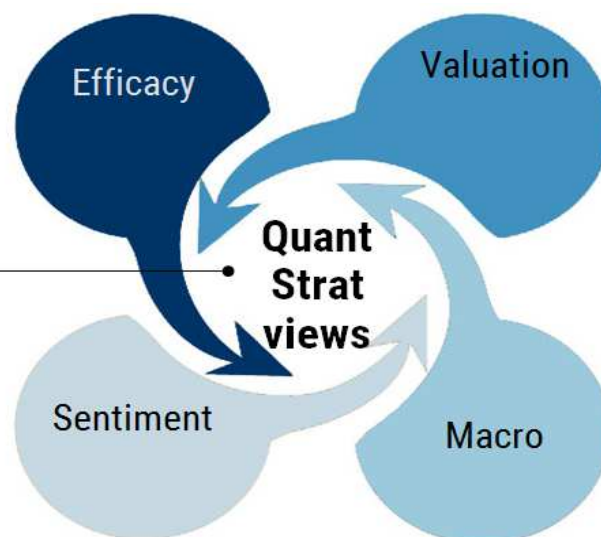
Strategic or Tactical Opportunities

Efficacy: Sharpe Ratio/ Performance

Valuation: Cheap vs Expensive

Sentiment: Earnings Revision Balance, Price ZSC

Macro: +/- Top Down environment, integration of strategy with factor exposure.



Source: Morgan Stanley Research.

Macro Regime Settings

Exhibit 75: Current Macro Regime Settings

	Europe	US	Japan	World
Economic Cycle	Slowdown	Expansion	Expansion	Expansion
Equity Risk Premium	Falling	Stable	Falling	Stable
Inflation	Stable	Stable	Falling	Falling
10-Year Yield	Flat	Flat	Rising	Flat
2-Year Yield	Rising	Rising	Rising	Rising
Yield Curve	Flattening	Flattening	Steepening	Flattening

Source: Bloomberg, Haver, LSEG Datastream, Morgan Stanley Research

Macro T-Stat Tables

Europe

Exhibit 76: Factor returns vs. long-term rates (10Y) – Europe

Long-term rates	All	Rising		Falling		Stable	
Europe	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	5.6	13.4	1.3	-6.3	-2.3	9.5	0.9
FCF Yield	6.3	6.0	-0.1	4.2	-0.8	7.8	0.7
Book/Price	1.8	9.5	1.3	-10.3	-2.2	5.8	0.8
Div Yield	7.5	6.8	-0.1	10.7	0.8	5.8	-0.5
Comp Value	5.9	11.9	1.1	-2.5	-1.7	8.3	0.6
GROWTH							
LT Growth	0.5	4.2	1.1	-4.9	-1.7	2.2	0.6
Internal Growth	2.5	-3.3	-1.6	11.0	2.4	0.0	-0.9
Comp Growth	3.2	1.4	-0.5	6.6	1.0	2.0	-0.4
MOMENTUM							
12x1M Mom	7.4	7.7	0.0	12.6	0.8	4.0	-0.6
Comp Momentum	6.2	5.3	-0.2	9.7	0.6	4.5	-0.4
Earnings Mom	3.2	3.4	0.0	3.2	0.0	3.2	0.0
QUALITY							
Accruals*	5.8	13.1	2.9	1.0	-1.9	5.3	-0.2
Net Debt/Equity*	2.8	3.7	0.2	5.9	0.9	0.3	-0.8
Capex/Sales*	2.6	-2.2	-1.6	0.9	-0.6	5.8	1.5
ROE	4.4	-4.4	-2.1	13.9	2.5	2.6	-0.6
Comp Quality	5.8	4.8	-0.2	14.8	2.2	0.6	-1.7
RISK							
Low Vol*	2.6	-3.6	-1.2	17.5	2.3	-3.8	-1.2
SIZE							
Size*	-1.5	-2.7	-0.3	-5.9	-1.2	1.9	1.1

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI Europe. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to 10-year Bund rates as proxy of the long-term rates.

Exhibit 77: Factor returns vs. short-term rates (2Y) – Europe

Short-term rates	All		Rising		Falling		Stable	
Europe	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat	
VALUE								
Fwd E/P	5.6	10.9	0.9	-1.9	-1.4	7.2	0.4	
FCF Yield	6.3	5.3	-0.3	8.1	0.7	5.9	-0.2	
Book/Price	1.8	10.2	1.5	-9.1	-1.8	3.7	0.4	
Div Yield	7.5	4.9	-0.6	14.0	1.6	5.3	-0.7	
Comp Value	5.9	10.4	0.8	2.2	-0.7	5.8	0.0	
GROWTH								
LT Growth	0.5	8.6	2.4	-7.8	-2.4	1.3	0.3	
Internal Growth	2.5	-7.0	-2.7	11.5	2.3	2.0	-0.2	
Comp Growth	3.2	2.1	-0.4	6.0	0.8	2.3	-0.3	
MOMENTUM								
12x1M Mom	7.4	7.8	0.1	11.1	0.5	5.3	-0.4	
Comp Momentum	6.2	5.3	-0.2	6.8	0.1	6.4	0.0	
Earnings Mom	3.2	2.8	-0.1	0.2	-0.7	5.0	0.5	
QUALITY								
Accruals*	5.8	10.7	1.9	3.8	-0.8	4.7	-0.6	
Net Debt/Equity*	2.8	0.3	-0.6	8.5	1.5	0.9	-0.7	
Capex/Sales*	2.6	-3.6	-2.2	3.0	0.1	5.0	1.1	
ROE	4.4	-4.0	-2.3	15.5	2.6	2.5	-0.6	
Comp Quality	5.8	2.3	-0.8	19.6	3.1	0.2	-1.9	
RISK								
Low Vol*	2.6	-10.7	-2.5	18.2	2.1	0.4	-0.5	
SIZE								
Size*	-1.5	-1.3	0.0	-6.8	-1.2	1.2	0.9	

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI Europe. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to 2-year Bund rates as proxy of the short-term rates.

Exhibit 78: Factor returns vs. yield curve (2s10s) – Europe

Yield curve	All	Steepening		Flattening		Unchanged	
Europe	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	5.6	7.7	0.4	2.9	-0.6	6.6	0.2
FCF Yield	6.3	6.6	0.1	6.7	0.2	5.7	-0.2
Book/Price	1.8	8.9	1.2	-3.1	-1.1	0.4	-0.3
Div Yield	7.5	10.3	0.8	7.5	0.0	5.0	-0.6
Comp Value	5.9	12.1	1.4	1.7	-1.0	4.6	-0.2
GROWTH							
LT Growth	0.5	-1.0	-0.5	0.3	-0.1	2.0	0.5
Internal Growth	2.5	1.0	-0.4	4.7	0.8	1.7	-0.2
Comp Growth	3.2	-1.1	-1.2	5.2	0.8	5.0	0.5
MOMENTUM							
12x1M Mom	7.4	-3.1	-1.4	10.4	0.6	13.5	1.1
Comp Momentum	6.2	-4.5	-1.6	9.6	0.8	12.2	1.2
Earnings Mom	3.2	-4.4	-1.5	5.3	0.6	7.9	1.4
QUALITY							
Accruals*	5.8	10.6	1.9	2.6	-1.6	4.6	-0.5
Net Debt/Equity*	2.8	2.1	-0.2	1.2	-0.6	4.9	0.6
Capex/Sales*	2.6	2.3	-0.1	1.7	-0.4	3.7	0.4
ROE	4.4	0.3	-0.9	7.0	0.9	5.5	0.3
Comp Quality	5.8	6.1	0.1	8.5	0.7	2.8	-0.9
RISK							
Low Vol*	2.6	-1.2	-0.5	7.5	1.1	1.0	-0.3
SIZE							
Size*	-1.5	0.3	0.4	-4.7	-1.0	0.1	0.5

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI Europe. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to 2-year and 10-year Bund rates to calculate the 2s10s.

Exhibit 80: Factor returns vs. equity risk premium – Europe

Equity risk premium	All	Rising		Falling		Stable	
Europe	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	5.6	9.7	0.8	7.3	0.3	-2.6	-1.6
FCF Yield	6.3	9.0	1.1	6.0	-0.1	2.9	-1.4
Book/Price	1.8	4.7	0.5	1.8	0.0	-2.7	-0.9
Div Yield	7.5	12.2	1.2	6.5	-0.3	1.8	-1.5
Comp Value	5.9	10.4	0.9	6.3	0.1	-1.4	-1.5
GROWTH							
LT Growth	0.5	-1.3	-0.6	2.0	0.5	1.2	0.2
Internal Growth	2.5	1.9	-0.2	3.0	0.2	2.9	0.1
Comp Growth	3.2	1.5	-0.6	2.9	-0.1	6.2	1.0
MOMENTUM							
12x1M Mom	7.4	2.2	-0.8	9.8	0.5	11.9	0.8
Comp Momentum	6.2	1.3	-0.8	8.5	0.5	10.7	0.9
Earnings Mom	3.2	-0.5	-0.8	5.3	0.7	6.1	0.8
QUALITY							
Accruals*	5.8	5.9	0.1	6.9	0.5	4.1	-0.7
Net Debt/Equity*	2.8	3.3	0.1	-0.4	-1.0	6.2	1.1
Capex/Sales*	2.6	2.6	0.0	1.3	-0.5	4.2	0.6
ROE	4.4	5.3	0.2	3.6	-0.2	4.4	0.0
Comp Quality	5.8	7.0	0.4	4.3	-0.4	5.9	0.0
RISK							
Low Vol*	2.6	-0.1	-0.4	4.0	0.3	4.8	0.4
SIZE							
Size*	-1.5	1.4	0.8	-4.0	-0.8	-2.4	-0.2

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI Europe. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. Equity risk premium is calculated as MSCI Europe earnings yield net of 10-year Bund rate.

Exhibit 79: Factor returns vs. inflation – Europe

Inflation	All	Rising		Falling		Stable	
Europe	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	5.2	3.4	-0.3	7.6	0.5	4.5	-0.2
FCF Yield	6.4	10.4	1.2	5.4	-0.4	4.6	-1.0
Book/Price	1.8	-4.8	-1.1	4.8	0.5	3.7	0.5
Div Yield	7.2	9.3	0.4	5.9	-0.4	6.8	-0.1
Comp Value	5.7	2.5	-0.5	8.3	0.5	5.6	0.0
GROWTH							
LT Growth	0.6	-1.6	-0.6	4.3	1.1	-1.0	-0.6
Internal Growth	2.5	8.7	1.9	-2.6	-1.4	2.6	0.0
Comp Growth	3.2	7.6	1.3	2.0	-0.3	1.2	-0.8
MOMENTUM							
12x1M Mom	7.6	16.9	1.5	5.5	-0.3	3.0	-0.9
Comp Momentum	6.3	13.5	1.4	4.9	-0.2	2.4	-0.8
Earnings Mom	3.3	8.8	1.5	2.5	-0.2	0.3	-0.9
QUALITY							
Accruals*	6.0	9.4	1.3	4.5	-0.6	4.9	-0.5
Net Debt/Equity*	2.9	3.9	0.2	2.3	-0.2	2.6	-0.1
Capex/Sales*	2.6	3.1	0.2	3.5	0.4	1.3	-0.5
ROE	4.3	8.8	1.1	2.4	-0.4	2.7	-0.5
Comp Quality	5.3	8.1	0.8	0.9	-1.2	7.3	0.6
RISK							
Low Vol*	2.2	15.1	2.2	-2.7	-0.8	-2.6	-0.9
SIZE							
Size*	-1.3	-7.6	-1.9	1.0	0.6	1.1	0.8

Source: Haver, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI Europe. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to HICP for the inflation.

Exhibit 81: Factor returns vs. economic cycles – Europe

Economic cycles	All	Expansion		Slowdown		Recovery		Recession	
Europe	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE									
Fwd E/P	6.1	5.6	-0.1	7.4	0.2	9.2	0.5	3.8	-0.4
FCF Yield	6.2	5.2	-0.4	9.7	1.2	2.0	-1.2	7.3	0.4
Book/Price	1.9	4.5	0.5	0.5	-0.3	14.4	1.8	-7.2	-1.6
Div Yield	7.7	4.1	-1.0	15.1	1.5	2.2	-1.3	9.8	0.5
Comp Value	6.0	6.5	0.1	5.2	-0.1	10.6	0.8	3.1	-0.5
GROWTH									
LT Growth	0.9	4.7	1.4	-3.0	-1.0	4.7	1.1	-2.5	-1.0
Internal Growth	2.7	-1.9	-1.6	9.9	2.2	-8.7	-2.5	9.4	2.0
Comp Growth	3.4	2.2	-0.4	6.6	1.2	-3.0	-1.4	6.2	0.8
MOMENTUM									
12x1M Mom	7.4	5.6	-0.4	9.3	0.4	-6.4	-1.6	16.3	1.2
Comp Momentum	6.4	2.7	-0.9	9.9	0.7	-4.6	-1.4	14.3	1.2
Earnings Mom	3.2	1.4	-0.6	6.8	1.0	-3.5	-1.1	6.8	0.9
QUALITY									
Accruals*	5.8	7.9	1.0	3.4	-0.9	3.9	-0.7	6.6	0.3
Net Debt/Equity*	2.9	-0.1	-0.9	2.8	0.0	-2.1	-1.3	8.9	1.6
Capex/Sales*	2.6	0.7	-0.8	1.5	-0.4	9.3	2.3	1.1	-0.5
ROE	4.4	-1.7	-1.8	10.8	1.7	-6.3	-2.2	12.7	2.0
Comp Quality	5.7	4.2	-0.5	4.2	-0.4	-7.1	-3.1	15.8	2.2
RISK									
Low Vol*	2.9	-2.8	-1.4	15.0	2.2	-15.3	-2.3	11.8	1.3
SIZE									
Size*	-1.7	-0.7	0.3	-2.8	-0.3	12.8	3.0	-10.8	-2.3

Source: OECD, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI Europe. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. The economic cycle is derived from the OECD Composite Leading Indicator. The regimes are defined from the level and the change in the signal: Expansion: level>=100 and spread>=0; Slowdown: level>=100 and spread<0; Recovery: level<100 and spread>=0; Recession: Level<100 and spread<0.

US

Exhibit 82: Factor returns vs. long-term rates (10Y) – US

Long-term rates	All	Rising		Falling		Stable	
US	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	5.1	6.9	0.3	5.2	0.0	3.7	-0.3
FCF Yield	7.4	5.9	-0.4	11.0	0.9	5.6	-0.6
Book/Price	0.5	9.2	1.7	-7.3	-1.6	0.9	0.1
Div Yield	0.6	-7.2	-1.6	5.9	1.0	1.8	0.3
Comp Value	3.7	6.2	0.5	1.8	-0.3	3.4	-0.1
GROWTH							
LT Growth	0.3	12.0	2.3	-6.9	-1.2	-1.9	-0.5
Internal Growth	-1.1	-11.1	-2.3	4.4	1.1	1.3	0.7
Comp Growth	1.9	-1.7	-1.1	7.3	1.6	0.0	-0.7
MOMENTUM							
12x1M Mom	3.6	-5.0	-1.1	14.7	1.5	0.3	-0.6
Comp Momentum	3.5	-2.1	-0.8	11.7	1.3	0.5	-0.6
Earnings Mom	2.0	1.6	-0.1	4.5	0.5	0.3	-0.5
QUALITY							
Accruals*	4.6	4.7	0.0	3.1	-0.5	5.8	0.5
Net Debt/Equity*	1.5	9.1	1.4	-2.9	-0.9	-0.3	-0.4
Capex/Sales*	4.7	2.2	-0.8	7.1	0.7	4.3	-0.1
ROE	4.3	-3.0	-1.7	13.3	2.1	2.1	-0.7
Comp Quality	5.6	6.5	0.2	8.0	0.5	2.9	-0.8
RISK							
Low Vol*	-1.6	-18.5	-2.2	15.6	2.1	-3.9	-0.4
SIZE							
Size*	0.9	7.1	1.2	-4.9	-1.2	1.2	0.1

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI US. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to 10-year treasury yield as proxy of the long-term rates.

Exhibit 84: Factor returns vs. yield curve (2s10s) – US

Yield curve	All	Steepening		Flattening		Unchanged	
US	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	5.1	10.0	0.9	3.3	-0.4	1.9	-0.6
FCF Yield	7.4	8.0	0.2	8.4	0.3	5.7	-0.5
Book/Price	0.5	3.7	0.6	-0.3	-0.2	-2.0	-0.6
Div Yield	0.6	1.0	0.1	1.5	0.2	-0.7	-0.3
Comp Value	3.7	7.7	0.7	3.0	-0.1	0.1	-0.7
GROWTH							
LT Growth	0.3	-3.0	-0.5	1.3	0.2	2.8	0.5
Internal Growth	-1.1	-2.0	-0.2	-1.1	0.0	-0.3	0.2
Comp Growth	1.9	-0.4	-0.6	3.7	0.7	2.3	0.1
MOMENTUM							
12x1M Mom	3.6	-1.1	-0.5	5.6	0.4	6.2	0.5
Comp Momentum	3.5	-1.1	-0.6	5.1	0.3	6.6	0.6
Earnings Mom	2.0	-2.0	-0.7	3.2	0.3	4.9	0.8
QUALITY							
Accruals*	4.6	8.3	1.2	3.5	-0.5	2.0	-1.1
Net Debt/Equity*	1.5	1.6	0.0	1.2	-0.1	1.7	0.0
Capex/Sales*	4.7	6.7	0.6	4.8	0.0	2.3	-0.7
ROE	4.3	2.4	-0.4	4.9	0.2	5.8	0.4
Comp Quality	5.6	8.6	0.5	4.3	-0.5	3.8	-0.5
RISK							
Low Vol*	-1.6	-5.7	-0.5	-1.3	0.1	2.5	0.7
SIZE							
Size*	0.9	8.0	1.3	-2.3	-0.9	-3.1	-1.0

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI US. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to 2-year and 10-year treasury yields to calculate the 2s10s.

Exhibit 83: Factor returns vs. short-term rates (2Y) – US

Short-term rates	All	Rising		Falling		Stable	
US	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	5.1	1.7	-0.6	8.0	0.4	5.3	0.0
FCF Yield	7.4	6.7	-0.2	11.1	0.9	5.8	-0.6
Book/Price	0.5	2.2	0.3	-4.1	-0.8	2.2	0.4
Div Yield	0.6	-4.5	-1.0	11.5	1.8	-2.7	-0.9
Comp Value	3.7	1.4	-0.4	5.8	0.3	3.7	0.0
GROWTH							
LT Growth	0.3	9.6	1.8	-13.3	-1.8	2.9	0.6
Internal Growth	-1.1	-6.5	-1.3	10.2	1.9	-4.6	-1.1
Comp Growth	1.9	1.9	0.0	6.5	1.2	-0.6	-1.0
MOMENTUM							
12x1M Mom	3.6	5.3	0.3	13.5	1.1	-2.9	-1.2
Comp Momentum	3.5	5.6	0.4	8.9	0.7	-0.7	-0.9
Earnings Mom	2.0	4.5	0.6	-0.3	-0.4	2.0	0.0
QUALITY							
Accruals*	4.6	3.0	-0.6	6.4	0.5	4.5	0.0
Net Debt/Equity*	1.5	10.4	1.8	-6.2	-1.2	1.0	-0.1
Capex/Sales*	4.7	4.4	-0.1	6.7	0.5	3.7	-0.4
ROE	4.3	0.8	-0.8	13.6	1.8	1.1	-1.0
Comp Quality	5.6	3.7	-0.5	11.3	1.0	3.4	-0.7
RISK							
Low Vol*	-1.6	-9.7	-1.1	21.6	2.4	-10.1	-1.5
SIZE							
Size*	0.9	2.2	0.3	-5.6	-1.1	3.7	0.8

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI US. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to 2-year treasury yield as proxy of the short-term rates.

Exhibit 85: Factor returns vs. inflation – US

Inflation	All	Rising		Falling		Stable	
US	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	5.1	1.5	-0.5	3.6	-0.3	9.0	0.8
FCF Yield	7.4	6.9	-0.1	6.2	-0.4	8.9	0.5
Book/Price	0.5	1.7	0.2	2.1	0.4	-1.8	-0.6
Div Yield	0.6	3.6	0.6	-5.7	-1.5	4.0	0.8
Comp Value	3.7	3.7	0.0	2.0	-0.4	5.1	0.3
GROWTH							
LT Growth	0.3	-0.7	-0.2	7.7	1.5	-5.4	-1.1
Internal Growth	-1.1	1.8	0.7	-6.1	-1.2	1.0	0.5
Comp Growth	1.9	2.4	0.1	0.5	-0.5	2.8	0.3
MOMENTUM							
12x1M Mom	3.6	5.7	0.3	-2.3	-0.8	7.2	0.6
Comp Momentum	3.5	4.4	0.2	-1.1	-0.7	6.8	0.6
Earnings Mom	2.0	2.0	0.0	0.6	-0.3	3.3	0.3
QUALITY							
Accruals*	4.6	6.6	0.6	2.6	-0.8	4.8	0.1
Net Debt/Equity*	1.5	5.2	0.7	4.1	0.6	-3.6	-1.2
Capex/Sales*	4.7	1.3	-1.0	8.5	1.1	3.9	-0.2
ROE	4.3	4.0	-0.1	0.4	-1.0	8.1	1.1
Comp Quality	5.6	6.8	0.4	6.3	0.2	4.0	-0.4
RISK							
Low Vol*	-1.6	1.4	0.4	-12.6	-1.5	5.8	1.2
SIZE							
Size*	0.9	0.3	-0.1	7.2	1.3	-4.2	-1.3

Source: Haver, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI US. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to PCE for the inflation.

Exhibit 86: Factor returns vs. equity risk premium – US

Equity risk premium	All	Rising		Falling		Stable	
US	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	5.1	9.0	0.6	4.4	-0.1	2.1	-0.7
FCF Yield	7.4	5.0	-0.6	6.6	-0.3	10.6	0.9
Book/Price	0.5	8.0	1.4	-1.3	-0.4	-4.6	-1.3
Div Yield	0.6	-4.8	-1.1	3.6	0.7	2.7	0.5
Comp Value	3.7	6.2	0.4	4.0	0.1	0.9	-0.6
GROWTH							
LT Growth	0.3	6.2	0.9	-3.3	-0.7	-1.4	-0.4
Internal Growth	-1.1	-8.3	-1.4	1.9	0.8	2.4	0.9
Comp Growth	1.9	-2.4	-1.3	3.5	0.5	4.4	0.9
MOMENTUM							
12x1M Mom	3.6	-6.7	-1.2	8.5	0.8	8.1	0.9
Comp Momentum	3.5	-5.0	-1.1	7.5	0.8	7.3	0.8
Earnings Mom	2.0	-2.2	-0.8	5.0	0.8	2.9	0.2
QUALITY							
Accruals*	4.6	6.4	0.6	3.2	-0.6	4.3	-0.1
Net Debt/Equity*	1.5	1.2	-0.1	2.1	0.1	1.2	-0.1
Capex/Sales*	4.7	4.8	0.0	4.1	-0.2	5.1	0.1
ROE	4.3	0.1	-0.9	5.4	0.3	7.2	0.9
Comp Quality	5.6	1.6	-1.1	6.9	0.3	7.9	0.6
RISK							
Low Vol*	-1.6	-10.2	-1.0	0.9	0.4	3.9	0.9
SIZE							
Size*	0.9	13.4	2.4	-2.2	-0.8	-7.6	-2.2

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI US. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. Equity risk premium is calculated as MSCI US earnings yield net of 10-year treasury yield.

Exhibit 87: Factor returns vs. economic cycles – US

Economic cycles	All	Expansion		Slowdown		Recovery		Recession	
US	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE									
Fwd E/P	5.1	1.1	-0.9	-0.9	-0.8	4.8	-0.1	12.1	1.1
FCF Yield	7.6	5.9	-0.5	4.1	-0.7	4.2	-0.8	13.3	1.5
Book/Price	-0.1	-3.7	-1.0	0.4	0.1	6.8	1.3	-2.0	-0.4
Div Yield	0.5	-1.6	-0.5	-0.1	-0.1	-7.4	-1.9	8.4	1.5
Comp Value	3.3	-1.0	-0.9	-0.1	-0.5	3.3	0.0	9.1	0.9
GROWTH									
LT Growth	0.6	7.3	1.5	2.3	0.2	2.1	0.3	-7.1	-1.2
Internal Growth	-0.9	-3.0	-0.6	-1.2	0.0	-8.8	-2.1	6.7	1.6
Comp Growth	2.3	4.7	1.0	4.8	0.7	-6.6	-2.3	5.1	0.8
MOMENTUM									
12x1M Mom	3.7	9.7	1.2	1.8	-0.2	-5.3	-1.1	6.0	0.3
Comp Momentum	3.8	9.4	1.3	4.3	0.1	-4.8	-1.2	4.6	0.1
Earnings Mom	2.2	6.2	1.2	5.0	0.5	-2.4	-0.9	0.5	-0.4
QUALITY									
Accruals*	4.4	0.3	-2.0	5.3	0.3	4.7	0.1	7.2	0.9
Net Debt/Equity*	1.7	3.5	0.4	5.5	0.6	2.4	0.2	-2.4	-0.7
Capex/Sales*	4.9	2.7	-0.7	-2.6	-1.7	7.8	0.8	8.7	1.1
ROE	4.8	5.8	0.4	4.0	-0.2	-4.9	-2.0	11.3	1.4
Comp Quality	6.4	4.1	-0.8	4.5	-0.5	5.8	-0.1	9.9	0.7
RISK									
Low Vol*	-1.3	-4.8	-0.6	-3.4	-0.2	-20.6	-2.4	16.7	2.1
SIZE									
Size*	0.4	-3.3	-1.1	-1.8	-0.4	9.7	1.9	-2.0	-0.5

Source: OECD, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI US. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. The economic cycle is derived from the OECD Composite Leading Indicator. The regimes are defined from the level and the change in the signal: Expansion: level>=100 and spread>=0; Slowdown: level>=100 and spread<0; Recovery: level<100 and spread>=0; Recession: Level<100 and spread<0.

Japan

Exhibit 88: Factor returns vs. long-term rates (10Y) – Japan

Long-term rates	All	Rising		Falling		Stable	
Japan	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	6.8	16.2	1.6	1.1	-1.3	6.2	-0.2
FCF Yield	6.5	7.2	0.2	4.8	-0.6	7.2	0.3
Book/Price	7.4	22.0	2.4	-3.6	-2.5	7.7	0.1
Div Yield	7.6	12.9	1.0	3.5	-1.0	7.7	0.0
Comp Value	9.3	21.2	2.1	1.0	-2.0	9.1	0.0
GROWTH							
LT Growth	-1.3	-1.2	0.0	1.4	0.7	-3.1	-0.6
Internal Growth	2.0	-3.8	-1.8	3.6	0.6	3.5	0.7
Comp Growth	0.6	-4.6	-1.3	3.2	0.9	1.2	0.2
MOMENTUM							
12x1M Mom	-0.9	-13.1	-1.8	4.8	1.0	0.9	0.4
Comp Momentum	-0.2	-9.7	-1.6	4.8	1.1	0.8	0.2
Earnings Mom	-0.4	-2.2	-0.4	0.2	0.2	0.0	0.1
QUALITY							
Accruals*	3.1	5.6	0.8	1.4	-0.7	3.1	0.0
Net Debt/Equity*	1.9	-8.7	-2.1	8.1	1.7	2.7	0.3
Capex/Sales*	1.0	2.4	0.4	-6.1	-2.4	4.7	1.4
ROE	-0.4	-6.5	-1.3	4.5	1.2	-0.7	-0.1
Comp Quality	2.4	-8.2	-1.9	13.9	3.0	0.0	-0.7
RISK							
Low Vol*	-3.3	-15.7	-1.8	7.1	1.6	-4.3	-0.2
SIZE							
Size*	0.4	4.3	0.9	-4.6	-1.2	1.8	0.4

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI Japan. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to 10-year Japan government bond yield as proxy of the long-term rates.

Exhibit 90: Factor returns vs. yield curve (2s10s) – Japan

Yield curve	All	Steepening		Flattening		Unchanged	
Japan	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	6.8	5.2	-0.4	8.2	0.3	6.8	0.0
FCF Yield	6.5	4.0	-0.9	8.3	0.7	6.9	0.2
Book/Price	7.4	13.6	1.3	5.3	-0.4	3.5	-0.8
Div Yield	7.6	6.8	-0.2	8.9	0.3	6.7	-0.2
Comp Value	9.3	10.4	0.2	10.4	0.2	6.9	-0.5
GROWTH							
LT Growth	-1.3	-2.9	-0.4	-0.8	0.2	-0.4	0.3
Internal Growth	2.0	-1.2	-1.2	5.1	1.2	1.5	-0.2
Comp Growth	0.6	-1.7	-0.7	1.1	0.2	2.2	0.5
MOMENTUM							
12x1M Mom	-0.9	-3.0	-0.4	0.8	0.3	-0.8	0.0
Comp Momentum	-0.2	-1.7	-0.3	1.6	0.4	-0.7	-0.1
Earnings Mom	-0.4	-2.3	-0.5	0.1	0.2	1.0	0.4
QUALITY							
Accruals*	3.1	4.4	0.5	2.5	-0.2	2.5	-0.3
Net Debt/Equity*	1.9	0.9	-0.2	4.9	0.9	-0.6	-0.6
Capex/Sales*	1.0	2.4	0.5	-1.1	-0.7	2.1	0.4
ROE	-0.4	-5.1	-1.2	1.9	0.5	1.8	0.6
Comp Quality	2.4	1.3	-0.2	5.6	0.8	-0.3	-0.7
RISK							
Low Vol*	-3.3	-9.6	-1.0	5.3	1.4	-7.1	-0.7
SIZE							
Size*	0.4	1.9	0.4	0.3	0.0	-1.1	-0.4

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI Japan. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to 2-year and 10-year Japan government bond yields to calculate the 2s10s.

Exhibit 89: Factor returns vs. short-term rates (2Y) – Japan

Short-term rates	All	Rising		Falling		Stable	
Japan	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	6.8	14.7	1.2	5.7	-0.2	5.4	-0.4
FCF Yield	6.5	5.8	-0.2	4.7	-0.5	7.1	0.3
Book/Price	7.4	19.6	1.3	0.3	-1.3	6.8	-0.2
Div Yield	7.6	18.8	1.5	1.8	-1.1	6.8	-0.3
Comp Value	9.3	20.7	1.3	4.8	-1.0	8.1	-0.3
GROWTH							
LT Growth	-1.3	-6.5	-0.8	0.4	0.3	-0.8	0.2
Internal Growth	2.0	0.6	-0.4	0.1	-0.5	2.8	0.4
Comp Growth	0.6	-5.6	-1.3	3.2	0.7	1.1	0.2
MOMENTUM							
12x1M Mom	-0.9	-12.1	-1.0	6.7	1.0	-0.7	0.1
Comp Momentum	-0.2	-15.0	-1.8	6.0	1.0	1.1	0.4
Earnings Mom	-0.4	-10.2	-2.1	-0.2	0.1	1.6	0.7
QUALITY							
Accruals*	3.1	3.0	0.0	2.3	-0.2	3.4	0.1
Net Debt/Equity*	1.9	-3.9	-1.3	5.7	0.8	2.1	0.1
Capex/Sales*	1.0	3.6	0.4	-3.9	-1.3	1.8	0.4
ROE	-0.4	-6.7	-0.9	3.2	0.6	0.0	0.1
Comp Quality	2.4	-5.7	-1.1	11.9	1.9	1.5	-0.3
RISK							
Low Vol*	-3.3	-19.3	-1.7	2.2	0.6	-1.5	0.4
SIZE							
Size*	0.4	4.3	0.5	-8.6	-1.7	2.0	0.6

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI Japan. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to 2-year Japan government bond yield as proxy of the short-term rates.

Exhibit 91: Factor returns vs. inflation – Japan

Inflation	All	Rising		Falling		Stable	
Japan	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	6.8	7.3	0.1	3.1	-0.8	10.4	0.8
FCF Yield	6.5	8.8	1.0	4.9	-0.5	5.6	-0.4
Book/Price	7.4	10.1	0.6	7.6	0.0	3.8	-0.7
Div Yield	7.6	12.4	1.3	3.2	-1.0	6.8	-0.2
Comp Value	9.3	13.5	1.1	8.5	-0.2	5.2	-0.8
GROWTH							
LT Growth	-1.3	1.5	0.9	-6.6	-1.3	1.4	0.8
Internal Growth	2.0	2.8	0.3	-0.1	-0.9	3.5	0.5
Comp Growth	0.6	0.2	-0.1	-1.7	-0.7	3.7	0.9
MOMENTUM							
12x1M Mom	-0.9	0.5	0.3	-2.1	-0.2	-1.2	0.0
Comp Momentum	-0.2	-1.5	-0.3	0.3	0.1	0.7	0.2
Earnings Mom	-0.4	-1.4	-0.3	-0.6	-0.1	1.1	0.4
QUALITY							
Accruals*	3.1	4.6	0.6	2.7	-0.2	1.7	-0.5
Net Debt/Equity*	1.9	0.6	-0.4	2.5	0.2	2.8	0.2
Capex/Sales*	1.0	-0.3	-0.4	0.7	-0.1	2.9	0.7
ROE	-0.4	-2.2	-0.4	0.4	0.2	0.9	0.3
Comp Quality	2.4	0.1	-0.6	0.6	-0.5	7.4	1.1
RISK							
Low Vol*	-3.3	2.6	1.0	-8.5	-0.9	-4.4	-0.2
SIZE							
Size*	0.4	-0.4	-0.2	-0.1	-0.1	2.0	0.4

Source: Haver, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI Japan. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to Japan general consumer price index for the inflation.

Exhibit 92: Factor returns vs. equity risk premium – Japan

Equity risk premium	All	Rising		Falling		Stable	
Japan	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	6.8	7.8	0.3	5.5	-0.3	6.1	-0.1
FCF Yield	6.5	7.7	0.5	5.3	-0.4	5.2	-0.5
Book/Price	7.4	12.4	1.1	0.7	-1.2	4.2	-0.6
Div Yield	7.6	10.6	0.9	2.9	-1.1	6.5	-0.2
Comp Value	9.3	13.2	1.0	3.0	-1.2	8.4	-0.2
GROWTH							
LT Growth	-1.3	-1.7	-0.1	0.2	0.4	-2.6	-0.3
Internal Growth	2.0	0.4	-0.7	3.2	0.4	4.2	0.7
Comp Growth	0.6	-1.3	-0.7	1.5	0.3	3.8	1.0
MOMENTUM							
12x1M Mom	-0.9	-5.6	-0.9	3.0	0.7	5.0	1.0
Comp Momentum	-0.2	-4.0	-0.9	3.5	0.8	4.0	0.8
Earnings Mom	-0.4	-2.7	-0.8	2.8	0.9	1.0	0.4
QUALITY							
Accruals*	3.1	4.3	0.6	4.7	0.6	-2.0	-1.6
Net Debt/Equity*	1.9	1.0	-0.3	3.1	0.3	2.5	0.2
Capex/Sales*	1.0	1.3	0.1	2.8	0.6	-2.2	-1.1
ROE	-0.4	-3.2	-0.8	1.3	0.4	4.3	1.2
Comp Quality	2.4	-1.1	-0.9	3.4	0.3	9.6	1.7
RISK							
Low Vol*	-3.3	-4.6	-0.2	-4.8	-0.3	1.9	0.9
SIZE							
Size*	0.4	4.7	1.1	-3.2	-0.9	-5.1	-1.3

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI Japan. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. Equity risk premium is calculated as MSCI Japan earnings yield net of 10-year Japan government bond yield.

Exhibit 93: Factor returns vs. economic cycles – Japan

Economic cycles	All	Expansion		Slowdown		Recovery		Recession	
Japan	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE									
Fwd E/P	7.1	14.0	1.7	9.6	0.5	-1.2	-1.6	4.5	-0.5
FCF Yield	6.6	8.1	0.5	7.7	0.4	-0.6	-2.3	9.2	0.9
Book/Price	7.8	9.4	0.3	12.8	0.9	7.9	0.0	3.1	-0.8
Div Yield	7.1	8.3	0.3	16.2	1.4	5.9	-0.3	0.7	-1.5
Comp Value	9.5	12.8	0.8	16.8	1.3	6.4	-0.6	3.7	-1.2
GROWTH									
LT Growth	-1.5	1.2	0.8	-4.6	-0.6	-1.2	0.1	-1.9	-0.1
Internal Growth	1.4	0.0	-0.5	5.4	1.0	0.5	-0.3	0.4	-0.4
Comp Growth	0.1	1.6	0.5	-1.6	-0.5	-0.4	-0.2	0.4	0.1
MOMENTUM									
12x1M Mom	-1.0	1.9	0.6	-5.0	-0.6	-3.5	-0.4	0.8	0.3
Comp Momentum	-0.1	4.6	1.2	-3.3	-0.6	-6.0	-1.0	1.5	0.3
Earnings Mom	-0.4	5.6	1.8	-1.1	-0.2	-6.1	-1.3	-1.5	-0.3
QUALITY									
Accruals*	3.7	5.2	0.6	3.6	0.0	2.2	-0.5	3.3	-0.1
Net Debt/Equity*	2.3	-4.5	-1.7	0.5	-0.4	5.2	0.7	7.7	1.4
Capex/Sales*	1.3	5.5	1.5	2.4	0.3	-4.3	-1.5	0.3	-0.3
ROE	-0.5	-2.9	-0.6	-0.3	0.0	-6.1	-1.2	4.9	1.1
Comp Quality	2.3	-6.5	-2.2	3.0	0.1	3.6	0.3	8.9	1.5
RISK									
Low Vol*	-4.2	-5.9	-0.3	-1.0	0.4	-15.5	-1.6	2.3	1.0
SIZE									
Size*	0.7	2.4	0.4	3.1	0.5	-1.9	-0.6	-0.8	-0.3

Source: OECD, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI Japan. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. The economic cycle is derived from the OECD Composite Leading Indicator. The regimes are defined from the level and the change in the signal: Expansion: level>=100 and spread>=0; Slowdown: level>=100 and spread<0; Recovery: level<100 and spread>=0; Recession: Level<100 and spread<0.

World

Exhibit 94: Factor returns vs. long-term rates (10Y) – World

Long-term rates	All	Rising		Falling		Stable	
World	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	6.8	16.6	1.8	0.0	-1.1	5.5	-0.3
FCF Yield	7.8	8.9	0.4	7.0	-0.3	7.7	0.0
Book/Price	0.5	10.4	2.0	-0.1	-0.1	-4.4	-1.2
Div Yield	5.3	1.2	-0.9	13.1	1.6	3.2	-0.7
Comp Value	6.6	16.0	1.8	5.0	-0.3	2.5	-1.1
GROWTH							
LT Growth	0.0	6.5	2.1	-7.7	-2.0	0.9	0.3
Internal Growth	2.7	-5.3	-2.8	7.9	1.7	4.0	0.6
Comp Growth	3.2	-1.7	-1.6	2.7	-0.2	6.0	1.2
MOMENTUM							
12x1M Mom	5.7	-3.7	-1.4	8.6	0.4	9.1	0.6
Comp Momentum	5.0	-2.6	-1.3	7.1	0.3	7.9	0.7
Earnings Mom	1.8	-1.2	-0.8	2.2	0.1	3.1	0.5
QUALITY							
Accruals*	4.6	7.3	1.1	1.4	-1.3	5.0	0.2
Net Debt/Equity*	1.9	2.6	0.2	2.6	0.2	1.1	-0.3
Capex/Sales*	3.3	0.4	-1.1	3.1	-0.1	5.0	0.8
ROE	5.5	-0.5	-1.6	8.6	0.8	6.9	0.5
Comp Quality	4.8	1.1	-1.5	8.6	1.3	4.7	-0.1
RISK							
Low Vol*	1.3	-11.7	-2.1	14.9	1.9	0.7	-0.1
SIZE							
Size*	-1.2	1.0	0.6	3.8	1.2	-5.0	-1.2

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI World. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to market cap weighted 10-year government bond yields as proxy of the long-term rates.

Exhibit 96: Factor returns vs. yield curve (2s10s) – World

Yield curve	All	Steepening		Flattening		Unchanged	
World	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	6.8	4.7	-0.4	9.9	0.6	5.6	-0.3
FCF Yield	7.8	6.2	-0.6	10.9	1.1	6.0	-0.9
Book/Price	0.5	7.0	1.3	-0.9	-0.3	-3.5	-1.0
Div Yield	5.3	0.9	-1.0	9.6	1.1	4.7	-0.2
Comp Value	6.6	8.1	0.3	7.9	0.3	4.0	-0.6
GROWTH							
LT Growth	0.0	-1.1	-0.3	-1.7	-0.6	2.6	0.8
Internal Growth	2.7	-2.0	-1.6	6.2	1.3	3.0	0.1
Comp Growth	3.2	-4.7	-2.6	6.2	1.1	6.5	1.4
MOMENTUM							
12x1M Mom	5.7	-5.0	-1.4	7.8	0.3	12.3	1.4
Comp Momentum	5.0	-4.1	-1.4	7.6	0.5	9.8	1.2
Earnings Mom	1.8	-4.6	-1.4	3.9	0.7	4.8	1.1
QUALITY							
Accruals*	4.6	9.1	1.6	3.6	-0.5	2.0	-1.4
Net Debt/Equity*	1.9	4.7	0.7	-0.2	-0.7	1.6	-0.1
Capex/Sales*	3.3	2.5	-0.3	0.8	-1.0	6.6	1.4
ROE	5.5	-3.3	-2.3	8.8	1.0	9.2	1.3
Comp Quality	4.8	5.5	0.2	5.0	0.1	4.1	-0.4
RISK							
Low Vol*	1.3	-8.9	-1.3	5.6	0.7	5.4	0.9
SIZE							
Size*	-1.2	4.7	1.5	-4.4	-0.8	-2.7	-0.5

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI World. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to market cap weighted 2-year and 10-year government bond yields to calculate the 2s10s.

Exhibit 95: Factor returns vs. short-term rates (2Y) – World

Short-term rates	All	Rising		Falling		Stable	
World	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	6.8	12.2	1.0	-4.8	-1.6	9.2	0.6
FCF Yield	7.8	8.1	0.1	6.7	-0.3	8.1	0.1
Book/Price	0.5	0.6	0.0	-0.6	-0.2	0.8	0.1
Div Yield	5.3	0.3	-1.1	14.0	1.6	4.0	-0.4
Comp Value	6.6	8.8	0.4	2.9	-0.6	7.1	0.1
GROWTH							
LT Growth	0.0	7.3	2.2	-9.6	-2.1	0.8	0.3
Internal Growth	2.7	-0.9	-1.3	9.4	1.8	1.5	-0.5
Comp Growth	3.2	3.8	0.2	1.5	-0.5	3.6	0.2
MOMENTUM							
12x1M Mom	5.7	8.2	0.4	11.8	0.8	2.3	-0.6
Comp Momentum	5.0	7.4	0.5	8.9	0.6	2.5	-0.6
Earnings Mom	1.8	4.6	0.9	0.7	-0.3	1.1	-0.2
QUALITY							
Accruals*	4.6	6.9	1.0	6.0	0.4	3.1	-0.8
Net Debt/Equity*	1.9	2.8	0.3	4.7	0.7	0.4	-0.6
Capex/Sales*	3.3	0.4	-1.1	1.9	-0.4	5.1	0.9
ROE	5.5	4.9	-0.2	6.3	0.2	5.4	0.0
Comp Quality	4.8	3.1	-0.7	11.3	1.9	3.0	-1.0
RISK							
Low Vol*	1.3	-3.2	-0.8	19.8	2.2	-4.0	-1.1
SIZE							
Size*	-1.2	-5.0	-1.1	1.7	0.6	-0.7	0.1

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI World. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to market cap weighted 2-year government bond yields as proxy of the long-term rates.

Exhibit 97: Factor returns vs. inflation – World

Inflation	All	Rising		Falling		Stable	
World	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	1.5	-1.2	-0.4	4.2	0.6	1.6	0.0
FCF Yield	6.1	9.6	1.0	4.5	-0.7	4.1	-0.8
Book/Price	-2.5	-1.4	0.2	-3.1	-0.1	-3.0	-0.1
Div Yield	1.0	5.8	1.1	-4.9	-1.4	2.1	0.3
Comp Value	1.5	3.3	0.3	0.4	-0.2	0.8	-0.2
GROWTH							
LT Growth	0.4	-3.5	-1.2	5.7	1.8	-1.0	-0.5
Internal Growth	0.6	2.7	0.8	-3.8	-1.9	3.1	0.9
Comp Growth	2.3	-0.1	-0.7	4.8	0.9	2.2	0.0
MOMENTUM							
12x1M Mom	4.0	2.8	-0.2	0.2	-0.5	9.7	1.0
Comp Momentum	4.4	1.8	-0.6	1.2	-0.5	10.8	1.2
Earnings Mom	2.6	-0.2	-0.8	0.6	-0.4	7.9	1.4
QUALITY							
Accruals*	3.4	6.8	1.1	-1.0	-2.1	4.4	0.5
Net Debt/Equity*	2.7	1.8	-0.2	4.1	0.4	1.9	-0.2
Capex/Sales*	4.0	3.1	-0.3	7.9	1.2	0.8	-1.0
ROE	4.5	3.8	-0.2	6.0	0.5	3.7	-0.3
Comp Quality	3.8	6.4	0.9	1.0	-1.1	3.8	0.0
RISK							
Low Vol*	-0.2	6.7	1.1	-8.2	-1.2	0.8	0.2
SIZE							
Size*	-2.0	-4.1	-0.6	-0.1	0.5	-1.7	0.1

Source: Haver, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI World. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. We refer to the Composite CPI for the Advanced Economies per Haver.

Exhibit 98: Factor returns vs. equity risk premium – World

Equity risk premium	All	Rising		Falling		Stable	
World	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE							
Fwd E/P	6.8	8.6	0.3	7.8	0.2	4.0	-0.6
FCF Yield	7.8	6.4	-0.5	8.5	0.3	8.5	0.3
Book/Price	0.5	5.5	1.0	0.1	-0.1	-4.4	-1.1
Div Yield	5.3	6.0	0.1	5.1	-0.1	4.9	-0.1
Comp Value	6.6	8.9	0.5	6.8	0.0	3.9	-0.6
GROWTH							
LT Growth	0.0	-0.8	-0.3	1.0	0.3	0.1	0.0
Internal Growth	2.7	-0.1	-0.9	2.8	0.0	5.5	1.1
Comp Growth	3.2	-0.5	-1.4	3.9	0.2	6.4	1.2
MOMENTUM							
12x1M Mom	5.7	-1.8	-1.1	5.6	0.0	13.8	1.4
Comp Momentum	5.0	-1.5	-1.1	5.5	0.1	11.4	1.3
Earnings Mom	1.8	-3.2	-1.3	3.3	0.6	5.6	1.2
QUALITY							
Accruals*	4.6	4.8	0.1	3.8	-0.4	5.2	0.3
Net Debt/Equity*	1.9	1.5	-0.1	-0.5	-0.8	4.5	1.0
Capex/Sales*	3.3	-0.3	-1.4	5.2	0.8	5.4	0.8
ROE	5.5	1.8	-1.1	5.8	0.1	9.0	1.1
Comp Quality	4.8	4.6	-0.1	3.1	-0.8	6.7	0.8
RISK							
Low Vol*	1.3	-6.4	-1.1	0.3	-0.2	10.5	1.8
SIZE							
Size*	-1.2	5.0	1.6	-4.2	-0.9	-4.7	-1.0

Source: Datastream, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI World. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. Equity risk premium is calculated as MSCI World earnings yield net of market cap weighted 10-year government bond yields.

Exhibit 99: Factor returns vs. economic cycles – World

Economic cycles	All	Expansion		Slowdown		Recovery		Recession	
World	Ret.	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat	Ret.	t-stat
VALUE									
Fwd E/P	8.1	7.9	-0.1	11.7	0.5	20.0	2.4	-3.0	-2.0
FCF Yield	7.5	6.9	-0.3	13.5	1.8	4.6	-1.0	5.9	-0.6
Book/Price	1.8	-3.2	-1.4	2.2	0.1	15.7	2.9	-3.4	-1.2
Div Yield	5.8	1.1	-1.5	15.7	1.8	9.6	1.0	0.9	-1.2
Comp Value	7.5	4.1	-1.0	11.8	0.7	20.4	2.7	-1.5	-1.9
GROWTH									
LT Growth	0.7	4.9	1.7	-4.9	-1.3	0.4	-0.1	0.5	-0.1
Internal Growth	2.5	1.9	-0.3	8.8	1.6	1.3	-0.4	-0.2	-0.9
Comp Growth	3.4	6.4	1.3	7.1	1.0	-1.8	-1.4	1.7	-0.6
MOMENTUM									
12x1M Mom	5.9	11.9	1.3	6.3	0.1	-3.1	-1.2	6.2	0.0
Comp Momentum	5.4	9.9	1.1	6.7	0.2	-2.3	-1.2	5.4	0.0
Earnings Mom	2.2	4.3	0.8	4.8	0.8	-0.5	-0.7	0.1	-0.5
QUALITY									
Accruals*	5.0	4.0	-0.5	5.9	0.3	4.7	-0.1	5.5	0.2
Net Debt/Equity*	2.1	2.7	0.2	-1.3	-0.9	-3.3	-1.6	7.7	1.8
Capex/Sales*	3.4	5.3	0.8	0.9	-0.8	3.4	0.0	3.4	0.0
ROE	5.4	6.3	0.3	10.3	1.2	-0.6	-1.4	5.4	0.0
Comp Quality	5.4	3.5	-1.0	6.1	0.2	0.2	-1.6	10.8	1.8
RISK									
Low Vol*	0.7	-0.5	-0.3	12.7	1.6	-13.3	-2.1	3.7	0.5
SIZE									
Size*	-0.3	-2.8	-0.8	-2.9	-0.5	9.6	2.5	-3.3	-0.8

Source: OECD, FactSet, Morgan Stanley Research. Data as of Dec 2024. Welch t-test is performed to test two samples having unequal variances and unequal sample sizes. Factor performance refers to long/short returns of the top vs. bottom quintile from MSCI World. The portfolios are rebalanced monthly and returns are equal weighted. * prefers low values. The economic cycle is derived from the OECD Composite Leading Indicator. The regimes are defined from the level and the change in the signal: Expansion: level>=100 and spread>=0; Slowdown: level>=100 and spread<0; Recovery: level<100 and spread>=0; Recession: Level<100 and spread<0.

The performance data provided is a hypothetical illustration of mathematical principles, it does not predict or project the performance of an investment or investment strategy. Past performance is no guarantee of future results.

Quant-Driven Sector Recommendation

As of end-April 2026, our systematic sector rotation framework ([Quantitative Equity Research: Rotation With Traction – Launching Our Systematic Sector Rotation Framework, 6 Jul 2023](#) and [Quantitative Equity Research: Quant Matters – Unlocking Systematic Sector Rotation Opportunities in Japan, 11 Nov 2024](#)) prefers and disfavours the following sectors in the three regions:

US

- (+) IT: tailwind from sentiment implied by Consumer Confidence and improving earnings revision breadth
- (+) Energy: tailwind from sentiment implied by copper price and improving residual momentum
- (–) Utilities: headwind from sentiment implied by 10Y treasury yield and unattractive dividend yield
- (–) Real Estate: headwind from sentiment implied by Consumer Confidence and deteriorating earnings revision breadth

EUROPE

- (+) Financials: tailwind from sentiment implied by EC CSI and improving earnings revision breadth
- (+) Energy: tailwind from sentiment implied by copper price and improving earnings revision breadth
- (–) Materials: headwind from sentiment implied by unemployment number and deteriorating profitability in terms of gross profit-to-assets
- (–) Industrials: headwind from sentiment implied by HICP inflation and increasing leverage in terms of change in long-term debt-to-assets

JAPAN

- (+) Energy: tailwind from sentiment implied by copper price and improving revenue revision breadth
- (+) Utilities: tailwind from sentiment implied by employment number and attractive dividend yield
- (–) Materials: headwind from sentiment implied by Economy Watchers Survey and deteriorating residual momentum
- (–) IT: headwind from sentiment implied by high beta and unattractive free cash flow yield

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Underweight/Sell	555	15%	84	9%	15%	202	12%
Total	3,673		909			1627	

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